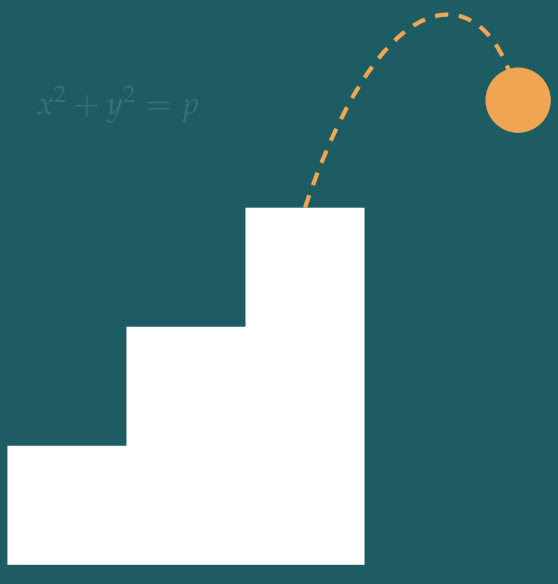
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2018 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2018 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2018 Worked Solutions

Worked solutions to all 39 questions of the 2018 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

First published online November 2026.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2018, Question 1

Pure • Area Between Curves

STEP I 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q1

The key idea

Setting the cubic equal to the line gives a factor of x and then a difference of two squares, so all three intersections come out cleanly. After that everything is routine calculus, and the substitution $c = b - a$ (the x-coordinate of P) keeps the algebra tidy. The final inequality looks daunting until both areas are written in terms of c: then it collapses to $b > a$.

Solution

The intersections and the sketch

At an intersection, $x(b-x)^2 = a^2x$, so either $x = 0$ (the origin) or

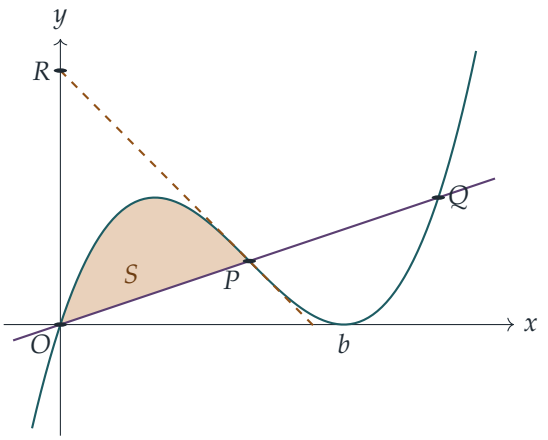
$$(b-x)^2 = a^2 \quad \Longrightarrow \quad b - x = \pm a \quad \Longrightarrow \quad x = b - a \text{ or } x = b + a.$$

Since $a > 0$, $b - a < b + a$, so

$$P = (b-a,\ a^2(b-a)), \qquad Q = (b+a,\ a^2(b+a)).$$

Because $0 < a < b$, both points lie in the first quadrant.

For the sketch: the cubic $y = x(b-x)^2$ has a single root at O and a repeated root at $x = b$, where it touches the x-axis. It is negative for $x < 0$, positive for $x > 0$ (apart from $x = b$), and tends to $+\infty$ as $x \to \infty$. Its derivative (found below) is $(b-x)(b-3x)$, so it has a local maximum at $x = b/3$ and a local minimum at $(b, 0)$. The line through O cuts the rising branch near O, crosses again at P (between $b/3$ and b when $a < \frac{2}{3}b$, or before $b/3$ otherwise), and meets the curve a third time at Q, beyond the touching point $x = b$. The figure uses $a = 1$, $b = 3$; it also shows the tangent at P, the point R, and the region of area S.



The tangent at P

Differentiating with the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (b-x)^2 - 2x(b-x) = (b-x)(b-3x).$$

At P, $x = b-a$, so $b-x = a$ and $b-3x = 3a-2b$. The gradient there is $m = a(3a-2b)$.

The tangent is $y = mx + k$, where k is chosen so that the line passes through P:

$$\begin{aligned} k &= a^2(b-a) - a(3a-2b)(b-a) \\ &= a(b-a)(a-3a+2b) \\ &= 2a(b-a)^2. \end{aligned}$$

Hence the tangent at P is $y = a(3a-2b)x + 2a(b-a)^2$, as required. In particular $R = (0,\, 2a(b-a)^2)$, which lies on the positive y-axis.

The area S

Write $c = b-a > 0$ for the x-coordinate of P. On $0 < x < c$,

$$x(b-x)^2 - a^2x = x\,(b-x-a)(b-x+a) > 0,$$

since all three factors are positive there. So the curve lies above the line between O and P, and

$$S = \int_0^c (x(b-x)^2 - a^2x)\,\mathrm{d}x = \int_0^c ((b^2-a^2)x - 2bx^2 + x^3)\,\mathrm{d}x = \frac{(b^2-a^2)c^2}{2} - \frac{2bc^3}{3} + \frac{c^4}{4}.$$

Now substitute $b^2 - a^2 = c(b+a) = c(c+2a)$ and $b = c+a$:

$$\begin{aligned} S &= \frac{c^3(c+2a)}{2} - \frac{2c^3(c+a)}{3} + \frac{c^4}{4} \\ &= c^3\Big[\,c(\tfrac{1}{2} - \tfrac{2}{3} + \tfrac{1}{4}) + a(1-\tfrac{2}{3})\Big] \\ &= c^3\Big(\frac{c}{12} + \frac{a}{3}\Big) = \frac{c^3(c+4a)}{12}. \end{aligned}$$

Since $c + 4a = b + 3a$, this is

$$S = \frac{1}{12}(b-a)^3(3a+b),$$

as required.

Ask yourself Why not just expand everything in a and b?

You can, but the target has a factor $(b-a)^3$, and pulling it out of a quartic in a and b is painful. Writing $c = b - a$ makes that factor appear automatically, because every term of the integral already carries c^2 or more.

The triangle T and the inequality

Take OR as the base of triangle OPR: it lies along the y-axis with length $2ac^2$. The perpendicular height is the distance from P to the y-axis, namely c. So

$$T = \tfrac{1}{2} \cdot 2ac^2 \cdot c = ac^3 = a(b-a)^3.$$

Therefore, since $(b-a)^3 > 0$,

$$S - \tfrac{1}{3}T = \frac{(b-a)^3}{12}(3a + b - 4a) = \frac{(b-a)^4}{12} > 0,$$

because $b \neq a$. Hence $S > \frac{1}{3}T$, as required.

Checks

- With $a = 1$, $b = 3$ (the figure): $P = (2,2)$, $Q = (4,4)$, and the curve gives $y = 2$ at $x = 2$ and $y = 4$ at $x = 4$. The tangent formula gives $y = -3x + 8$; the gradient formula $(3-x)(3-3x)$ at $x = 2$ is -3. Integrating directly, $S = \int_0^2 (x^3 - 6x^2 + 8x)\,\mathrm{d}x = 4 - 16 + 16 = 4$, matching $\frac{1}{12} \cdot 8 \cdot 6 = 4$. Also $T = 8$, and $4 > \frac{8}{3}$.
- The ratio $S/(\frac{1}{3}T) = (3a+b)/(4a)$ tends to 1 as $b \to a$, so the constant $\frac{1}{3}$ is the best possible. That makes sense: for a parabola $y = px - qx^2$ the same construction gives $S = qc^3/6$ and $T = qc^3/2$, so $S = \frac{1}{3}T$ exactly, and when P is very close to O the cubic between them is almost a parabola.
- For thousands of random pairs $0 < a < b$, a computer check of the intersections, the tangent, the area (by numerical integration) and the inequality agreed with the formulas.

Common slips

- Dividing $x(b-x)^2 = a^2 x$ by x without noting the root $x = 0$, or taking only $b - x = a$ and losing Q.

- Integrating line minus curve and getting a negative "area": check which graph is on top before integrating.
- In the sketch, drawing the cubic crossing the x-axis at $x = b$ rather than touching it there.

Beyond STEP

The limiting ratio of 1 in the last part is in the spirit of Archimedes' quadrature of the parabola: for a parabola the segment has exactly a third of the area of triangle OPR, and near O the cubic is locally a parabola, so the inequality sharpens to equality in the limit. Understanding *why* local behaviour is governed by the first few Taylor terms, and making "almost a parabola" precise, is the business of first-year analysis. Abbott's *Understanding Analysis* is a friendly place to see Taylor's theorem done properly.

WORKED SOLUTION

STEP I 2018, Question 2

Pure • Logarithmic Inequalities

> **STEP I 2018, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/18-s1-q2

The key idea

Everything rests on the change-of-base rule: a logarithm in any base is a ratio of logarithms in a common base, and $\frac{1}{\log_b c} = \log_c b$. Once every logarithm is rewritten in terms of ln, each part becomes a short piece of algebra with inequalities. The given facts such as $e^2 < 8$ are just disguised bounds on $\ln 2$ and $\ln 10$. All the while we use the fact that ln is increasing, so taking logarithms keeps the direction of an inequality between positive numbers.

Solution

The opening result

By the definition of a logarithm, $x = \log_b c$ means

$$c = b^x.$$

Taking logarithms to base a of both sides and using $\log_a(b^x) = x \log_a b$,

$$\log_a c = x \log_a b.$$

Here $b \neq 1$ (so that $\log_b$ makes sense), so $\log_a b \neq 0$ and we may divide:

$$\frac{\log_a c}{\log_a b} = x = \log_b c,$$

as required. Two consequences are used below. Taking $a = c$ gives $\log_b c = \dfrac{1}{\log_c b}$, and taking $a = \mathrm{e}$ gives $\log_b c = \dfrac{\ln c}{\ln b}$.

Part (i)

By the reciprocal rule, $\dfrac{1}{\log_2 \pi} = \log_\pi 2$ and $\dfrac{1}{\log_5 \pi} = \log_\pi 5$. So

$$\frac{1}{\log_2 \pi} + \frac{1}{\log_5 \pi} = \log_\pi 2 + \log_\pi 5 = \log_\pi 10.$$

Since $\pi > 1$, the function $\log_\pi$ is increasing, and $10 > \pi^2$ gives

$$\log_\pi 10 > \log_\pi \pi^2 = 2,$$

which is the required inequality.

Part (ii)

Write the first given fact in natural logarithms:

$$\log_2 \frac{\pi}{e} = \frac{\ln \pi - 1}{\ln 2} > \frac{1}{5} \quad \Longrightarrow \quad \ln \pi > 1 + \frac{\ln 2}{5},$$

where we multiplied by $\ln 2 > 0$, so the inequality keeps its direction.

Ask yourself What is $e^2 < 8$ telling us?

It is a lower bound for $\ln 2$. Taking ln of both sides (allowed, as ln is increasing) gives $2 < \ln 8 = 3 \ln 2$, so $\ln 2 > \frac{2}{3}$. Substituting,

$$\ln \pi > 1 + \frac{1}{5} \cdot \frac{2}{3} = 1 + \frac{2}{15} = \frac{17}{15},$$

as required.

Part (iii)

From $\pi^2 < 10$, taking logarithms gives $2 \ln \pi < \ln 10$. So it is enough to show $\ln 10 < \frac{30}{13}$. Translate the other two facts. Since $e^3 > 20$, we have $3 > \ln 20 = \ln 2 + \ln 10$, that is,

$$\ln 2 < 3 - \ln 10.$$

Also $\log_{10} 2 = \dfrac{\ln 2}{\ln 10} > \dfrac{3}{10}$, and since $\ln 10 > 0$,

$$\ln 10 < \frac{10}{3} \ln 2.$$

The first is an upper bound for $\ln 2$, and $\ln 2$ appears in the second with a positive coefficient, so we may substitute it:

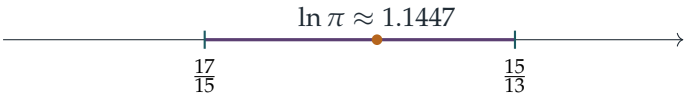
$$\ln 10 < \frac{10}{3}(3 - \ln 10) = 10 - \frac{10}{3} \ln 10.$$

Collecting the $\ln 10$ terms, $\frac{13}{3}\ln 10 < 10$, so $\ln 10 < \frac{30}{13}$. Therefore

$$\ln \pi < \tfrac{1}{2}\ln 10 < \tfrac{1}{2}\cdot\tfrac{30}{13} = \frac{15}{13},$$

as required.

Parts (ii) and (iii) together trap $\ln \pi$ in a narrow interval:



Checks

Numerically, $\ln \pi \approx 1.1447$, which does lie between $\frac{17}{15} \approx 1.1333$ and $\frac{15}{13} \approx 1.1538$. In part (i), $\log_\pi 10 \approx 2.0115$, only just above 2, which matches how close $\pi^2 \approx 9.87$ is to 10. The bound $\ln 10 < \frac{30}{13} \approx 2.3077$ is also sound against $\ln 10 \approx 2.3026$. It is worth confirming the given facts too: $\log_2(\pi/\mathrm{e}) \approx 0.209$, $\mathrm{e}^2 \approx 7.39$, $\mathrm{e}^3 \approx 20.09$ and $\log_{10} 2 \approx 0.3010$.

Common slips

- Writing $\dfrac{1}{\log_2 \pi}$ as $\log_2 \frac{1}{\pi}$. The reciprocal swaps the base and the argument; it does not invert the argument.
- In part (iii), using a lower bound for $\ln 2$ where an upper bound is needed. Chaining inequalities only works when each bound points the same way.
- Forgetting to say that $\ln 2$, $\ln 10$ and $\log_\pi$ behave well (positive, or increasing) before multiplying or taking logarithms across an inequality.

Beyond STEP

Proving numerical bounds from a handful of crude facts is a small taste of how analysts estimate constants: much of analysis consists of trapping an unknown quantity between two estimates you can control. Sharper bounds on $\ln 2$ or π come from series and integral inequalities, such as $\ln 2 = \int_1^2 \frac{1}{t}\,\mathrm{d}t$ estimated by rectangles. Abbott's *Understanding Analysis* is a friendly introduction to this way of thinking.

WORKED SOLUTION

STEP I 2018, Question 3

Pure • Angles and Hyperbolas

STEP I 2018, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q3

The key idea

Both angles are tied to the coordinates of P through their tangents: the slope of RP gives $\tan\alpha$ and the slope of SP gives $\tan\beta$. The condition $\beta = 2\alpha$ then becomes a double-angle identity for tan, and clearing fractions gives the curve. Part (ii) runs the same algebra backwards. The catch is that knowing $\tan\beta = \tan 2\alpha$ only fixes β up to a multiple of π, so we have to track which quadrant 2α is in.

Solution

Setting up the angles

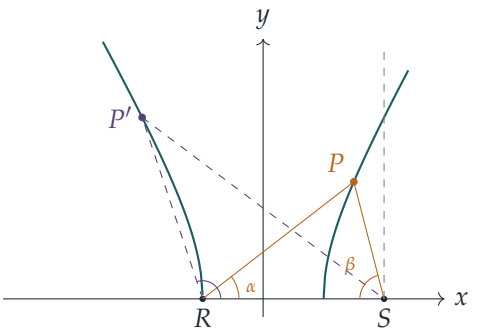
Since $y > 0$, the angle α at R is measured from the positive x-direction (along RS) round to RP, so $0 < \alpha < \pi$, and

$$\tan\alpha = \frac{y}{x+a} \qquad (x \neq -a).$$

This holds whether P is to the right of R (acute α, positive tangent) or to the left (obtuse α, negative tangent). At S, the angle β is measured from the negative x-direction (along SR) to SP. Because $x < 2a$, the point P is to the left of S, so β is acute and

$$\tan\beta = \frac{y}{2a-x} > 0, \qquad 0 < \beta < \tfrac{\pi}{2}.$$

Write $t = \tan\alpha$ when $x \neq -a$.



The curve $y^2 = 3(x^2 - a^2)$ with $y > 0$, $x < 2a$ (drawn for $a = 1$) has two pieces: an arc of the right branch with $a < x < 2a$, and the whole upper left branch $x < -a$. A point P' on the left branch makes α obtuse.

Part (i)

Suppose $\beta = 2\alpha$. Since $\beta < \frac{\pi}{2}$, we have $0 < \alpha < \frac{\pi}{4}$, so $x > -a$, $0 < t < 1$, and the double-angle formula applies with $1 - t^2 \neq 0$:

$$\frac{y}{2a - x} = \tan 2\alpha = \frac{2t}{1 - t^2} = \frac{2y/(x+a)}{1 - y^2/(x+a)^2} = \frac{2y(x+a)}{(x+a)^2 - y^2}.$$

Cancel y (allowed, as $y > 0$) and cross-multiply:

$$\begin{aligned} &(x+a)^2 - y^2 = 2(x+a)(2a - x) \\ \iff \quad &y^2 = (x+a)\big[(x+a) - 2(2a - x)\big] = (x+a)(3x - 3a) = 3(x^2 - a^2). \end{aligned}$$

So P lies on $y^2 = 3(x^2 - a^2)$, as required.

Part (ii)

Now suppose P lies on the curve, with $y > 0$ and $x < 2a$. Then $x^2 - a^2 = \frac{1}{3}y^2 > 0$, so $|x| > a$: either $a < x < 2a$ or $x < -a$. In particular $x \neq -a$, so $t = \tan\alpha$ is defined, and $t \neq 0$.

The steps of part (i) reverse. The curve's equation is equivalent to $(x+a)^2 - y^2 = 2(x+a)(2a - x)$. Dividing by $(x+a)^2$ and using $2a - x = y/\tan\beta$,

$$1 - t^2 = \frac{2(2a - x)}{x + a} = \frac{2y}{(x+a)\tan\beta} = \frac{2t}{\tan\beta}, \qquad \text{so} \qquad (1 - t^2)\tan\beta = 2t.$$

If $t^2 = 1$, this would give $t = 0$, which is impossible; so $t^2 \neq 1$ and

$$\tan\beta = \frac{2t}{1 - t^2} = \tan 2\alpha.$$

Ask yourself Does $\tan\beta = \tan 2\alpha$ mean $\beta = 2\alpha$?

Not necessarily: it means β and 2α differ by a multiple of π. We use what we know about the sizes of the angles, splitting into the two pieces of the curve.

Case $a < x < 2a$. Here $x + a > 0$, so $0 < \alpha < \frac{\pi}{2}$ and $0 < 2\alpha < \pi$. Since $\tan 2\alpha = \tan\beta > 0$, 2α lies in $(0, \frac{\pi}{2})$, the same interval as β. As tan is one-to-one on $(0, \frac{\pi}{2})$,

$$\beta = 2\alpha.$$

Case $x < -a$. Here $x + a < 0$, so $\frac{\pi}{2} < \alpha < \pi$ and $\pi < 2\alpha < 2\pi$. Since $\tan 2\alpha > 0$, $2\alpha \in (\pi, \frac{3\pi}{2})$, so $2\alpha - \pi \in (0, \frac{\pi}{2})$ and $\tan(2\alpha - \pi) = \tan 2\alpha = \tan\beta$. As before, tan is one-to-one on $(0, \frac{\pi}{2})$, so

$$\beta = 2\alpha - \pi.$$

So the possible relationships are $\beta = 2\alpha$ (when P is on the right-hand branch) and $\beta = 2\alpha - \pi$ (when P is on the left-hand branch). Both occur, since every point of each piece satisfies the conditions.

Checks

- Take $a = 1$ and $P = (-2, 3)$, which is on the curve since $3(4 - 1) = 9$. Then $\tan\alpha = 3/(-1) = -3$, so $\alpha \approx 108.43°$, and $\tan\beta = \frac{3}{4}$, so $\beta \approx 36.87°$. Indeed $2\alpha - 180° \approx 36.87°$.
- As $x \to 2a$ on the right branch, $y \to 3a$, so $\tan\alpha \to 1$ and $\tan\beta \to \infty$: $\alpha \to \frac{\pi}{4}$ and $\beta \to \frac{\pi}{2} = 2\alpha$.
- Far out on the left branch, P approaches the asymptote $y = -\sqrt{3}\,x$, so $\alpha \to \frac{2\pi}{3}$ and $\beta \to \frac{\pi}{3} = 2\alpha - \pi$.

A computer check of both relationships at many random points on each branch, for random a, also agreed.

Common slips

- Writing $\tan\alpha = y/|x + a|$, or assuming α is acute. On the left branch α is obtuse and $\tan\alpha < 0$.
- Concluding $\beta = 2\alpha$ from $\tan\beta = \tan 2\alpha$ without checking the ranges of the angles. This misses the second relationship.
- Forgetting that $x < 2a$ and $y > 0$ cut the curve down to two pieces, and that $|x| > a$ is needed to rule out $x = -a$.

Beyond STEP

The curve is a hyperbola with eccentricity 2, focus S and directrix $x = \frac{a}{2}$. The property in part (i) gives the classical way to trisect an angle using a hyperbola (known to Pappus), which gets round the impossibility of trisection with ruler and compasses alone. The impossibility proof itself is a highlight of a first course in field theory. Allenby's *Rings, Fields and Groups* covers it, and Needham's *Visual Complex Analysis* is a lovely companion on the geometry of angles and conics.

WORKED SOLUTION

STEP I 2018, Question 4

Pure • Curve Sketching Integrals

STEP I 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q4

The key idea

Everything in f is built from $\ln x$, and the awkward factor $\frac{1}{x}$ is exactly the derivative of $\ln x$. So the substitution $u = \ln x$ turns the integral into the integral of a simple rational function of u. In terms of u, replacing x by x^{-1} just changes the sign of u, so the symmetry comes down to the answer being an even function of u.

Solution

Throughout, write $u = \ln x$, so $u \neq 0$ because $x \neq 1$, and $u > 0$ exactly when $x > 1$.

The zeros of f and f′

Let

$$g(u) = \frac{(1-u^2)^2}{u} = \frac{1}{u} - 2u + u^3, \qquad \text{so that} \qquad \mathrm{f}(x) = \frac{g(\ln x)}{x}.$$

When $(\ln x)^2 = 1$, the factor $(1-u^2)^2$ is zero, so $\mathrm{f}(x) = 0$. By the quotient rule and the chain rule,

$$\mathrm{f}'(x) = \frac{g'(u) \cdot \frac{1}{x} \cdot x - g(u)}{x^2} = \frac{g'(u) - g(u)}{x^2}.$$

We already have $g(\pm 1) = 0$. Also, because $(1-u^2)^2 = (1-u)^2(1+u)^2$ has a *repeated* factor at $u = \pm 1$, its derivative $-4u(1-u^2)$ vanishes there too, and so

$$g'(u) = \frac{-4u(1-u^2) \cdot u - (1-u^2)^2}{u^2} = 0 \quad \text{at } u = \pm 1.$$

Hence $\mathrm{f}'(x) = 0$ as well whenever $(\ln x)^2 = 1$, that is, at $x = e$ and $x = e^{-1}$.

Part (i)

Substitute $u = \ln t$, so $\mathrm{d}u = \frac{1}{t}\,\mathrm{d}t$. Then

$$\int \mathrm{f}(t)\,\mathrm{d}t = \int \frac{(1-u^2)^2}{u}\,\mathrm{d}u = \int \left(\frac{1}{u} - 2u + u^3\right)\mathrm{d}u = \ln|u| - u^2 + \tfrac{1}{4}u^4 + C.$$

Define $G(u) = \ln|u| - u^2 + \frac{1}{4}u^4$. The two pieces of F never let the range of integration cross $t = 1$, so u keeps one sign on each and the antiderivative is valid there.

For $x > 1$, the lower limit $t = e$ gives $u = 1$ and $G(1) = 0 - 1 + \frac{1}{4} = -\frac{3}{4}$. For $0 < x < 1$, the lower limit $t = e^{-1}$ gives $u = -1$ and $G(-1) = -\frac{3}{4}$ too. So both cases give the same formula:

$$\boxed{\mathrm{F}(x) = \ln|\ln x| - (\ln x)^2 + \tfrac{1}{4}(\ln x)^4 + \tfrac{3}{4} \qquad (x > 0,\ x \neq 1).}$$

Now $\ln(x^{-1}) = -\ln x$. Every appearance of $\ln x$ above is inside $|\cdot|$ or raised to an even power, so

$$\mathrm{F}(x^{-1}) = \ln|-\ln x| - (-\ln x)^2 + \tfrac{1}{4}(-\ln x)^4 + \tfrac{3}{4} = \mathrm{F}(x),$$

as required. (The constant $\frac{3}{4}$ is the same on both sides only because the two lower limits e and e^{-1} are themselves a reciprocal pair.)

Part (ii)

? **Ask yourself** Where is F increasing, and what happens next to $x = 1$?

Gradient. $\mathrm{F}'(x) = \mathrm{f}(x)$. The factor $(1-u^2)^2$ is never negative and $x > 0$, so $\mathrm{f}(x)$ has the sign of $u = \ln x$. So F is decreasing on $(0,1)$ and increasing on $(1,\infty)$. Its only stationary points are at $x = e^{-1}$ and $x = e$, where $\mathrm{F} = 0$ by definition.

These are not turning points. Near $x = e$ the gradient is positive on both sides, and near $x = e^{-1}$ it is negative on both sides. Since $\mathrm{f}' = \mathrm{F}''$ is also zero there, they are *stationary points of inflection*. It follows that

$$\mathrm{F}(x) > 0 \text{ for } 0 < x < e^{-1} \text{ and } x > e, \qquad \mathrm{F}(x) < 0 \text{ for } e^{-1} < x < 1 \text{ and } 1 < x < e.$$

Behaviour at the ends.

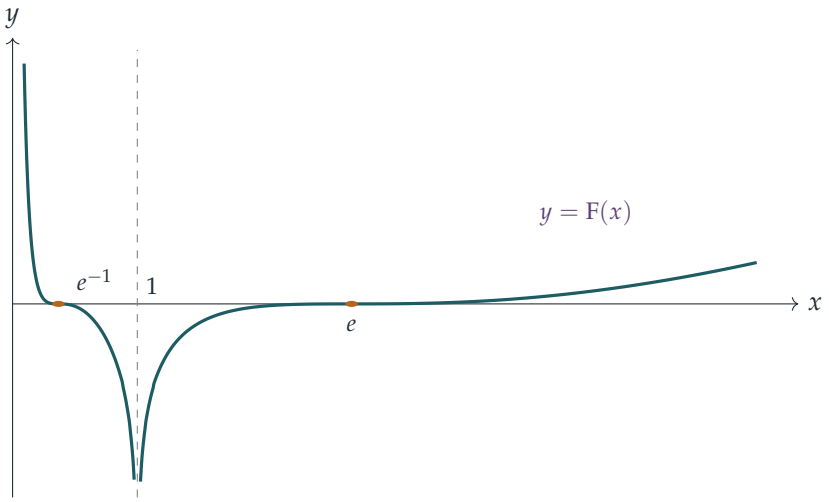
- As $x \to 1$, $u \to 0$ and $\ln|u| \to -\infty$, while the other terms tend to $\frac{3}{4}$. So $\mathrm{F}(x) \to -\infty$ from both sides, and $x = 1$ is a vertical asymptote.
- As $x \to \infty$, $u \to \infty$ and the $\frac{1}{4}u^4$ term dominates, so $\mathrm{F}(x) \to \infty$. It does so very slowly: the gradient $\mathrm{f}(x) \approx u^3/x \to 0$.
- As $x \to 0^+$, $u \to -\infty$, so again $\mathrm{F}(x) \to \infty$, by the symmetry. Here it does so steeply: the gradient $\mathrm{f}(x) \approx u^3/x \to -\infty$. The curve does not meet the y-axis.

Concavity (for a more careful sketch). Using $\mathrm{F}'' = \mathrm{f}' = (g' - g)/x^2$ and simplifying,

$$\mathrm{F}''(x) = -\frac{(u^2-1)(u^3 - 3u^2 - u - 1)}{u^2x^2}.$$

The cubic $u^3 - 3u^2 - u - 1$ has a single real root, at $u \approx 3.38$ (its local maximum value, at $u = 1 - \frac{2}{\sqrt{3}}$, is negative). It is negative for every smaller u. So the curve is concave up on $(0, e^{-1})$ and concave down on $(e^{-1}, 1)$ and on $(1, e)$. It is concave up from e until a further inflection at $x \approx 29.5$, and concave down after that. That last inflection is far to the right and needn't appear on a sketch.

The symmetry $\mathrm{F}(x^{-1}) = \mathrm{F}(x)$ says the height at x matches the height at $1/x$. For example $\mathrm{F}(2) = \mathrm{F}(\frac{1}{2}) \approx -0.04$ and $\mathrm{F}(5) = \mathrm{F}(\frac{1}{5}) \approx 0.31$. So the part of the curve on $(1, \infty)$ is the part on $(0, 1)$ squashed horizontally towards the asymptote.



The curve comes down steeply from $+\infty$ near the y-axis. It levels off to touch the x-axis at a stationary inflection at $(e^{-1}, 0)$, then plunges to $-\infty$ at the asymptote $x = 1$. On the right it climbs back from $-\infty$, flattens to a second stationary inflection at $(e, 0)$, and then rises slowly to $+\infty$.

Checks

- Differentiate the boxed formula: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln|u| - u^2 + \frac{1}{4}u^4\right) = \frac{1}{x}\left(\frac{1}{u} - 2u + u^3\right) = \mathrm{f}(x)$, and the formula gives 0 at $x = e$ and at $x = e^{-1}$.
- Numerically, $\int_e^5 \mathrm{f}(t)\,\mathrm{d}t \approx 0.3130$ and $\int_{1/e}^{1/5} \mathrm{f}(t)\,\mathrm{d}t \approx 0.3130$, matching the formula and the symmetry.
- The signs fit: on $(1, e)$ the function increases to 0, so it must be negative there, and indeed $\mathrm{F}(2) \approx -0.039$.

Common slips

- Writing $\ln(\ln x)$ for $0 < x < 1$, where $\ln x < 0$. The antiderivative of $\frac{1}{u}$ is $\ln|u|$, and the modulus is exactly what makes the symmetry work.

- Drawing turning points at $x = e^{\pm 1}$. Because f has a squared factor, it does not change sign there, so these are stationary points of inflection.
- Missing the vertical asymptote at $x = 1$, or joining the two branches across it.

Beyond STEP

F is defined by an integral, and the fundamental theorem of calculus says $\mathrm{F}' = \mathrm{f}$ on each interval where f is continuous. That is why the two pieces need separate base points: f blows up at $x = 1$, so no single integral can be taken across it. The symmetry $x \mapsto x^{-1}$, which becomes $u \mapsto -u$ under $u = \ln x$, is a small example of turning a multiplicative symmetry into an additive one, an idea that runs through analysis. For a careful account of the fundamental theorem and improper integrals, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2018, Question 5

Pure • Polynomials and Factor Theorem

STEP I 2018, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q5

The key idea

"Takes the value 1 at these points" is really a statement about roots: if $P(k) = 1$ then k is a root of $P(x) - 1$, so $x - k$ is a factor of it. Every part of the question becomes easy once we subtract the common value and factorise. In part (iii) the factorisation turns questions about S into questions about writing a small number as a product of four *distinct* integers, and distinctness is what does all the work.

Solution

Part (i)

By the remainder theorem, the remainder on dividing $P(x)$ by $x - k$ is $P(k)$. So we need $P(1) = P(2) = P(3) = P(4) = 1$, that is, $1, 2, 3, 4$ are roots of $P(x) - 1$. Since $P(x) - 1$ has degree at most 4 and these four distinct roots, it is a constant multiple of $(x-1)(x-2)(x-3)(x-4)$. Hence the most general such polynomial is

$$P(x) = 1 + k(x-1)(x-2)(x-3)(x-4), \qquad k \neq 0,$$

where $k \neq 0$ is needed for the degree to be exactly 4.

Part (ii)

Exactly as in part (i), $1, 2, \ldots, N$ are N distinct roots of $P(x) - 1$, which has degree N (because $N \geq 1$, subtracting 1 does not change the degree). So

$$P(x) - 1 = k(x-1)(x-2)\cdots(x-N)$$

for some constant k, and $k \neq 0$ since P has degree $N \geq 1$. Then

$$P(N+1) - 1 = k \cdot N(N-1)\cdots 1 = k\,N! \neq 0,$$

as both k and $N!$ are non-zero. So $P(N+1) \neq 1$.

If $P(N+1)=2$, then $k\,N!=1$, so $k=\frac{1}{N!}$. For a positive integer r,

$$\begin{aligned}P(N+r) &= 1+\frac{(N+r-1)(N+r-2)\cdots r}{N!}\\ &= 1+\frac{(N+r-1)!}{(r-1)!\,N!}=1+\binom{N+r-1}{N}.\end{aligned}$$

(The product has N factors, running down from $N+r-1$ to r.)

We now want $1+\binom{N+r-1}{N}=N+r$ for every N. Try small r. With $r=1$ we get $P(N+1)=2$, which equals $N+1$ only when $N=1$. With $r=2$,

$$P(N+2)=1+\binom{N+1}{N}=1+(N+1)=N+2,$$

so $r=2$ works whatever N is.

Part (iii)

The four distinct integers a,b,c,d are roots of $S(x)-2001$, which has degree 4 and leading coefficient 1. So

$$S(x)=2001+(x-a)(x-b)(x-c)(x-d).$$

(a) Suppose $S(e)=2018$ for some integer e. Then

$$(e-a)(e-b)(e-c)(e-d)=17.$$

The four factors are integers, and they are distinct because a,b,c,d are. None of them is 0, since the product is not.

? **Ask yourself** How can four distinct integers fail to multiply to a prime?

Among distinct non-zero integers, at most two can have absolute value 1 (namely 1 and -1). So at least two of the four factors have absolute value at least 2, and the product of those two has absolute value at least 4. But that product divides 17, so it must be ± 17; since 17 is prime, it cannot be written as a product of two integers each of absolute value at least 2. This contradiction shows that no integer e has $S(e)=2018$.

(b) Now $S(0)=2001+(-a)(-b)(-c)(-d)=2001+abcd$, so $S(0)=2017$ exactly when

$$abcd=16.$$

Conversely, any four distinct integers with $abcd=16$ give a valid S (the formula above is monic with integer coefficients), so we must count sets $\{a,b,c,d\}$ of four distinct integers with product 16; each set can be put in increasing order in exactly one way.

Each of a,b,c,d divides 16, so its absolute value is one of $1,2,4,8,16$, and the four absolute values multiply to 16. Because the integers are distinct, each absolute value

can occur at most twice (as $\pm m$). Writing the absolute values as powers 2^p with the four exponents adding to 4 and no exponent used more than twice, the possibilities are

$$\{1,1,2,8\}, \qquad \{1,1,4,4\}, \qquad \{1,2,2,4\}.$$

(Using 16 would force the other three to be 1, 1, 1, which is not allowed; and $\{2,2,2,2\}$ uses 2 four times.) Now choose signs so that the product is $+16$:

- $\{1,1,2,8\}$: the pair is $1,-1$, contributing -1, so exactly one of 2, 8 is negative. This gives $\{-2,-1,1,8\}$ and $\{-8,-1,1,2\}$.
- $\{1,1,4,4\}$: both pairs are $\pm$, and $(1)(-1)(4)(-4)=16$. This gives $\{-4,-1,1,4\}$.
- $\{1,2,2,4\}$: the pair $2,-2$ contributes -4, so exactly one of 1, 4 is negative. This gives $\{-4,-2,1,2\}$ and $\{-2,-1,2,4\}$.

So there are **5** ways to choose $a<b<c<d$.

Checks

- In part (ii) with $N=1$, $P(x)=x$, so $P(N+r)=1+r$ agrees with $1+\binom{r}{1}$, and $P(3)=3$. With $N=2$, $P(x)=1+\frac{1}{2}(x-1)(x-2)$ gives $P(4)=4$, as claimed for $r=2$.
- For (iii)(b), multiply out each of the five sets: for example $(-2)(-1)(2)(4)=16$. A computer search over all quadruples of distinct integers in $[-16,16]$ finds exactly these five.
- For (iii)(a), note that $2018-2001=17$ is prime; with 2001 replaced by a value giving a difference like $4=(-2)(-1)(1)(2)$, such an e could exist, so primality really matters.

Common slips

- Forgetting $k\neq 0$ in part (i), or writing $P(x)=(x-1)(x-2)(x-3)(x-4)+1$ only, which is not the most general polynomial.
- In part (iii)(b), counting only positive divisors, or forgetting that 1 and -1 can both be used.
- Writing $S(0)=2001-abcd$: there are four minus signs, so the product is $+abcd$.

Beyond STEP

The idea that a polynomial of degree N is determined by its values at $N+1$ points underlies Lagrange interpolation and finite differences; part (ii) is the polynomial $1+\binom{x-1}{N}$ in disguise. Part (iii) is a small instance of a standard tool in number theory: if P has integer coefficients then $m-n$ divides $P(m)-P(n)$, which severely limits the

values a polynomial can take at integers. Davenport's *The Higher Arithmetic* is a good place to read more about divisibility arguments of this kind.

WORKED SOLUTION

STEP I 2018, Question 6

Pure • Midpoint and Trapezium Rules

STEP I 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q6

The key idea

Both approximations are sums of sines of equally spaced angles, and the given product-to-sum identity makes such sums telescope once they are multiplied by a well-chosen $2\sin\theta$. The midpoint rule samples at odd multiples of $\frac{\pi}{2n}$ and the trapezium rule at even multiples, so with $\theta = \frac{\pi}{2n}$ each part becomes a telescoping sum. Once A_n and B_n are in closed form, part (iii) is just half-angle formulae.

Solution

The preliminary identity

Take $P = \theta$ and $Q = (2k-1)\theta$ in the identity:

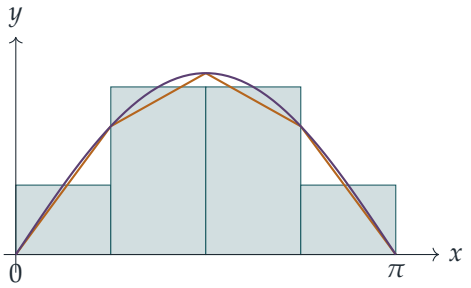
$$2\sin\theta\,\sin(2k-1)\theta = \cos(2k-2)\theta - \cos 2k\theta.$$

Summing from $k = 1$ to $k = n$, each $\cos 2k\theta$ subtracted in one term is added back in the next, so the sum telescopes:

$$\begin{aligned} 2\sin\theta\sum_{k=1}^{n}\sin(2k-1)\theta &= (\cos 0 - \cos 2\theta) + (\cos 2\theta - \cos 4\theta) + \cdots \\ &\quad + \big(\cos(2n-2)\theta - \cos 2n\theta\big) \\ &= 1 - \cos 2n\theta, \end{aligned}$$

as required.

Throughout the rest, write $h = \frac{\pi}{n}$ for the strip width and $\theta = \frac{\pi}{2n} = \frac{h}{2}$.



For $n = 4$: the midpoint rectangles (shaded) overestimate the area, and the trapezium chords (bold) underestimate it, because $\sin x$ is concave on $[0, \pi]$.

Part (i)

The kth strip runs from $(k-1)h$ to kh, so its midpoint is at $x = (k - \frac{1}{2})h = (2k-1)\theta$, and its height is $\sin(2k-1)\theta$. Hence

$$A_n = h \sum_{k=1}^{n} \sin(2k-1)\theta.$$

Multiplying by $2\sin\theta$ and using the identity, with $2n\theta = \pi$,

$$2\sin\theta\, A_n = h\,(1 - \cos\pi) = 2h.$$

Dividing by 2 gives $A_n \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{n}$, as required.

Part (ii)

The ordinates are at $x = kh = 2k\theta$ for $k = 0, 1, \ldots, n$, and the end ordinates are $\sin 0 = \sin\pi = 0$. So the trapezium rule gives

$$B_n = \frac{h}{2}\left(\sin 0 + 2\sum_{k=1}^{n-1} \sin 2k\theta + \sin\pi\right) = h \sum_{k=1}^{n-1} \sin 2k\theta.$$

These are now even multiples of θ, so we use the identity once more, this time with $P = \theta$, $Q = 2k\theta$:

$$2\sin\theta\, \sin 2k\theta = \cos(2k-1)\theta - \cos(2k+1)\theta.$$

Summing from $k = 1$ to $n-1$ again telescopes:

$$2\sin\theta \sum_{k=1}^{n-1} \sin 2k\theta = \cos\theta - \cos(2n-1)\theta.$$

Ask yourself How do we simplify $\cos(2n-1)\theta$?

Since $2n\theta = \pi$, we have $(2n-1)\theta = \pi - \theta$, and $\cos(\pi - \theta) = -\cos\theta$. So the right-hand side is $2\cos\theta$, and

$$2\sin\theta\, B_n = 2h\cos\theta, \qquad \text{that is,} \qquad B_n \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{n}\cos\left(\frac{\pi}{2n}\right).$$

(When $n = 1$ the sum is empty and $B_1 = 0$; the formula still holds, since $\cos\frac{\pi}{2} = 0$.)

Part (iii)

Since $0 < \theta \leq \frac{\pi}{2}$, $\sin\theta \neq 0$, so parts (i) and (ii) give

$$A_n = \frac{h}{\sin\theta}, \qquad B_n = \frac{h\cos\theta}{\sin\theta}.$$

Replacing n by $2n$ halves both h and θ:

$$A_{2n} = \frac{h/2}{\sin(\theta/2)}, \qquad B_{2n} = \frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)}.$$

For the first result, use $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$:

$$\tfrac{1}{2}(A_n + B_n) = \frac{h}{2}\cdot\frac{1+\cos\theta}{\sin\theta} = \frac{h}{2}\cdot\frac{2\cos^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)} = \frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)} = B_{2n}.$$

For the second, using the same double-angle formula for $\sin\theta$ (and $\cos\frac{\theta}{2} \neq 0$),

$$A_n B_{2n} = \frac{h}{2\sin(\theta/2)\cos(\theta/2)}\cdot\frac{(h/2)\cos(\theta/2)}{\sin(\theta/2)} = \frac{h^2}{4\sin^2(\theta/2)} = \left(\frac{h/2}{\sin(\theta/2)}\right)^2 = A_{2n}^2.$$

Both results are shown.

Checks

- $n = 1$: one midpoint rectangle of width π and height $\sin\frac{\pi}{2} = 1$ gives $A_1 = \pi$, matching $\pi/\sin\frac{\pi}{2}$; and $B_1 = 0$ as noted.
- $n = 2$: $A_2 = \frac{\pi}{2}(\sin\frac{\pi}{4} + \sin\frac{3\pi}{4}) = \frac{\pi}{\sqrt{2}} \approx 2.2214$ and $B_2 = \frac{\pi}{2} \approx 1.5708$; then $\frac{1}{2}(A_1 + B_1) = \frac{\pi}{2} = B_2$ and $A_1 B_2 = \frac{\pi^2}{2} = A_2^2$.
- $n = 10$: summing directly gives $A_{10} \approx 2.00825$ and $B_{10} \approx 1.98352$, either side of the true area 2. As $n \to \infty$, $\frac{\pi}{n}\big/\sin\frac{\pi}{2n} \to 2$ and $\cos\frac{\pi}{2n} \to 1$, so both tend to 2, as they should. All four results were also confirmed by computer for every n up to 200.

Common slips

- Putting the midpoints at kh rather than $(k - \frac{1}{2})h$, which turns the midpoint rule into a sum at the trapezium nodes.
- Forgetting that the trapezium rule halves the end ordinates. Here they vanish, which is why B_n is simply h times the interior ordinates.
- Leaving $\cos(2n-1)\theta$ unsimplified, rather than using $(2n-1)\theta = \pi - \theta$.

Beyond STEP

Part (iii) says that B_{2n} is the arithmetic mean of A_n and B_n, and A_{2n} is the geometric mean of A_n and B_{2n}. Taking reciprocals turns this into the harmonic-then-geometric mean pattern of Archimedes' method of squeezing π between polygons around and inside a circle, and the iteration is known as Borchardt's algorithm. More generally, comparing the midpoint and trapezium errors (the midpoint error is about half the trapezium error, with opposite sign) leads to Simpson's rule and to Richardson extrapolation, which are central ideas in numerical analysis. For the error analysis behind these rules, see Bartle and Sherbert, *Introduction to Real Analysis*, on the Riemann integral.

WORKED SOLUTION

STEP I 2018, Question 7

Pure • Solving the Cubic

STEP I 2018, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q7

The key idea

The substitution is a Möbius transformation chosen so that, after clearing denominators, the z^2 and z terms cancel and only z^3 and a constant are left. A cubic of the form $az^3 + b = 0$ is easy to solve, so the whole job is to bring a general reduced cubic $x^3 - 3cx + d = 0$ into the special shape. Doing that means finding two numbers from their sum and product, which is a quadratic, and the condition for it to have distinct real roots is where $d^2 > 4c^3$ comes from.

Solution

Part (i)

Put $x = \dfrac{pz+q}{z+1}$ and multiply the equation through by $(z+1)^3$:

$$(pz+q)^3 - 3pq(pz+q)(z+1)^2 + pq(p+q)(z+1)^3 = 0.$$

We need $(pz+q)(z+1)^2 = pz^3 + (2p+q)z^2 + (p+2q)z + q$. Now collect powers of z.

$$\begin{aligned}
z^3: &\quad p^3 - 3p^2q + p^2q + pq^2 = p(p^2 - 2pq + q^2) = p(p-q)^2,\\
z^2: &\quad 3p^2q - 3pq(2p+q) + 3pq(p+q) = 3p^2q - 6p^2q - 3pq^2 + 3p^2q + 3pq^2 = 0,\\
z^1: &\quad 3pq^2 - 3pq(p+2q) + 3pq(p+q) = 3pq^2 - 3p^2q - 6pq^2 + 3p^2q + 3pq^2 = 0,\\
z^0: &\quad q^3 - 3pq^2 + p^2q + pq^2 = q(q^2 - 2pq + p^2) = q(p-q)^2.
\end{aligned}$$

So the equation reduces to $az^3 + b = 0$ with

$$a = p(p-q)^2, \qquad b = q(p-q)^2.$$

Since $p \neq q$ we may divide by $(p-q)^2$, giving the simpler form $pz^3 + q = 0$.

The substitution can be inverted, $z = \dfrac{x-q}{p-x}$, for every $x \neq p$, and $x = p$ is a root of the cubic only when $p(p-q)^2 = 0$. Also $z = -1$ never solves $pz^3 + q = 0$, because

that would need $p = q$. So the roots of the cubic correspond exactly to the roots of $pz^3 + q = 0$ (when $p = 0$ the cubic is just $x^3 = 0$).

Part (ii)

Comparing $x^3 - 3cx + d$ with $x^3 - 3pqx + pq(p+q)$, we need

$$pq = c, \qquad pq(p+q) = d, \qquad \text{that is} \qquad pq = c, \quad p+q = \frac{d}{c},$$

where dividing by c is allowed because $c \neq 0$. Two numbers with sum $\frac{d}{c}$ and product c are the roots of

$$t^2 - \frac{d}{c}\,t + c = 0.$$

This quadratic has distinct real roots exactly when its discriminant is positive:

$$\frac{d^2}{c^2} - 4c > 0 \iff d^2 > 4c^3,$$

multiplying by $c^2 > 0$. So when $d^2 > 4c^3$, taking p and q to be the two roots of the quadratic writes the equation in the required form, with p, q real and distinct.

Part (iii)

Here $c = -2$ and $d = -2$, so $d^2 = 4 > -32 = 4c^3$ and part (ii) applies. We need $p + q = \frac{d}{c} = 1$ and $pq = c = -2$, so p, q are the roots of $t^2 - t - 2 = (t-2)(t+1) = 0$. Take $p = 2$, $q = -1$. By part (i) the equation becomes

$$2z^3 - 1 = 0, \qquad z^3 = \tfrac{1}{2},$$

whose only real root is $z = 2^{-1/3}$. Write $u = 2^{1/3}$, so $z = \frac{1}{u}$. Then

$$x = \frac{2z-1}{z+1} = \frac{2-u}{1+u}.$$

Ask yourself Can this be tidied up, given that the denominator involves a cube root?

Yes. Rationalise using $1 + u^3 = (1+u)(1-u+u^2)$, which equals 3 since $u^3 = 2$:

$$x = \frac{(2-u)(1-u+u^2)}{3} = \frac{2-3u+3u^2-u^3}{3} = \frac{3u^2-3u}{3} = u^2 - u.$$

So the real root is

$$x = 2^{2/3} - 2^{1/3} = \sqrt[3]{4} - \sqrt[3]{2}.$$

It is the only real root: $x^3 + 6x - 2$ has derivative $3x^2 + 6 > 0$, so it is strictly increasing. The other two values of z are complex and give the two complex roots. (Choosing $p = -1$, $q = 2$ instead gives $z^3 = 2$ and the same root.)

Part (iv)

Since $x = p$ gives $p^3 - 3p^3 + 2p^3 = 0$, $x - p$ is a factor, and

$$x^3 - 3p^2x + 2p^3 = (x-p)(x^2 + px - 2p^2) = (x-p)^2(x+2p).$$

So the roots are $x = p$ (repeated) and $x = -2p$.

Notice that this is the form of part (i) with $q = p$, which is exactly the case part (ii) left out. Now take $d^2 = 4c^3$ with $c, d \neq 0$. Then $c^3 = \frac{1}{4}d^2 > 0$, so $c > 0$. We look for p with $p^2 = c$ and $2p^3 = d$. Dividing the second by the first suggests

$$p = \frac{d}{2c},$$

and this works: $p^2 = \dfrac{d^2}{4c^2} = \dfrac{4c^3}{4c^2} = c$, and then $2p^3 = 2p \cdot p^2 = 2 \cdot \dfrac{d}{2c} \cdot c = d$. So $x^3 - 3cx + d = x^3 - 3p^2x + 2p^3$, and its roots are

$$x = \frac{d}{2c} \text{ (twice)} \qquad \text{and} \qquad x = -\frac{d}{c}.$$

Checks

- Part (iii): $\sqrt[3]{4} - \sqrt[3]{2} \approx 0.32748$, and $x^3 + 6x - 2$ at this value is 0 to fifteen decimal places. Cardano's formula gives the same root, $\sqrt[3]{1+3} + \sqrt[3]{1-3}$.
- Part (iv): with $c = 1$, $d = 2$ the cubic is $x^3 - 3x + 2 = (x-1)^2(x+2)$, which matches the double root $\frac{d}{2c} = 1$ and the simple root $-\frac{d}{c} = -2$.
- Part (i): with $p = 1$, $q = 0$ the cubic is $x^3 = 0$ and the substitution gives $x = \frac{z}{z+1}$, so $x^3(z+1)^3 = z^3$, which matches $a = 1$, $b = 0$.

Common slips

- Forgetting that c may be negative in part (ii): multiplying the inequality by c^2 is safe, but multiplying by c is not.
- In part (iii), giving all three cube roots of $\frac{1}{2}$ as real roots. Only $z = 2^{-1/3}$ is real.
- In part (iv), taking $p = \sqrt{c}$. That has the wrong sign when $d < 0$; $p = \frac{d}{2c}$ gets the sign right automatically.

Beyond STEP

This is a neat route to Cardano's formula for the cubic. The quadratic $t^2 - \frac{d}{c}t + c = 0$ plays the part of the "resolvent", and the sign of $d^2 - 4c^3$ is (up to a factor) the discriminant of the cubic, which decides whether it has one real root or three. Why cubics and quartics can be solved by radicals while general quintics cannot is the subject

of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to start reading about it.

WORKED SOLUTION

STEP I 2018, Question 8

Pure • Coupled Differential Equations

STEP I 2018, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q8

The key idea

The two derivative rules are built to cancel against each other: whenever s' appears it brings in c^2, and whenever c' appears it brings in $-\mathrm{s}^2$. So the natural move throughout is to differentiate products and quotients and watch the terms combine, then use the identity from (i) to turn everything into a single function. For the integrals, read each integrand as "something times s^2" or "something times c^2", because those are exactly the derivatives we know. The u-integrals in the last two parts are the same integrals in disguise.

Solution

Throughout, write s and c for $\mathrm{s}(x)$ and $\mathrm{c}(x)$ when there is no ambiguity.

Part (i)

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{s}^3+\mathrm{c}^3\right)=3\mathrm{s}^2\,\mathrm{s}'+3\mathrm{c}^2\,\mathrm{c}'=3\mathrm{s}^2\mathrm{c}^2-3\mathrm{c}^2\mathrm{s}^2=0,$$

so $\mathrm{s}^3+\mathrm{c}^3$ is constant. Its value at $x=0$ is $0^3+1^3=1$, hence

$$\mathrm{s}(x)^3+\mathrm{c}(x)^3=1 \quad \text{for all } x,$$

as required. This identity is the engine of the rest of the question: it lets us swap s^3 for $1-\mathrm{c}^3$ and vice versa.

Part (ii)

By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{sc})=\mathrm{s}'\mathrm{c}+\mathrm{sc}'=\mathrm{c}^2\cdot\mathrm{c}+\mathrm{s}\cdot(-\mathrm{s}^2)=\mathrm{c}^3-\mathrm{s}^3.$$

Using (i), $\mathrm{s}^3 = 1 - \mathrm{c}^3$, so this equals $\mathrm{c}^3 - (1 - \mathrm{c}^3) = 2\mathrm{c}^3 - 1$, which is the result we wanted.

For the quotient (wherever $\mathrm{c}(x) \neq 0$), the quotient rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{s}}{\mathrm{c}}\right) = \frac{\mathrm{s}'\mathrm{c} - \mathrm{s}\mathrm{c}'}{\mathrm{c}^2} = \frac{\mathrm{c}^3 + \mathrm{s}^3}{\mathrm{c}^2} = \frac{1}{\mathrm{c}(x)^2},$$

again by (i). Notice the pleasing parallel with $\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \frac{1}{\cos^2 x}$.

Part (iii)

Since $\mathrm{c}' = -\mathrm{s}^2$, the first integral is immediate:

$$\int \mathrm{s}(x)^2 \,\mathrm{d}x = -\mathrm{c}(x) + C.$$

Ask yourself How can s^5 be written so that the known derivative $\mathrm{c}' = -\mathrm{s}^2$ appears?

Split off a factor s^2 and use (i) on what is left: $\mathrm{s}^5 = \mathrm{s}^3 \cdot \mathrm{s}^2 = (1 - \mathrm{c}^3)\,\mathrm{s}^2 = -(1 - \mathrm{c}^3)\,\mathrm{c}'$. This is of the form $g(\mathrm{c})\,\mathrm{c}'$, which integrates by the reverse chain rule:

$$\int \mathrm{s}(x)^5 \,\mathrm{d}x = -\int (1 - \mathrm{c}^3)\,\mathrm{c}' \,\mathrm{d}x = -\mathrm{c}(x) + \tfrac{1}{4}\mathrm{c}(x)^4 + C.$$

(Differentiating back: $-\mathrm{c}' + \mathrm{c}^3\mathrm{c}' = -\mathrm{c}'(1 - \mathrm{c}^3) = \mathrm{s}^2 \cdot \mathrm{s}^3$, as it should be.)

Part (iv)

Put $u = \mathrm{s}(x)$, so $x = \mathrm{s}^{-1}(u)$ and $\mathrm{d}u = \mathrm{s}'(x)\,\mathrm{d}x = \mathrm{c}(x)^2\,\mathrm{d}x$. From (i), $\mathrm{c}^3 = 1 - u^3$, so $\mathrm{c} = (1 - u^3)^{1/3}$ (the real cube root, which is valid whatever the sign of c), and therefore

$$(1 - u^3)^{2/3} = \mathrm{c}(x)^2.$$

Hence

$$\int \frac{1}{(1 - u^3)^{2/3}}\,\mathrm{d}u = \int \frac{1}{\mathrm{c}^2}\,\mathrm{c}^2\,\mathrm{d}x = \int 1\,\mathrm{d}x = x + C = \mathrm{s}^{-1}(u) + C,$$

which is what we had to show.

Part (v)

Use the same substitution, with $(1 - u^3)^{1/3} = \mathrm{c}$ and $\mathrm{d}u = \mathrm{c}^2\,\mathrm{d}x$.

For the first integral, $(1 - u^3)^{4/3} = \mathrm{c}^4$, so, by the quotient result in (ii),

$$\int \frac{\mathrm{d}u}{(1 - u^3)^{4/3}} = \int \frac{\mathrm{c}^2}{\mathrm{c}^4}\,\mathrm{d}x = \int \frac{\mathrm{d}x}{\mathrm{c}^2} = \frac{\mathrm{s}(x)}{\mathrm{c}(x)} + C = \frac{u}{(1 - u^3)^{1/3}} + C.$$

For the second,

$$\int (1-u^3)^{1/3}\,\mathrm{d}u = \int \mathrm{c}\cdot\mathrm{c}^2\,\mathrm{d}x = \int \mathrm{c}^3\,\mathrm{d}x.$$

The product result in (ii) says $\mathrm{c}^3 = \frac{1}{2}\left(1 + \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{sc})\right)$, so

$$\int \mathrm{c}^3\,\mathrm{d}x = \tfrac{1}{2}\big(x + \mathrm{s}(x)\mathrm{c}(x)\big) + C.$$

Returning to u, with $x = \mathrm{s}^{-1}(u)$ and $\mathrm{sc} = u(1-u^3)^{1/3}$:

$$\int (1-u^3)^{1/3}\,\mathrm{d}u = \tfrac{1}{2}\Big(\mathrm{s}^{-1}(u) + u(1-u^3)^{1/3}\Big) + C.$$

Checks

- The answers in (v) can be checked by differentiating with respect to u. For example,

$$\frac{\mathrm{d}}{\mathrm{d}u}\,\frac{u}{(1-u^3)^{1/3}} = \frac{1}{(1-u^3)^{1/3}} + \frac{u^3}{(1-u^3)^{4/3}} = \frac{1}{(1-u^3)^{4/3}}.$$

- Solving the differential equations numerically up to $x = 1.2$ gives $\mathrm{s}(1.2) \approx 0.9411$ and $\mathrm{c}(1.2) \approx 0.5500$, and indeed $0.9411^3 + 0.5500^3 \approx 1$. With $U = \mathrm{s}(1.2)$, numerical integration gives $\int_0^U (1-u^3)^{-2/3}\,\mathrm{d}u \approx 1.2000$, as (iv) predicts, and $\int_0^U (1-u^3)^{1/3}\,\mathrm{d}u \approx 0.8588 = \frac{1}{2}(1.2 + 0.9411\times 0.5500)$, matching (v).
- Numerically, $\int_0^{1.2} \mathrm{s}^2\,\mathrm{d}x \approx 0.4500 = 1 - \mathrm{c}(1.2)$, agreeing with (iii).

Common slips

- Writing $\int \mathrm{s}^2\,\mathrm{d}x = +\mathrm{c} + C$: the minus sign in $\mathrm{c}' = -\mathrm{s}^2$ is easy to lose.
- In (iv) and (v), forgetting to convert $\mathrm{d}u$ into $\mathrm{c}^2\,\mathrm{d}x$, or leaving the final answers in terms of x rather than u.
- Trying to integrate c^3 directly in (v) instead of spotting that (ii) already expresses it as a derivative plus a constant.

Beyond STEP

The pair s, c behaves like sin and cos for the curve $x^3 + y^3 = 1$ in place of the circle $x^2 + y^2 = 1$; such "generalised trigonometric functions" are close relatives of the elliptic functions, which arise by inverting integrals like the one in (iv), just as $\sin^{-1} u = \int (1-u^2)^{-1/2}\,\mathrm{d}u$. Defining functions through a differential equation with initial conditions, and deriving all their properties from it, is a standard first-year technique, and the assumption that the functions are "uniquely defined" is the existence and uniqueness theorem for ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers both the theory and many examples of coupled systems.

WORKED SOLUTION

STEP I 2018, Question 9

Mechanics • Motion on Inclined Planes

STEP I 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

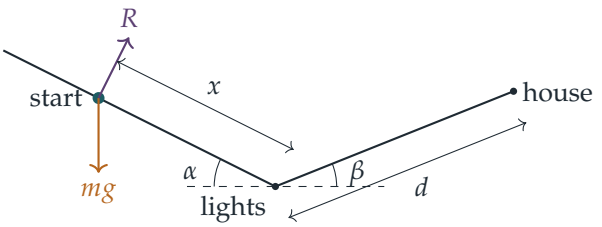
step.maths.org/questions/18-s1-q9

The key idea

With no resistance, the go-kart has constant acceleration on each straight section: $g\sin\alpha$ down the first slope and a deceleration of $g\sin\beta$ up the second. So each section is a constant-acceleration problem, and the two are glued together by the speed at the traffic lights. Energy (or $v^2 = u^2 + 2as$) settles whether I get home at all; the timing comes from $s = ut + \frac{1}{2}at^2$ on each section, and the rest is careful algebra in terms of k.

Solution

Model the go-kart and rider as a particle moving along each straight line of greatest slope, with no friction or air resistance. The only forces are the weight mg and the normal reaction R, which does no work.



Reaching the house

Resolving along the first slope, the component of the weight down the slope is $mg\sin\alpha$, so the acceleration is $g\sin\alpha$. Starting from rest and travelling a distance x, the speed V at the lights satisfies

$$V^2 = 2gx\sin\alpha.$$

Since there is no sudden change of speed, I start up the second section with speed V, and now the weight gives a deceleration $g\sin\beta$ along the slope. At distance s beyond the lights my speed v satisfies $v^2 = V^2 - 2gs\sin\beta$. To reach the house, v^2 must stay

non-negative all the way to $s = d$, i.e.

$$2gx\sin\alpha - 2gd\sin\beta \geq 0 \quad \Longrightarrow \quad x\sin\alpha \geq d\sin\beta,$$

as required. (Equivalently: by conservation of energy I can never rise above my starting height, which is $x\sin\alpha$ above the lights, while the house is $d\sin\beta$ above them.)

The total time

To keep the algebra tidy, write $c = \left(\frac{1}{2}g\sin\alpha\right)^{1/2}$, so $g\sin\alpha = 2c^2$ and, since $\sin\beta = \sin\alpha/k$, $g\sin\beta = 2c^2/k$.

First section. From $x = \frac{1}{2}(g\sin\alpha)t_1^2 = c^2t_1^2$ the time to the lights is $t_1 = \sqrt{x}/c$, and the speed there is $V = (g\sin\alpha)t_1 = 2c\sqrt{x}$.

Second section. Measuring s from the lights, $s = Vt - \frac{1}{2}(g\sin\beta)t^2$, so the time t_2 to reach $s = d$ satisfies

$$\frac{c^2}{k}t_2^2 - 2c\sqrt{x}\,t_2 + d = 0.$$

? **Ask yourself** This quadratic has two roots. Which one is the time I arrive?

The larger root is the time at which a particle on an endlessly long slope would pass $s = d$ again on the way back down, after stopping. I arrive the first time I reach $s = d$, so we want the smaller root:

$$t_2 = \frac{2c\sqrt{x} - \sqrt{4c^2x - 4c^2d/k}}{2c^2/k} = \frac{k}{c}\left(\sqrt{x} - \sqrt{x - \tfrac{d}{k}}\right) = \frac{1}{c}\left(k\sqrt{x} - \sqrt{k^2x - kd}\right),$$

using $k\sqrt{x - d/k} = \sqrt{k^2x - kd}$ as $k > 0$. The square root is real precisely because $x \geq d/k$, which is the condition $x\sin\alpha \geq d\sin\beta$ from the first part.

Adding, $T = t_1 + t_2$ gives

$$\left(\frac{g\sin\alpha}{2}\right)^{1/2} T = cT = (1+k)\sqrt{x} - \sqrt{k^2x - kd},$$

which is the required result.

Minimising T

Since c is a positive constant, we minimise $f(x) = (1+k)\sqrt{x} - \sqrt{k^2x - kd}$ over $x \geq d/k$. Differentiating,

$$f'(x) = \frac{1+k}{2\sqrt{x}} - \frac{k^2}{2\sqrt{k^2x - kd}}.$$

Setting $f'(x) = 0$ gives $(1+k)\sqrt{k^2x - kd} = k^2\sqrt{x}$. Both sides are positive, so squaring loses nothing:

$$\begin{aligned}(1+k)^2(k^2x - kd) &= k^4x\\ x\left[(1+k)^2k^2 - k^4\right] &= (1+k)^2kd\\ xk^2(1+2k) &= (1+k)^2kd,\end{aligned}$$

using $(1+k)^2 - k^2 = 1 + 2k$. Dividing by $k^2(1+2k) > 0$,

$$x = \frac{(1+k)^2}{k(1+2k)}\,d.$$

This is an allowed starting point: $(1+k)^2 = 1 + 2k + k^2 > 1 + 2k$, so this x is strictly greater than d/k and I do reach the house.

It makes sense that there is a best x: start too close and I crawl over the last stretch of the hill; start too far and I spend a long time on the first slope.

Checks

- At the optimum, $\sqrt{k^2x - kd} = k^2\sqrt{x}/(1+k)$, so $cT = \sqrt{x}\,\dfrac{1+2k}{1+k} = \sqrt{\dfrac{(1+2k)d}{k}}$, a pleasingly simple minimum time.
- Take $\alpha = \beta$, so $k = 1$. Then the optimum is $x = \frac{4}{3}d$, and the formula gives $cT = 2\sqrt{4d/3} - \sqrt{d/3} = \sqrt{3d}$, agreeing with the previous check.
- At the threshold $x = d/k$ I just creep to the door, $f'(x) \to -\infty$, so moving the start back a little certainly saves time.
- Stepping the motion forward numerically in small time intervals, for several random angles and distances, reproduced the formula for T to five significant figures; a computer search over x located the minimum at the value found above.

Common slips

- Taking the larger root of the quadratic for t_2: that is the time of coming back down past the house.
- Forgetting that the speed at the lights is not given and must be carried over from the first section.
- Dropping the condition $x \geq d/k$, which is exactly what makes $\sqrt{k^2x - kd}$ real.

Beyond STEP

The tidy answer for the minimum time hints at a broader theme: choosing a path or starting point to minimise travel time is the subject of the *calculus of variations*, whose

most famous problem, the brachistochrone, asks for the curve of fastest descent under gravity. In first-year mechanics you'll also meet the idea that, for conservative forces, energy alone tells you where a particle can go, while time requires integrating the equation of motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP I 2018, Question 10

Mechanics • Connected Particles

STEP I 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

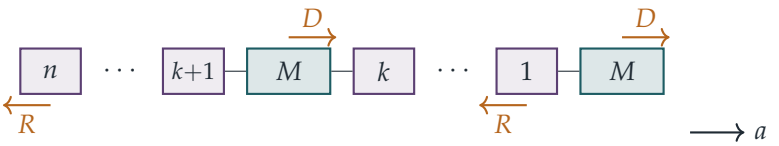
step.maths.org/questions/18-s1-q10

The key idea

Every vehicle has the same acceleration, so we find it once from the whole train and then apply Newton's second law to whichever piece of the train is most convenient: cutting the train at a coupling exposes that coupling's tension as an external force. The trick that makes the later parts easy is to work out the front tension T in two ways, from the front engine and from everything behind it. Doing so ties T to a single quantity, $R + ma$, which also governs every other coupling.

Solution

Model. Treat each engine and carriage as a particle moving along the horizontal track, and the couplings as light rods that can be in tension or compression. The engines feel no resistance, and the vertical forces on each vehicle balance, so only horizontal forces matter. Measure the acceleration a in the direction of motion. Number the carriages $1, 2, \ldots, n$ from the front, so that carriages $1, \ldots, k$ are between the engines and $k+1, \ldots, n$ are behind the second engine (with $1 \le k \le n-1$).



(Each carriage has a resistance R acting backwards; only two are drawn.)

The acceleration. The whole train has mass $nm + 2M$, and the couplings are internal, so

$$(nm + 2M)a = 2D - nR, \qquad a = \frac{2D - nR}{nm + 2M} > 0,$$

positive because $2D > nR$. In particular $R + ma > 0$, a fact we use repeatedly.

Part (i)

The front engine is pulled forward by D and backward by the tension T in its coupling, so

$$D - T = Ma.$$

Hence

$$\begin{aligned} T = D - \frac{M(2D - nR)}{nm + 2M} &= \frac{D(nm + 2M) - 2MD + nMR}{nm + 2M} \\ &= \frac{n(mD + MR)}{nm + 2M}, \end{aligned}$$

as required.

A neater form of T. Now look instead at everything behind the front engine: one engine and n carriages, of total mass $M + nm$. It is pulled forward by T and by the second engine's D, and resisted by nR:

$$T + D - nR = (M + nm)a.$$

Subtracting $D - T = Ma$ from this eliminates D:

$$2T - nR = nma, \qquad \text{so} \qquad T = \tfrac{1}{2}n\,(R + ma).$$

Part (ii)

Every coupling other than the front one is of one of two kinds.

Couplings in front of the second engine. Let T_j be the tension in the coupling just behind carriage j, for $1 \le j \le k$ (so T_k is the coupling between carriage k and the second engine). The front engine and carriages $1, \ldots, j$ have mass $M + jm$ and are pulled back by T_j:

$$D - jR - T_j = (M + jm)a \quad \Longrightarrow \quad T_j = (D - Ma) - j(R + ma) = T - j(R + ma).$$

Since $R + ma > 0$ and $j \ge 1$, we have $T_j < T$ for every such coupling, whatever k is.

Couplings behind the second engine. Let S_j be the tension in the coupling just in front of carriage j, for $k + 1 \le j \le n$. It is easiest to look at the vehicles behind it: carriages $j, \ldots, n$, of which there are $n - j + 1$, pulled forward only by S_j:

$$S_j - (n - j + 1)R = (n - j + 1)ma \quad \Longrightarrow \quad S_j = (n - j + 1)(R + ma).$$

? **Ask yourself** Which of these rear couplings carries the largest tension, and how does it compare with T?

The tension S_j decreases as j increases, so the largest is $S_{k+1} = (n-k)(R+ma)$, in the coupling directly behind the second engine. Comparing with $T = \frac{1}{2}n(R+ma)$, and dividing by $R+ma > 0$,

$$S_{k+1} < T \iff n-k < \tfrac{1}{2}n \iff k > \tfrac{1}{2}n.$$

So if $k > \frac{1}{2}n$, every rear coupling has tension at most $S_{k+1} < T$, and every front coupling has $T_j < T$. Therefore T is greater than the tension in any other coupling, as required. (The condition is also necessary: if $k \leq \frac{1}{2}n$ then $S_{k+1} \geq T$.)

Part (iii)

Take the coupling between carriage k and the second engine. From part (ii) and $T = \frac{1}{2}n(R+ma)$,

$$T_k = T - k(R+ma) = \left(\tfrac{1}{2}n - k\right)(R+ma).$$

If $k > \frac{1}{2}n$, the first factor is negative and the second is positive, so $T_k < 0$: this coupling is in compression. Physically, the second engine is pushing the k carriages in front of it, which is more than half the train, while the front engine pulls them.

Checks

- Take $n = 5$, $k = 3$, $M = 10$, $m = 2$, $D = 20$, $R = 1$. Then $a = \frac{35}{30} = \frac{7}{6}$ and the formula gives $T = \frac{5(40+10)}{30} = \frac{25}{3}$, matching $D - Ma = 20 - \frac{35}{3}$. Going back along the train the tensions are $\frac{25}{3}, 5, \frac{5}{3}, -\frac{5}{3}$ (carriage 3 to engine), then $\frac{20}{3}, \frac{10}{3}$. So T is the largest, and exactly one coupling is in compression, as predicted.
- If $M = 0$ the engines are just forces, and $T = D$, as it must be for a massless front engine.
- Each tension is a force times a dimensionless ratio, so the units are correct.

Common slips

- Giving the engines a resistance as well: the question puts resistance only on the carriages.
- Forgetting that the second engine's driving force acts on any section of train that contains it.
- Assuming every coupling must be in tension; a coupling is a rod, so a negative tension is a perfectly physical push.

Beyond STEP

Splitting a system into pieces and drawing a separate free-body diagram for each, with internal forces appearing in equal and opposite pairs, is the basis of the mechanics

of connected bodies. In first-year courses this becomes the analysis of trusses and frameworks, where you solve for internal forces in many members at once and some members turn out to be struts in compression rather than ties in tension. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats free-body diagrams and Newton's third law carefully.

WORKED SOLUTION

STEP I 2018, Question 11

Mechanics • Friction on Inclined Plane

STEP I 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q11

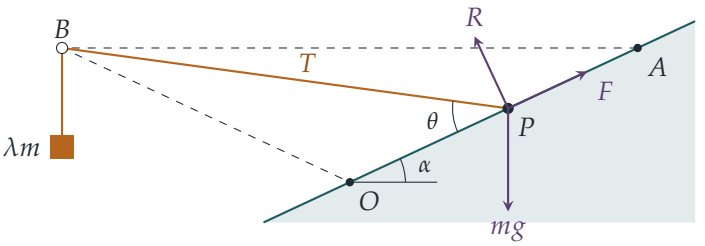
The key idea

Start with the geometry: $OA = OB = d$, so triangle OAB is isosceles, and because AB is horizontal its base angles are both α. The string runs from P back over O's side towards B, so it pulls P partly *down* the slope and partly *off* the plane. Friction must then hold the particle up, while the normal reaction is reduced by the tension. Each condition for equilibrium is an inequality in θ, and it has to hold for every θ in the range from (i), so we look for the worst position.

Solution

Model P and the hanging mass as particles, the string as light and inextensible, and the pulley as smooth, so the tension is the same throughout the string. The hanging particle is in equilibrium, so

$$T = \lambda mg.$$



Part (i)

Put O at the origin with the x-axis horizontal. Then $A = (d\cos\alpha,\ d\sin\alpha)$. The point B is at the same height as A and at distance d from O, so $B = (\pm d\cos\alpha,\ d\sin\alpha)$. The plus sign gives A itself, so

$$B = (-d\cos\alpha,\ d\sin\alpha).$$

This lies above the line of the plane, on the far side of O from A. In triangle OAB we have $OA = OB = d$, and $\angle OAB = \alpha$ because AB is horizontal and AO makes angle α with the horizontal. So $\angle OBA = \alpha$ too, and $\angle AOB = 180^\circ - 2\alpha$, which is obtuse because $\alpha < 45^\circ$.

Let P lie on OA. The string runs along PB, and the acute angle it makes with L is $\theta = \angle BPO$. This angle is acute because it is at most $2\alpha < 90^\circ$, as we now show. It is an exterior angle of triangle PAB, so

$$\theta = \angle BPO = \angle PAB + \angle PBA = \alpha + \angle PBA.$$

As P moves from O to A, the ray BP turns from BO towards BA, so $\angle PBA$ decreases from $\angle OBA = \alpha$ to 0. Therefore

$$\alpha \leq \theta \leq 2\alpha,$$

with $\theta = 2\alpha$ when P is at O and $\theta \to \alpha$ as P approaches A.

Note what this means for the forces. Measured from the down-slope direction PO, the string makes an acute angle θ and points away from the plane. So the tension has a component $T\cos\theta$ down the slope and a component $T\sin\theta$ perpendicular to the plane, pulling P away from it.

Part (ii)

Resolving perpendicular to the plane:

$$R = mg\cos\alpha - T\sin\theta = mg(\cos\alpha - \lambda\sin\theta).$$

Ask yourself Where is friction in this part, when we don't know μ?

We don't need it. Whatever μ is, P must stay in contact with the plane, so we need $R \geq 0$. (In fact $R > 0$: if $R = 0$ there would be no friction, and nothing would balance $mg\sin\alpha + T\cos\theta > 0$ along the slope.) So equilibrium at every point requires

$$\lambda\sin\theta \leq \cos\alpha \quad \text{for all } \theta \text{ that occur.}$$

The largest value of $\sin\theta$ occurs at $\theta = 2\alpha$ (or arbitrarily close to it, as P approaches O), since sin is increasing on $[\alpha, 2\alpha] \subset [0^\circ, 90^\circ)$. Hence $\lambda\sin 2\alpha \leq \cos\alpha$, that is $2\lambda\sin\alpha\cos\alpha \leq \cos\alpha$. Dividing by $\cos\alpha > 0$ gives

$$2\lambda\sin\alpha \leq 1,$$

as required.

Part (iii)

Gravity and the tension both pull P down the slope, so friction F acts up the slope. Resolving along the slope,

$$F = mg\sin\alpha + T\cos\theta = mg(\sin\alpha + \lambda\cos\theta).$$

Equilibrium at this position is possible if and only if $F \leq \mu R$ (this also forces $R > 0$, since $F > 0$). With $\mu = \tan\beta$, the condition is

$$\sin\alpha + \lambda\cos\theta \leq \frac{\sin\beta}{\cos\beta}(\cos\alpha - \lambda\sin\theta).$$

Multiplying by $\cos\beta > 0$ and collecting the λ terms:

$$\lambda(\cos\theta\cos\beta + \sin\theta\sin\beta) \leq \sin\beta\cos\alpha - \cos\beta\sin\alpha,$$
$$\lambda\cos(\beta - \theta) \leq \sin(\beta - \alpha).$$

Here $0 < \beta < 90^\circ$ and $0 < \theta \leq 2\alpha < 90^\circ$, so $-90^\circ < \beta - \theta < 90^\circ$ and $\cos(\beta - \theta) > 0$. So equilibrium at the position with angle θ is possible exactly when

$$\lambda \leq \frac{\sin(\beta - \alpha)}{\cos(\beta - \theta)}.$$

For this to hold at every position, it must hold where $\cos(\beta - \theta)$ is largest, that is, where θ is as close to β as the range $\alpha \leq \theta \leq 2\alpha$ allows.

Case $\beta \geq 2\alpha$. Every θ in the range is at most β, so $0 \leq \beta - 2\alpha \leq \beta - \theta \leq \beta - \alpha < 90^\circ$. Cosine decreases on $[0^\circ, 90^\circ)$, so $\cos(\beta - \theta)$ is largest at $\theta = 2\alpha$, the position at (or next to) O. A necessary condition is therefore

$$\lambda \leq \frac{\sin(\beta - \alpha)}{\cos(\beta - 2\alpha)}.$$

It is also sufficient, because at every other position the bound on λ is weaker.

Case $\alpha \leq \beta \leq 2\alpha$. Now $\theta = \beta$ lies in the range: by part (i), some position of P on L between O and A gives exactly this angle (approached as $P \to A$ if $\beta = \alpha$). There $\cos(\beta - \theta) = 1$, its largest possible value, and the condition becomes

$$\lambda \leq \sin(\beta - \alpha).$$

At that position the string is perpendicular to the total contact force, which makes angle β with the normal when friction is limiting. If $\beta = \alpha$, this forces $\lambda = 0$: the particle is then only just in limiting equilibrium under its own weight, and any string tension would make it slip.

Checks

- The two answers agree when $\beta = 2\alpha$: then $\cos(\beta - 2\alpha) = 1$ and both give $\lambda \leq \sin\alpha$.
- With $\lambda = 0$ the condition becomes $\sin(\beta - \alpha) \geq 0$, i.e. $\mu \geq \tan\alpha$, the familiar condition for a block to rest on a rough slope.
- For $\alpha = 20^\circ$, $\beta = 50^\circ$ the bound is $\sin 30^\circ / \cos 10^\circ \approx 0.508$. A computer check placed P at many points of OA, resolved the forces directly, and found that the largest λ giving equilibrium everywhere matched both formulae for thousands of random α and β. It also confirmed that $\alpha \leq \theta \leq 2\alpha$.

Common slips

- Putting B on A's side of O. Since $OB = OA$ and $B \neq A$, B is on the other side of O, so the tension pulls P *down* the slope, not up it.
- Forgetting that the tension reduces the normal reaction, which reduces the friction available.
- Checking only the end positions in the second case of (iii). The worst position is where $\theta = \beta$, which is in the interior.

Beyond STEP

The combination $\lambda\cos(\beta - \theta) \leq \sin(\beta - \alpha)$ is the algebraic form of the *angle of friction*: the total contact force must lie within angle β of the normal, the so-called cone of friction. Many friction problems collapse to a single line of geometry once you think in terms of that cone. Asking for equilibrium "for every position" is a small optimisation problem, and engineers reason the same way when they design for the worst case. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and the free-body diagram carefully and with many challenging problems.

WORKED SOLUTION

STEP I 2018, Question 12

Probability and Statistics • Sampling With Replacement

STEP I 2018, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q12

The key idea

Write each count of heads as a sum of two indicator variables, one for each toss. The expectation of a sum is the sum of expectations whether or not the tosses are independent, so the means come out easily. The variances turn on a single number, the probability that *both* tosses show heads, and that is exactly where drawing with replacement and drawing without replacement differ.

Solution

Throughout, let $p = \frac{1}{3}(p_1 + p_2 + p_3)$ and

$$q = p_1p_2 + p_2p_3 + p_3p_1.$$

We assume that each coin is equally likely to be chosen and that, once the coins have been chosen, their tosses are independent.

Part (i)

Let C_k be the event that coin k is chosen, so $\mathrm{P}(C_k) = \frac{1}{3}$. By the law of total probability,

$$\mathrm{P}(\text{head}) = \sum_{k=1}^{3} \mathrm{P}(C_k)\,\mathrm{P}(\text{head} \mid C_k) = \tfrac{1}{3}(p_1 + p_2 + p_3) = p.$$

Part (ii)

Let X_1 and X_2 be the indicators of a head on the first and second toss, so $N_1 = X_1 + X_2$. Because the first coin is returned, the second toss happens in exactly the same situation as the first, and by part (i) $\mathrm{P}(X_1 = 1) = \mathrm{P}(X_2 = 1) = p$. Hence

$$\mathrm{E}(N_1) = \mathrm{E}(X_1) + \mathrm{E}(X_2) = 2p.$$

For the variance we need $\mathrm{P}(X_1 = 1,\, X_2 = 1)$. The two choices are independent, each of the 9 ordered pairs of coins (j,k) has probability $\frac{1}{9}$, and given the pair the chance of two heads is $p_j p_k$. So

$$\mathrm{P}(X_1 = 1,\, X_2 = 1) = \frac{1}{9}\sum_{j=1}^{3}\sum_{k=1}^{3} p_j p_k = \left(\frac{p_1+p_2+p_3}{3}\right)^2 = p^2 = \mathrm{P}(X_1 = 1)\,\mathrm{P}(X_2 = 1).$$

The same factorisation works for the other outcomes, so X_1 and X_2 are independent and $N_1 \sim \mathrm{B}(2,p)$. Directly: since $X_i^2 = X_i$,

$$\begin{aligned}\mathrm{E}(N_1^2) &= \mathrm{E}(X_1) + \mathrm{E}(X_2) + 2\,\mathrm{E}(X_1X_2) = 2p + 2p^2,\\ \mathrm{Var}(N_1) &= 2p + 2p^2 - (2p)^2 = 2p(1-p),\end{aligned}$$

as required.

Part (iii)

Now pick the two coins one after the other without replacement, and let Y_1, Y_2 be the indicators of a head on the first and second coin picked, so $N_2 = Y_1 + Y_2$. Each coin is equally likely to be the first one picked, and by symmetry each is equally likely to be the second, so as in part (i) $\mathrm{P}(Y_1 = 1) = \mathrm{P}(Y_2 = 1) = p$ and

$$\mathrm{E}(N_2) = 2p.$$

Ask yourself Can we reuse the calculation of P(both heads) from part (ii)?

No: the same coin can no longer be picked twice. There are now 6 equally likely ordered pairs (j,k) with $j \neq k$, and each unordered pair $\{j,k\}$ appears twice, so

$$\mathrm{P}(Y_1 = 1,\, Y_2 = 1) = \frac{1}{6}\sum_{j\neq k} p_j p_k = \frac{2q}{6} = \frac{q}{3}.$$

In general this is not p^2, so Y_1 and Y_2 are not independent. Using $Y_i^2 = Y_i$ again,

$$\begin{aligned}\mathrm{E}(N_2^2) &= \mathrm{E}(Y_1) + \mathrm{E}(Y_2) + 2\,\mathrm{E}(Y_1Y_2) = 2p + \tfrac{2}{3}q,\\ \mathrm{Var}(N_2) &= 2p + \tfrac{2}{3}(p_1p_2 + p_2p_3 + p_3p_1) - 4p^2.\end{aligned}$$

Part (iv)

Subtracting,

$$\mathrm{Var}(N_1) - \mathrm{Var}(N_2) = \left(2p - 2p^2\right) - \left(2p + \tfrac{2}{3}q - 4p^2\right) = 2p^2 - \tfrac{2}{3}q.$$

Now $9p^2 = (p_1+p_2+p_3)^2 = p_1^2 + p_2^2 + p_3^2 + 2q$, so

$$\begin{aligned}2p^2 - \tfrac{2}{3}q &= \tfrac{2}{9}\left(p_1^2 + p_2^2 + p_3^2 + 2q\right) - \tfrac{6}{9}q = \tfrac{2}{9}\left(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1\right)\\ &= \tfrac{1}{9}\Big[(p_1-p_2)^2 + (p_2-p_3)^2 + (p_3-p_1)^2\Big],\end{aligned}$$

the last step because expanding the three squares gives $2(p_1^2+p_2^2+p_3^2)-2q$.

A sum of squares of real numbers is non-negative, so $\mathrm{Var}(N_2)\le\mathrm{Var}(N_1)$. Equality holds if and only if all three squares vanish, that is, if and only if $p_1=p_2=p_3$. This completes the proof.

As a by-product we get a tidy form for the answer to part (iii):

$$\mathrm{Var}(N_2)=2p(1-p)-\tfrac{1}{9}\Big[(p_1-p_2)^2+(p_2-p_3)^2+(p_3-p_1)^2\Big].$$

Checks

- If $p_1=p_2=p_3=p$, every coin is the same, so it cannot matter how the coins are chosen: both counts are $\mathrm{B}(2,p)$ and the variances agree, matching the equality case.
- Take $p_1=\frac{1}{2}$, $p_2=\frac{1}{3}$, $p_3=\frac{1}{6}$, so $p=\frac{1}{3}$ and $q=\frac{11}{36}$. Then $\mathrm{Var}(N_1)=\frac{4}{9}$ and $\mathrm{Var}(N_2)=\frac{2}{3}+\frac{11}{54}-\frac{4}{9}=\frac{23}{54}$, which is $\frac{1}{54}$ less, agreeing with $\frac{1}{9}\left(\frac{1}{36}+\frac{1}{36}+\frac{1}{9}\right)=\frac{1}{54}$. Listing all the outcomes by hand gives the same values.
- Extreme case $p_1=p_2=1$, $p_3=0$: in part (iii) exactly one of the two chosen coins is a sure head except when coins 1 and 2 are both chosen, so N_2 is 2 with probability $\frac{1}{3}$ and 1 otherwise, giving variance $\frac{2}{9}$. The formula gives $2p+\frac{2}{3}q-4p^2=\frac{4}{3}+\frac{2}{3}-\frac{16}{9}=\frac{2}{9}$.

Common slips

- Treating the two tosses in part (iii) as independent and getting $2p(1-p)$ again; the whole point of part (iv) is that they are not.
- Writing $\mathrm{Var}(N_1)$ as an average of the variances $2p_k(1-p_k)$ for each coin; that ignores the extra variability from the random choice of coin, and in part (ii) a different coin may be used for each toss.

Beyond STEP

The difference between parts (ii) and (iii) is that of sampling with and without replacement. Sampling without replacement makes the draws negatively correlated, which reduces the variance; the same effect gives the "finite population correction" in survey sampling and is why the hypergeometric distribution has a smaller variance than the binomial with the same mean. The algebra in part (iv) is the identity $\sum p_k^2-\sum_{j<k}p_jp_k=\frac{1}{2}\sum_{j<k}(p_j-p_k)^2$, a close cousin of the fact that a variance is never negative. Blitzstein and Hwang, *Introduction to Probability*, treats indicator variables, covariance and the hypergeometric distribution in exactly this spirit.

WORKED SOLUTION

STEP I 2018, Question 13

Probability and Statistics • Binomial and Bayes

STEP I 2018, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q13

The key idea

Everything rests on one count: if a candidate attempts k questions and gets j of them right, the score is $2j-(k-j)=3j-k$. So passing means getting at least $(k+5)/3$ right, and for each k this tells us exactly which binomial terms to add up. In part (iii) the neat move is to treat each question as having three outcomes (right, wrong, not attempted), which turns the problem into a single multinomial count.

Solution

Throughout, let $p=\frac{1}{n}$ be the probability that a randomly chosen answer is correct, and $q=1-p$. Since $n\geq 2$, we have $0<p\leq\frac{1}{2}$.

Part (i)

Let P_k be the probability that A passes when she attempts exactly k questions. Her choices are independent, so the number J she gets right has the binomial distribution $\mathrm{B}(k,p)$. Her score is $3J-k$, so she passes exactly when

$$3J-k\geq 5 \iff J\geq\frac{k+5}{3}.$$

Going through the values of k:

- $k=0,1,2$: we would need $J\geq\frac{5}{3},2,\frac{7}{3}$ respectively, so $J\geq 2$, $J\geq 2$ and $J\geq 3$, which are impossible with $J\leq k$ (the maximum score is $2k\leq 4$). So $P_0=P_1=P_2=0$.
- $k=3$: we need $J\geq\frac{8}{3}$, so $J=3$, and $P_3=p^3$.
- $k=4$: we need $J\geq 3$, so

$$P_4=\binom{4}{3}p^3q+p^4=4p^3(1-p)+p^4=p^3(4-3p).$$

- $k = 5$: we need $J \geq \frac{10}{3}$, so $J = 4$ or 5, and

$$P_5 = \binom{5}{4} p^4 q + p^5 = 5p^4(1-p) + p^5 = p^4(5-4p).$$

Now compare P_4 with its two rivals. Since $p < 1$, we have $4 - 3p > 1$, so $P_4 > p^3 = P_3$. Also

$$P_4 - P_5 = p^3\big[(4-3p) - p(5-4p)\big] = p^3\big(4p^2 - 8p + 4\big) = 4p^3(1-p)^2 > 0,$$

because $0 < p < 1$. As $P_4 > 0$ beats $P_0 = P_1 = P_2 = 0$ too, P_4 is strictly the largest, so A should choose $k = 4$, as required.

Ask yourself Isn't attempting all five questions bound to give more chances to score?

Not here. With five attempts she needs four right, because a fifth attempt costs a mark whenever it is wrong, and it is wrong far more often than right. With four attempts, three right is enough. The factorisation $4p^3(1-p)^2$ shows the advantage for every $n \geq 2$, not just for large n.

Part (ii)

Let K be the number of questions B attempts, so $\mathrm{P}(K = k) = \frac{1}{6}$ for $k = 0, \ldots, 5$, and let S be the event that B passes. Given $K = k$, B is in exactly A's position, so $\mathrm{P}(S \mid K = k) = P_k$. By Bayes' theorem,

$$\mathrm{P}(K = 4 \mid S) = \frac{\frac{1}{6}P_4}{\frac{1}{6}(P_0 + P_1 + \cdots + P_5)} = \frac{p^3(4-3p)}{p^3 + p^3(4-3p) + p^4(5-4p)}.$$

Cancelling p^3 (which is non-zero), the denominator becomes $1 + 4 - 3p + 5p - 4p^2 = 5 + 2p - 4p^2$, so

$$\mathrm{P}(K = 4 \mid S) = \frac{4-3p}{5+2p-4p^2}.$$

Putting $p = \frac{1}{n}$ and multiplying top and bottom by n^2:

$$\mathrm{P}(K = 4 \mid S) = \frac{4n^2 - 3n}{5n^2 + 2n - 4} = \frac{n(4n-3)}{5n^2 + 2n - 4}.$$

Part (iii)

For each question, C's coin toss and her choice of answer are independent of each other and of what happens on the other questions. So each question independently has one

of three outcomes:

$$\begin{aligned} \text{correct } (+2): &\quad \frac{n}{n+1}\cdot\frac{1}{n} = \frac{1}{n+1}, \\ \text{wrong } (-1): &\quad \frac{n}{n+1}\cdot\frac{n-1}{n} = \frac{n-1}{n+1}, \\ \text{not attempted } (0): &\quad \frac{1}{n+1}. \end{aligned}$$

Suppose she gets c right and w wrong, with $c+w \le 5$. She passes when $2c - w \ge 5$. This forces $c \ge 3$, and then:

- $c = 3$: we need $w \le 1$;
- $c = 4$: $w \le 1$ automatically, since $c + w \le 5$;
- $c = 5$: $w = 0$.

The number of ways of assigning c correct, w wrong and $5 - c - w$ unattempted questions is the multinomial coefficient $\frac{5!}{c!\,w!\,(5-c-w)!}$. Since a correct and an unattempted question have the same probability $\frac{1}{n+1}$, each arrangement has probability $\frac{(n-1)^w}{(n+1)^5}$. Adding up the five cases:

(c, w)	arrangements	probability $\times (n+1)^5$
$(3,0)$	10	10
$(3,1)$	20	$20(n-1)$
$(4,0)$	5	5
$(4,1)$	5	$5(n-1)$
$(5,0)$	1	1

The total is $16 + 25(n-1) = 25n - 9$, so

$$\mathrm{P}(\text{C passes}) = \frac{25n-9}{(n+1)^5}.$$

Checks

- With $n = 2$: $P_3 = \frac{1}{8}$, $P_4 = \frac{5}{16}$, $P_5 = \frac{3}{16}$, so $k = 4$ wins. In (ii) the formula gives $\frac{2\cdot 5}{20} = \frac{1}{2}$, matching $\frac{5/16}{10/16}$ directly.
- For (iii) with $n = 3$ the answer is $\frac{66}{1024} \approx 0.0645$, and a computer simulation of 200 000 tests gave 0.0644. All three formulas also agree with exact enumeration by computer for $n = 2, \ldots, 29$.
- As $n \to \infty$ the answer to (ii) tends to $\frac{4}{5}$: when guessing is hopeless, a pass almost always comes from four attempts (needing three right) rather than five (needing four right) or three (needing all three right, which has probability p^3, much less than $4p^3$).

Common slips

- Forgetting that with five attempts three right gives $6 - 2 = 4$, which fails: you need four right.
- In (ii), leaving P_3 out of the denominator: attempting three and getting all three right is also a way to pass.
- In (iii), treating it as "attempt k questions with probability $\binom{5}{k}h^k(1-h)^{5-k}$" and then using part (i) works, but it is easy to lose a term; the three-outcome count is cleaner.

Beyond STEP

Part (iii) is an example of *thinning*: each question is kept with some probability and the outcomes stay independent, so the counts of right, wrong and skipped answers have a multinomial distribution. Part (ii) is Bayesian updating, the idea behind much of modern statistics. Both are developed in Blitzstein and Hwang, *Introduction to Probability*, and in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 2018, Question 1

Pure • Roots of Polynomials

STEP II 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q1

The key idea

The coefficients read the same forwards and backwards, so dividing by x^2 turns the quartic into a quadratic in $t = x + \frac{1}{x}$. Each value of t gives a pair of roots k, $\frac{1}{k}$ of the quartic, and a real x can only give $|t| \geq 2$, with $t = \pm 2$ exactly when $x = \pm 1$. Counting distinct roots then comes down to asking which values of t occur and whether either of them is ± 2.

Solution

Roots come in reciprocal pairs

Write $P(x) = x^4 + ax^3 + bx^2 + ax + 1$. Since $P(0) = 1 \neq 0$, any root k is non-zero, and

$$k^4 P\left(\tfrac{1}{k}\right) = 1 + ak + bk^2 + ak^3 + k^4 = P(k) = 0.$$

As $k^4 \neq 0$, this gives $P(\frac{1}{k}) = 0$, so k^{-1} is a root.

The substitution $t = x + \frac{1}{x}$

Dividing $P(x) = 0$ by x^2 (allowed, as $x \neq 0$) and using $x^2 + \frac{1}{x^2} = t^2 - 2$:

$$\left(x^2 + \frac{1}{x^2}\right) + a\left(x + \frac{1}{x}\right) + b = 0 \iff t^2 + at + (b - 2) = 0.$$

Let t_1, t_2 be the roots of this quadratic, so $t_1 + t_2 = -a$ and $t_1 t_2 = b - 2$. Expanding shows

$$P(x) = (x^2 - t_1 x + 1)(x^2 - t_2 x + 1),$$

because the right-hand side is $x^4 - (t_1 + t_2)x^3 + (2 + t_1 t_2)x^2 - (t_1 + t_2)x + 1$. So the roots of P are the roots of $x^2 - t_1 x + 1$ together with those of $x^2 - t_2 x + 1$.

Now use the assumption that all the roots are real. Each factor $x^2 - t_i x + 1$ has two real roots k and $\frac{1}{k}$ (their product is 1), and $t_i = k + \frac{1}{k}$ is real. For real $k \neq 0$, $k + \frac{1}{k}$ has the

sign of k, and

$$\left|k+\tfrac{1}{k}\right|-2=\frac{(|k|-1)^2}{|k|}\geq 0,$$

with equality only when $k=\pm 1$. So:

- if $|t_i|>2$, the factor x^2-t_ix+1 has two distinct real roots, neither equal to ± 1;
- if $t_i=2$, the factor is $(x-1)^2$; if $t_i=-2$, it is $(x+1)^2$.

Also, different values of t cannot share a root x, since x determines $t=x+\frac{1}{x}$.

Part (i)

If there is only one distinct root k, then $\frac{1}{k}=k$, so $k=\pm 1$ and $P(x)=(x\mp 1)^4$. Expanding,

$$(x-1)^4=x^4-4x^3+6x^2-4x+1,\qquad (x+1)^4=x^4+4x^3+6x^2+4x+1,$$

both of the required form. So the values are $a=-4,\ b=6$ and $a=4,\ b=6$.

Part (ii)

Run through the possibilities for t_1, t_2 (each real with $|t_i|\geq 2$):

- $t_1\neq t_2$, both with $|t_i|>2$: four distinct roots.
- $t_1=t_2$: the quartic is a perfect square $(x^2-t_1x+1)^2$, so it has two distinct roots if $|t_1|>2$ and one if $t_1=\pm 2$.
- $\{t_1,t_2\}=\{2,-2\}$: $P(x)=(x-1)^2(x+1)^2$, two distinct roots.
- One of the t_i is ± 2 and the other has modulus greater than 2: roots ± 1 (double) and two others, which is three distinct roots.

? **Ask yourself** Why must one of the values of t be ± 2?

Three distinct roots out of four means some root is repeated, and the list above shows that this happens with three distinct roots only in the last case. Put more directly: the roots split into reciprocal pairs, and an odd number of distinct roots forces one root to be its own reciprocal, i.e. ± 1.

So $t=2$ or $t=-2$ is a root of $t^2+at+b-2=0$:

$$\begin{aligned} t=2:&\quad 4+2a+b-2=0 \iff b=-2a-2,\\ t=-2:&\quad 4-2a+b-2=0 \iff b=2a-2.\end{aligned}$$

Hence either $b=2a-2$ or $b=-2a-2$, as required.

Part (iii)

With $b=2a-2$ the quadratic in t is

$$t^2+at+2a-4=(t+2)(t+a-2),$$

so $t_1 = -2$ and $t_2 = 2 - a$. The first gives $(x+1)^2$, and the second gives $x^2 - (2-a)x + 1 = 0$, so

$$x = -1 \text{ (twice)}, \qquad x = \frac{2 - a \pm \sqrt{a^2 - 4a}}{2}.$$

These are real because we are told all roots are real; this amounts to $a^2 - 4a = a(a-4) \geq 0$, i.e. $a \leq 0$ or $a \geq 4$.

The final part

By (ii), three distinct roots needs $b = 2a - 2$ or $b = -2a - 2$. Conversely, suppose $b = 2a - 2$ (with all roots real). From (iii) the values of t are -2 and $2 - a$, and from the case list we get exactly three distinct roots unless $2 - a = \pm 2$:

- $a = 0$, $b = -2$: $t = \pm 2$, $P(x) = (x^2 - 1)^2$, only two distinct roots;
- $a = 4$, $b = 6$: $t = -2$ twice, $P(x) = (x+1)^4$, one root.

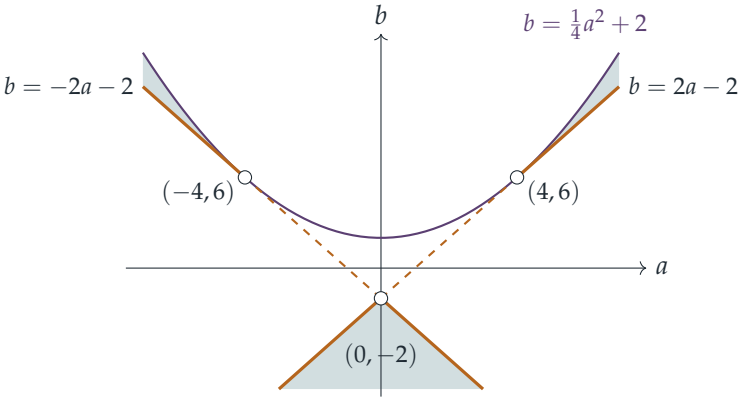
Replacing x by $-x$ turns P into the same form with a replaced by $-a$, which swaps the two lines. So for $b = -2a - 2$ the values of t are 2 and $-a - 2$, and the exceptions are $a = 0$, $b = -2$ and $a = -4$, $b = 6$.

Therefore, for real a, b with all roots real, $(*)$ has exactly three distinct roots if and only if

$$b = 2a - 2 \text{ or } b = -2a - 2, \qquad \text{with } (a,b) \neq (0,-2),\ (4,6),\ (-4,6).$$

Note that $(4,-10)$ and $(-4,-10)$ are allowed: for example $a = 4$, $b = -10$ gives $t = 2$ and $t = -6$, so three distinct roots.

The picture shows the (a,b)-plane. The shaded region is where all roots are real: below both lines (one value of t is at most -2 and the other at least 2), or between a line and the parabola $b = \frac{1}{4}a^2 + 2$ for $|a| \geq 4$ (both values of t on the same side). The parabola is where $t_1 = t_2$, and it touches the lines at $(\pm 4, 6)$. The answer is the solid parts of the two lines on the edge of the shaded region, minus the three open circles.



(The dashed parts of the lines, $0 < |a| < 4$, lie outside the region where all roots are real, which is why (iii) needed $a \leq 0$ or $a \geq 4$.)

Checks

- Take $a = -4$, $b = -10$ (on $b = 2a - 2$, allowed). Then (iii) gives $x = -1, -1, 3 \pm 2\sqrt{2}$, and indeed $(x+1)^2(x^2 - 6x + 1) = x^4 - 4x^3 - 10x^2 - 4x + 1$. The roots $3 \pm 2\sqrt{2} \approx 5.828,\ 0.172$ are reciprocals, as they should be.
- At the excluded point $a = 0$, $b = -2$ the quartic is $x^4 - 2x^2 + 1 = (x^2 - 1)^2$, with only two distinct roots.
- A computer found the roots numerically for every (a, b) on a grid of step $\frac{1}{4}$ with $|a| \leq 10$, $|b| \leq 15$; of the 1579 grid points with all roots real, exactly the 88 predicted by the condition above gave three distinct roots.

Common slips

- Forgetting the exceptional points: $b = 2a - 2$ alone is necessary but not sufficient, since $(0, -2)$ and $(4, 6)$ lie on it.
- Assuming that a repeated root must be ± 1: a double pair $k, k, \frac{1}{k}, \frac{1}{k}$ is also possible, but it gives two distinct roots, not three.

Beyond STEP

Polynomials whose coefficients read the same both ways are called *palindromic* (or reciprocal), and the trick $t = x + \frac{1}{x}$ halves the degree of any of even degree. The same substitution with $x = e^{i\theta}$ gives $t = 2\cos\theta$, which is how the minimal polynomials of $2\cos(2\pi/n)$ are found and how Gauss handled the regular 17-gon. The idea of describing where a polynomial's roots collide, as the curve in the (a, b)-plane above does, leads to the *discriminant*. Allenby's *Rings, Fields and Groups* is a good place to read more about polynomials and their roots.

WORKED SOLUTION

STEP II 2018, Question 2

Pure • Concave Functions

STEP II 2018, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q2

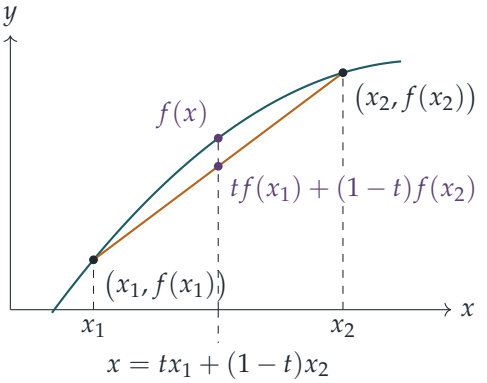
The key idea

The point $tx_1 + (1-t)x_2$ runs from x_2 to x_1 as t goes from 0 to 1, and $tf(x_1) + (1-t)f(x_2)$ is the height of the chord above that point. So concavity says that every chord lies on or below the curve. Once that two-point inequality is in hand, a three-point version comes from using it twice, and both triangle inequalities follow by applying it to a suitable concave function with $A + B + C = \pi$.

Solution

The sketch

Write $x = tx_1 + (1-t)x_2$. Then the point $\big(x,\ tf(x_1) + (1-t)f(x_2)\big)$ divides the chord from $\big(x_2, f(x_2)\big)$ to $\big(x_1, f(x_1)\big)$ in the ratio $(1-t) : t$, so it lies on the chord directly above (or below) x. The definition says the curve at x is at least as high as the chord.



Why $f'' < 0$ gives concavity

If $x_1 = x_2$, or $t = 0$ or $t = 1$, the inequality is an equality, so take $0 < t < 1$ and, without loss of generality, $x_1 < x_2$ (swapping x_1, x_2 just swaps t and $1 - t$). Then $x = tx_1 + (1-t)x_2$ satisfies $x_1 < x < x_2$, with

$$x - x_1 = (1-t)(x_2 - x_1), \qquad x_2 - x = t(x_2 - x_1).$$

By the mean value theorem there are $c_1 \in (x_1, x)$ and $c_2 \in (x, x_2)$ with

$$\frac{f(x) - f(x_1)}{x - x_1} = f'(c_1), \qquad \frac{f(x_2) - f(x)}{x_2 - x} = f'(c_2).$$

Since $f'' < 0$ on (a, b), f' is strictly decreasing there, and $c_1 < c_2$, so $f'(c_1) > f'(c_2)$. Substituting the lengths above and cancelling the positive factor $x_2 - x_1$,

$$\frac{f(x) - f(x_1)}{1-t} > \frac{f(x_2) - f(x)}{t}.$$

Multiplying by $t(1-t) > 0$ gives $tf(x) - tf(x_1) > (1-t)f(x_2) - (1-t)f(x)$, that is,

$$f(x) > tf(x_1) + (1-t)f(x_2).$$

So f is concave on (a, b). In words: the gradient of the curve keeps falling, so the slope from x_1 to x is bigger than the slope from x to x_2, and the curve bulges above the chord.

Part (i)

? **Ask yourself** Which two points, with which weight, average to $\frac{1}{3}(u+v+w)$?

Group v and w together: take $x_1 = u$, $x_2 = \frac{1}{2}(v+w)$ and $t = \frac{1}{3}$. Then x_2 lies in (a, b), since it is between v and w, and

$$tx_1 + (1-t)x_2 = \tfrac{1}{3}u + \tfrac{2}{3} \cdot \tfrac{1}{2}(v+w) = \tfrac{1}{3}(u+v+w).$$

The definition gives

$$f\Big(\frac{u+v+w}{3}\Big) \geq \tfrac{1}{3}f(u) + \tfrac{2}{3}f\Big(\frac{v+w}{2}\Big).$$

Now use the definition again with $x_1 = v$, $x_2 = w$, $t = \frac{1}{2}$: $f\big(\frac{1}{2}(v+w)\big) \geq \frac{1}{2}f(v) + \frac{1}{2}f(w)$. Since this term has the positive coefficient $\frac{2}{3}$, substituting keeps the inequality the same way round:

$$f\Big(\frac{u+v+w}{3}\Big) \geq \tfrac{1}{3}f(u) + \tfrac{2}{3} \cdot \tfrac{1}{2}\big(f(v) + f(w)\big) = \frac{f(u) + f(v) + f(w)}{3},$$

which is the required result.

Part (ii)

Take $f(x) = \sin x$ on $0 < x < \pi$. Here $f''(x) = -\sin x < 0$, so sin is concave on $(0, \pi)$. The angles of a triangle satisfy $0 < A, B, C < \pi$ and $A + B + C = \pi$, so part (i) with $u = A$, $v = B$, $w = C$ gives

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2},$$

so $\sin A + \sin B + \sin C \leq \dfrac{3\sqrt{3}}{2}$.

Part (iii)

For $0 < x < \pi$ we have $\sin x > 0$, so $g(x) = \ln(\sin x)$ is defined there, with

$$g'(x) = \frac{\cos x}{\sin x} = \cot x, \qquad g''(x) = -\operatorname{cosec}^2 x < 0.$$

So g is concave on $(0, \pi)$, and part (i) gives

$$\frac{\ln \sin A + \ln \sin B + \ln \sin C}{3} \leq \ln \sin \frac{\pi}{3} = \ln \frac{\sqrt{3}}{2}.$$

Multiplying by 3 and combining the logarithms, $\ln(\sin A \sin B \sin C) \leq \ln \left(\frac{\sqrt{3}}{2}\right)^3$. Since exp is increasing,

$$\sin A \times \sin B \times \sin C \leq \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{3\sqrt{3}}{8}.$$

Checks

Both bounds are attained by the equilateral triangle: $3 \sin \frac{\pi}{3} = \frac{3\sqrt{3}}{2} \approx 2.598$ and $\sin^3 \frac{\pi}{3} = \frac{3\sqrt{3}}{8} \approx 0.6495$. A right-angled isosceles triangle gives a sum of $1 + \sqrt{2} \approx 2.414$ and a product of $\frac{1}{2}$, both below the bounds. A computer check over 200 000 random triangles found no sum above 2.598076 and no product above 0.649519. A very flat triangle ($A, B \to 0$, $C \to \pi$) sends both towards 0, as it should.

Common slips

- In part (i), using $t = \frac{1}{3}$ with $x_1 = u$ and $x_2 = v$ gives $\frac{1}{3}u + \frac{2}{3}v$, not the mean of three numbers; the second point must be the midpoint of v and w.
- In part (iii), you can't apply part (ii)'s inequality to the product directly; it is the logarithm that turns the product into a sum. Check also that $\ln \sin x$ is defined, which needs $\sin x > 0$.
- Saying that $f'' < 0$ "means the curve bends down" is not by itself a proof of the chord inequality; the mean value theorem argument makes it precise.

Beyond STEP

Part (i) is the three-point case of Jensen's inequality: for concave f and weights $t_i \geq 0$ summing to 1, $f\left(\sum t_i x_i\right) \geq \sum t_i f(x_i)$, proved by induction exactly as here. With $f = \ln$ it gives the AM–GM inequality, and in probability it says $f(\mathrm{E}[X]) \geq \mathrm{E}[f(X)]$ for concave f. Convex and concave functions, and the mean value theorem behind the second paragraph, are treated carefully in Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP II 2018, Question 3

Pure • Rotational Symmetry and Integrals

STEP II 2018, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q3

The key idea

A half-turn about a point (a, b) sends (x, y) to $(2a - x, 2b - y)$, so a curve with that symmetry pairs each value of x with its reflection $2a - x$ and makes the two heights add to $2b$. For the tangent function the natural pairing is $x \leftrightarrow \frac{1}{2}\pi - x$, because $\tan(\frac{1}{2}\pi - x) = 1/\tan x$. That one identity gives the symmetry of the curve, and then an integral over an interval symmetric about the centre can be found without ever integrating.

Solution

Part (i)

Writing $f(x) = (1 + \tan x)^{-1}$ and using the chain rule,

$$f'(x) = -\frac{\sec^2 x}{(1+\tan x)^2} = -\frac{1}{\cos^2 x\,(1+\tan x)^2} = -\frac{1}{(\cos x + \sin x)^2}$$
$$= -\frac{1}{\cos^2 x + 2\sin x \cos x + \sin^2 x} = -\frac{1}{1+\sin 2x},$$

as required. (Multiplying $1 + \tan x$ by $\cos x$ is the step that tidies everything up.)

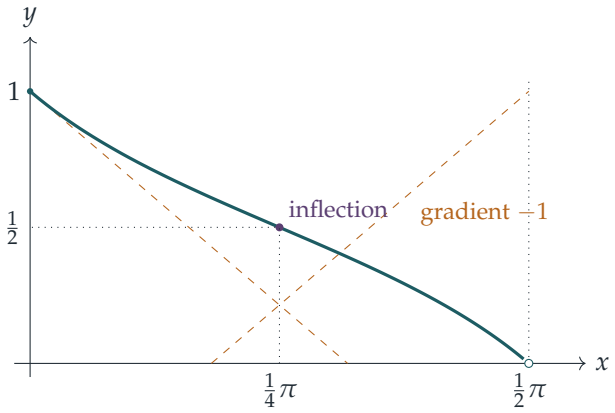
For the range, $0 \le x < \frac{1}{2}\pi$ gives $0 \le 2x < \pi$, so $\sin 2x$ takes every value in $[0, 1]$: the value 0 at $x = 0$ and the value 1 at $x = \frac{1}{4}\pi$. Hence $1 \le 1 + \sin 2x \le 2$, and taking reciprocals (which reverses the inequalities, as everything is positive) and then negatives,

$$-1 \le f'(x) \le -\tfrac{1}{2}.$$

Both ends are attained: $f'(0) = -1$ and $f'(\frac{1}{4}\pi) = -\frac{1}{2}$.

For the sketch: $f(0) = 1$, $f(\frac{1}{4}\pi) = \frac{1}{2}$, and $f(x) \to 0$ as $x \to \frac{1}{2}\pi$ because $\tan x \to \infty$. The curve is always decreasing, with gradient between -1 and $-\frac{1}{2}$. It is steepest at $x = 0$ (gradient -1), least steep at $x = \frac{1}{4}\pi$, and steepens again towards gradient -1 as

$x \to \frac{1}{2}\pi$. So f' has a maximum at $\frac{1}{4}\pi$, which is a point of inflection: the curve is concave up on $[0, \frac{1}{4}\pi)$ and concave down on $(\frac{1}{4}\pi, \frac{1}{2}\pi)$. (Indeed $f''(x) = 2\cos 2x/(1+\sin 2x)^2$, which changes sign at $\frac{1}{4}\pi$.)



The dashed lines have gradient -1: one is the tangent at $(0,1)$, and the other is the direction the curve approaches as $x \to \frac{1}{2}\pi$. The curve looks symmetric under a half-turn about $(\frac{1}{4}\pi, \frac{1}{2})$, which is what the rest of the question is about.

Part (ii)

A rotation through π about (a,b) sends (x,y) to $(2a-x,\, 2b-y)$, since (a,b) is the midpoint of a point and its image. The curve has rotational symmetry of order 2 about (a,b) exactly when this rotation maps the curve onto itself. (For that to be possible the domain must be symmetric about $x=a$, so that $2a-x$ is in the domain whenever x is.)

($\Rightarrow$) Suppose the curve has this symmetry. For any x in the domain, $(x, g(x))$ is on the curve, so its image $(2a-x,\, 2b-g(x))$ is on the curve. The only point of the curve with x-coordinate $2a-x$ is $(2a-x,\, g(2a-x))$, because the curve is a graph. So $g(2a-x) = 2b-g(x)$, that is,

$$g(x) + g(2a-x) = 2b.$$

($\Leftarrow$) Suppose $g(x)+g(2a-x) = 2b$ for every x in the domain. The image of a typical point $(x, g(x))$ is

$$(2a-x,\, 2b-g(x)) = (2a-x,\, g(2a-x)),$$

which is on the curve. So the rotation maps the curve into itself, and since doing it twice gives the identity, it maps the curve onto itself. Putting $x=a$ in the identity gives $g(a) = b$, consistent with (a,b) being on the curve.

For the integral, symmetry about the origin means $a = b = 0$, so $g(-x) = -g(x)$: g is an odd function. Substituting $x = -u$ in the left half,

$$\int_{-1}^{0} g(x)\,\mathrm{d}x = \int_{0}^{1} g(-u)\,\mathrm{d}u = -\int_{0}^{1} g(u)\,\mathrm{d}u,$$

so the two halves cancel and

$$\int_{-1}^{1} g(x)\,\mathrm{d}x = 0.$$

Part (iii)

Let $h(x) = \dfrac{1}{1+\tan^k x}$ for $0 < x < \frac{1}{2}\pi$. Guided by the sketch in (i), try $a = \frac{1}{4}\pi$, $b = \frac{1}{2}$. The domain $(0, \frac{1}{2}\pi)$ is symmetric about $\frac{1}{4}\pi$, and $2a - x = \frac{1}{2}\pi - x$.

> **Ask yourself** How do we simplify $h(\frac{1}{2}\pi - x)$?

Use $\tan(\frac{1}{2}\pi - x) = \cot x = 1/\tan x$, valid for $0 < x < \frac{1}{2}\pi$, where $\tan x > 0$. Then, multiplying top and bottom by $\tan^k x$,

$$h(\tfrac{1}{2}\pi - x) = \frac{1}{1+\tan^{-k} x} = \frac{\tan^k x}{\tan^k x + 1},$$

so

$$h(x) + h(\tfrac{1}{2}\pi - x) = \frac{1+\tan^k x}{1+\tan^k x} = 1 = 2\cdot\tfrac{1}{2}.$$

The point $(\frac{1}{4}\pi, \frac{1}{2})$ is on the curve, since $\tan\frac{1}{4}\pi = 1$ gives $h(\frac{1}{4}\pi) = \frac{1}{2}$. By (ii), the curve has rotational symmetry of order 2 about $(\frac{1}{4}\pi, \frac{1}{2})$, whatever the positive constant k.

For the integral, let $I = \displaystyle\int_{\pi/6}^{\pi/3} h(x)\,\mathrm{d}x$. The interval $[\frac{1}{6}\pi, \frac{1}{3}\pi]$ is symmetric about $\frac{1}{4}\pi$, so the substitution $x = \frac{1}{2}\pi - u$ maps it to itself (with the ends swapped, which absorbs the minus sign from $\mathrm{d}x = -\mathrm{d}u$):

$$\begin{aligned} I &= \int_{\pi/6}^{\pi/3} h(\tfrac{1}{2}\pi - u)\,\mathrm{d}u = \int_{\pi/6}^{\pi/3} \big(1 - h(u)\big)\,\mathrm{d}u \\ &= \left(\tfrac{1}{3}\pi - \tfrac{1}{6}\pi\right) - I. \end{aligned}$$

Hence $2I = \frac{1}{6}\pi$, and

$$\int_{\pi/6}^{\pi/3} \frac{1}{1+\tan^k x}\,\mathrm{d}x = \frac{\pi}{12}.$$

Geometrically, the half-turn swaps the region under the curve with the region above it inside the rectangle $[\frac{1}{6}\pi, \frac{1}{3}\pi] \times [0, 1]$, so the area is half the rectangle, $\frac{1}{2}\cdot\frac{1}{6}\pi$. This is the argument of (ii) with g replaced by $h - \frac{1}{2}$.

Checks

- With $k = 1$ and $x = \frac{1}{6}\pi$: $h = 1/(1+\frac{1}{\sqrt{3}}) = \frac{1}{2}(3-\sqrt{3}) \approx 0.634$, and with $x = \frac{1}{3}\pi$: $h = 1/(1+\sqrt{3}) = \frac{1}{2}(\sqrt{3}-1) \approx 0.366$. These add to 1, as the symmetry says.

- The answer does not depend on k. As $k \to \infty$, h is close to 1 for $x < \frac{1}{4}\pi$ and close to 0 for $x > \frac{1}{4}\pi$, so the integral tends to the length of $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, which is $\frac{1}{12}\pi$, in agreement.
- Numerical integration for $k = 0.3$, 1, 2, 3.7 and 10 gave 0.2617994 each time, matching $\pi/12 \approx 0.2617994$.

Common slips

- In (i), stating the range as $-1 < f'(x) \leq -\frac{1}{2}$: the value -1 is attained at $x = 0$, which is in the domain.
- In (ii), proving only one direction. "If and only if" needs both the symmetry $\Rightarrow$ identity and identity $\Rightarrow$ symmetry.
- In (iii), trying to integrate $1/(1 + \tan^k x)$ directly. For general k there is no elementary antiderivative to find, and the symmetry is the whole point.

Beyond STEP

The trick in (iii), adding the integral to itself after the substitution $x \mapsto a + b - x$, is often called "King's rule". It is a special case of using a symmetry of the domain to simplify an integral, an idea that becomes central in group theory and in physics, where symmetries of a problem predict which integrals vanish (as in (ii)) or take simple values. The same reasoning evaluates $\int_0^{\pi/2} \frac{\mathrm{d}x}{1+\tan^k x} = \frac{1}{4}\pi$ for every $k > 0$. Siklos's *Advanced Problems in Mathematics* has many more integrals that yield to this kind of symmetry argument.

WORKED SOLUTION

STEP II 2018, Question 4

Pure • Trigonometric Equations

STEP II 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q4

The key idea

The given identity turns a sum of cosines into a product, and a product is what we want when solving an equation equal to zero. In (i), pairing the outer terms and the inner terms produces a common factor at once. In (ii) and (iii) the useful move is to express everything through the half-sum and half-difference of the angles, after which (ii) factorises and (iii) becomes a sum of two squares equal to zero.

Solution

Part (i)

Pair the first and last terms, and the two middle terms. By the identity,

$$\cos x + \cos 4x = 2\cos\tfrac{5x}{2}\cos\tfrac{3x}{2},$$
$$3(\cos 2x + \cos 3x) = 6\cos\tfrac{5x}{2}\cos\tfrac{x}{2}.$$

(We may write $\cos(-\frac{3x}{2}) = \cos\frac{3x}{2}$, since cosine is even.) So the left-hand side is

$$2\cos\tfrac{5x}{2}\left(\cos\tfrac{3x}{2} + 3\cos\tfrac{x}{2}\right).$$

With $c = \cos\frac{x}{2}$, the triple-angle formula $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ gives $\cos\frac{3x}{2} = 4c^3 - 3c$, so the bracket is just $4c^3$. Hence the equation is

$$8\cos\tfrac{5x}{2}\cos^3\tfrac{x}{2} = 0.$$

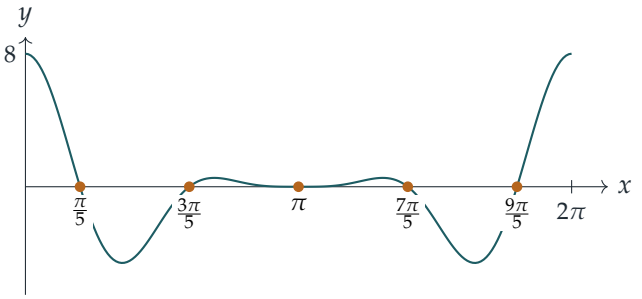
First factor. $\cos\frac{5x}{2} = 0$ means $\frac{5x}{2} = \frac{\pi}{2} + k\pi$ for an integer k, i.e. $x = \frac{(2k+1)\pi}{5}$. For $0 \le x \le 2\pi$ we need $0 \le 2k+1 \le 10$, so $k = 0, 1, 2, 3, 4$.

Second factor. $\cos\frac{x}{2} = 0$ with $0 \le \frac{x}{2} \le \pi$ gives only $\frac{x}{2} = \frac{\pi}{2}$, i.e. $x = \pi$, which is already in the list.

Therefore the solutions are

$$x = \frac{\pi}{5},\ \frac{3\pi}{5},\ \pi,\ \frac{7\pi}{5},\ \frac{9\pi}{5}.$$

The graph of the left-hand side shows them; at $x = \pi$ both $\cos\frac{5x}{2}$ and $\cos^3\frac{x}{2}$ vanish, so the curve only touches the axis there rather than crossing it.



Part (ii)

Using the identity on the first two terms, and $\cos 2x = 2\cos^2 x - 1$,

$$\cos(x+y) + \cos(x-y) - \cos 2x = 2\cos x \cos y - 2\cos^2 x + 1.$$

Setting this equal to 1 gives

$$2\cos x\,(\cos y - \cos x) = 0,$$

so $\cos x = 0$ or $\cos y = \cos x$.

- If $\cos x = 0$ with $0 \le x \le \pi$, then $x = \frac{\pi}{2}$ (and then any y works).
- If $\cos y = \cos x$, then $x = y$, because cosine is strictly decreasing on $[0, \pi]$ and so takes each value there only once.

So either $x = y$ or $x = \frac{\pi}{2}$, as required.

Part (iii)

Let $s = \frac{1}{2}(x+y)$ and $d = \frac{1}{2}(x-y)$. By the identity $\cos x + \cos y = 2\cos s \cos d$, and $\cos(x+y) = \cos 2s = 2\cos^2 s - 1$. The equation becomes

$$2\cos s\cos d - 2\cos^2 s + 1 = \tfrac{3}{2} \iff 4\cos^2 s - 4\cos s\cos d + 1 = 0.$$

Ask yourself One equation, two unknowns: how can it pin down both x and y?

Complete the square in $\cos s$. Since $1 = \cos^2 d + \sin^2 d$,

$$4\cos^2 s - 4\cos s\cos d + \cos^2 d + \sin^2 d = (2\cos s - \cos d)^2 + \sin^2 d = 0.$$

A sum of two real squares is zero only if both are zero, which gives two equations:

$$\sin d = 0 \qquad \text{and} \qquad 2\cos s = \cos d.$$

Since $x, y \in [0, \pi]$, we have $-\frac{\pi}{2} \le d \le \frac{\pi}{2}$, and the only zero of sin there is $d = 0$. So $x = y$, and then $s = x$ and $\cos d = 1$, giving $\cos x = \frac{1}{2}$. With $0 \le x \le \pi$ this means $x = \frac{\pi}{3}$.

So the only solution is

$$x = y = \frac{\pi}{3}.$$

Indeed, $\cos\frac{\pi}{3} + \cos\frac{\pi}{3} - \cos\frac{2\pi}{3} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$.

The same algebra shows more: for all x, y,

$$\tfrac{3}{2} - \big(\cos x + \cos y - \cos(x+y)\big) = \tfrac{1}{2}\big[(2\cos s - \cos d)^2 + \sin^2 d\big] \ge 0,$$

so $\frac{3}{2}$ is the largest value the left-hand side can take, and the equation asks exactly where that maximum is reached.

Checks

- In (i), at $x = \frac{\pi}{5}$: $\cos\frac{\pi}{5} + \cos\frac{4\pi}{5} = 0$ and $\cos\frac{2\pi}{5} + \cos\frac{3\pi}{5} = 0$, since $\cos(\pi - \theta) = -\cos\theta$. At $x = 0$ the left-hand side is $1 + 3 + 3 + 1 = 8$, matching $8\cos 0\cos^3 0$.
- A computer search on a fine grid over $[0, 2\pi]$ found exactly the five zeros above, and a grid over $[0, \pi]^2$ found the maximum of $\cos x + \cos y - \cos(x+y)$ to be 1.5, attained only at $x = y \approx 1.0472 = \frac{\pi}{3}$.
- In (ii), $x = \frac{\pi}{2}$ gives $\cos(\frac{\pi}{2} + y) + \cos(\frac{\pi}{2} - y) - \cos\pi = -\sin y + \sin y + 1 = 1$ for every y.

Common slips

- Listing $x = \pi$ twice, or missing it: it comes from both factors in (i) but is one solution.
- In (ii), dividing by $\cos x$ and so losing $x = \frac{\pi}{2}$.
- In (iii), stopping at a quadratic in $\cos s$ without using the discriminant (or the squares) to force $\cos^2 d = 1$.

Beyond STEP

The coefficients $1, 3, 3, 1$ in (i) are no accident: the left-hand side is the real part of $e^{ix}(1 + e^{ix})^3$, and since $1 + e^{ix} = 2\cos\frac{x}{2}\, e^{ix/2}$ this is $8\cos^3\frac{x}{2}\cos\frac{5x}{2}$ in one line. Part (iii) is the classical inequality $\cos A + \cos B + \cos C \le \frac{3}{2}$ for the angles of a triangle in disguise (put $A = x$, $B = y$, $C = \pi - x - y$), with equality only for the equilateral triangle; maximising a function of two variables like this is a standard first-year topic in multivariable calculus. Needham's *Visual Complex Analysis* shows how complex exponentials make trigonometric identities like these almost automatic.

WORKED SOLUTION

STEP II 2018, Question 5

Pure • Integration of Series

STEP II 2018, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q5

The key idea

Each part turns a function into a power series and integrates it one term at a time. In (i) that is just integrating a geometric series. In (ii), after expanding the exponential, every term becomes an integral of the form $\int_0^\infty x^m \mathrm{e}^{-x}\,\mathrm{d}x$, which is a factorial, and the factorials cancel to leave exactly the series from (i). For (iii), the $\ln x$ in the denominator is the clue: substituting $x = \mathrm{e}^{-t}$ turns it into $-t$ and turns the integral into two copies of the integral in (ii).

Solution

Part (i)

For $|x| < 1$ the binomial expansion (a geometric series with ratio $-x$) is

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n.$$

Integrate both sides from 0 to x. On the left, $\int_0^x \frac{\mathrm{d}u}{1+u} = \ln(1+x)$, since $\ln 1 = 0$. On the right, integrating term by term,

$$\begin{aligned}\ln(1+x) &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \qquad \text{(replacing } n+1 \text{ by } n\text{)} \\ &= -\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n} = -\sum_{n=1}^{\infty} \frac{(-x)^n}{n},\end{aligned}$$

which is the required result for $|x| < 1$.

Part (ii)

For all x,

$$\mathrm{e}^{-ax} = \sum_{n=0}^{\infty} \frac{(-ax)^n}{n!} = 1 - ax + \frac{a^2x^2}{2!} - \frac{a^3x^3}{3!} + \cdots.$$

The $n = 0$ term is 1, so

$$\frac{1 - \mathrm{e}^{-ax}}{x} = -\sum_{n=1}^{\infty} \frac{(-a)^n x^{n-1}}{n!}.$$

Dividing by x is harmless here: every remaining term has a factor x, so the integrand is perfectly well behaved at $x = 0$ (it tends to a). Multiplying by e^{-x} and integrating term by term (we are told to ignore convergence),

$$\int_0^{\infty} \frac{(1 - \mathrm{e}^{-ax})\,\mathrm{e}^{-x}}{x}\,\mathrm{d}x = -\sum_{n=1}^{\infty} \frac{(-a)^n}{n!} I_{n-1}, \qquad \text{where } I_m = \int_0^{\infty} x^m \mathrm{e}^{-x}\,\mathrm{d}x.$$

Ask yourself What is I_m?

Integrate by parts with $u = x^m$ and $\mathrm{d}v = \mathrm{e}^{-x}\,\mathrm{d}x$. For $m \geq 1$,

$$I_m = \Big[-x^m \mathrm{e}^{-x} \Big]_0^{\infty} + m \int_0^{\infty} x^{m-1} \mathrm{e}^{-x}\,\mathrm{d}x = m I_{m-1},$$

because $x^m \mathrm{e}^{-x} \to 0$ as $x \to \infty$ (the exponential wins) and is 0 at $x = 0$. Since $I_0 = \int_0^{\infty} \mathrm{e}^{-x}\,\mathrm{d}x = 1$, it follows that $I_m = m!$.

So $I_{n-1} = (n-1)!$, and $\dfrac{(n-1)!}{n!} = \dfrac{1}{n}$. Therefore

$$\int_0^{\infty} \frac{(1 - \mathrm{e}^{-ax})\,\mathrm{e}^{-x}}{x}\,\mathrm{d}x = -\sum_{n=1}^{\infty} \frac{(-a)^n}{n} = \ln(1 + a),$$

using part (i) with $x = a$, which is allowed because $|a| < 1$. This is the required result.

Part (iii)

Substitute $x = \mathrm{e}^{-t}$, so that $\ln x = -t$ and $\mathrm{d}x = -\mathrm{e}^{-t}\,\mathrm{d}t$. As x runs from 0 to 1, t runs from ∞ down to 0. Also $x^p = \mathrm{e}^{-pt}$ and $x^q = \mathrm{e}^{-qt}$. Hence

$$\int_0^1 \frac{x^p - x^q}{\ln x}\,\mathrm{d}x = \int_{\infty}^0 \frac{\mathrm{e}^{-pt} - \mathrm{e}^{-qt}}{-t}\left(-\mathrm{e}^{-t}\right)\mathrm{d}t = \int_0^{\infty} \frac{\left(\mathrm{e}^{-qt} - \mathrm{e}^{-pt}\right)\mathrm{e}^{-t}}{t}\,\mathrm{d}t,$$

where the two minus signs cancelled and swapping the limits produced the swap of p and q. Now write $\mathrm{e}^{-qt} - \mathrm{e}^{-pt} = \left(1 - \mathrm{e}^{-pt}\right) - \left(1 - \mathrm{e}^{-qt}\right)$ and split the integral. Each piece has the form in (ii), with $a = p$ and $a = q$ respectively, and both satisfy $|a| < 1$. So

$$\int_0^1 \frac{x^p - x^q}{\ln x}\,\mathrm{d}x = \ln(1 + p) - \ln(1 + q) = \ln\left(\frac{1 + p}{1 + q}\right).$$

Splitting is legitimate because each of the two integrals converges on its own: by (ii) each has a finite value.

Checks

- If $p = q$ the integrand is zero, and indeed $\ln \frac{1+p}{1+p} = 0$.

- Swapping p and q changes the sign of both the integrand and the answer.
- Sign check: for $0 < x < 1$, $\ln x < 0$; if $p < q$ then $x^p > x^q$, so the integrand is negative, and so is $\ln \frac{1+p}{1+q}$.
- With $p = \frac{1}{2}$, $q = 0$ the answer is $\ln \frac{3}{2} \approx 0.405465$. A numerical integration of $\int_0^1 \frac{\sqrt{x}-1}{\ln x}\,\mathrm{d}x$ gives the same value, and random choices of p and q in $(-1,1)$ agree with the formula to six decimal places.

Common slips

- Forgetting that the $n = 0$ term of e^{-ax} cancels the 1, and then trying to integrate e^{-x}/x, which diverges at 0.
- Losing a sign in the substitution in (iii): there are three minus signs to track (from $\ln x$, from $\mathrm{d}x$, and from reversing the limits), and getting one wrong gives $\ln \frac{1+q}{1+p}$.
- Writing $I_{n-1} = n!$ instead of $(n-1)!$; then the factorials no longer cancel to $\frac{1}{n}$.

Beyond STEP

The integral $I_m = \int_0^\infty x^m \mathrm{e}^{-x}\,\mathrm{d}x = m!$ is the gamma function $\Gamma(m+1)$, which extends the factorial to non-integers. The integral in (iii) is a classic of "differentiating under the integral sign": differentiating with respect to p removes the awkward $\ln x$ and leaves $\int_0^1 x^p\,\mathrm{d}x = \frac{1}{p+1}$. The result actually holds whenever $p, q > -1$. Swapping a sum and an integral, which we were told not to worry about, is justified in analysis courses by uniform or dominated convergence. For the analysis behind these steps, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2018, Question 6

Pure • Factorials and Primes

STEP II 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q6

The key idea

Both parts turn on one fact: a prime p divides $k!$ exactly when $p \leq k$. In (i), once $n \geq 5$ the factorial is a multiple of 5, and so is $n! + 5$. In (ii), the left-hand side is built only from numbers up to $2n-1$, so no prime bigger than $2n-1$ can divide it. If $m!$ is large enough to contain such a prime, the equation fails. The two theorems are there to make m large and to supply that prime.

Solution

Part (i)

For $n \geq 5$, 5 is one of the factors of $n!$, so $5 \mid n!$ and hence $5 \mid n! + 5$. But $n! + 5 > 5$, so $n! + 5$ has a factor 5 strictly between 1 and itself, and is not prime. So only $n \leq 4$ needs checking:

$$1! + 5 = 6, \qquad 2! + 5 = 7, \qquad 3! + 5 = 11, \qquad 4! + 5 = 29.$$

The number 6 is not prime, and the other three are. The solutions are

$$(n, p) = (2, 7),\ (3, 11),\ (4, 29).$$

Part (ii)

Write $P_n = 1! \times 3! \times \cdots \times (2n-1)!$. Every factor in every one of these factorials is an integer between 1 and $2n-1$. So if a prime q divides P_n then, since q is prime, it divides one of those integers, and so $q \leq 2n-1$. That gives the key fact:

$$\text{no prime } q > 2n-1 \text{ divides } P_n. \qquad (*)$$

On the other hand, every prime $q \leq m$ divides $m!$. So if $P_n = m!$, then m is smaller than every prime greater than $2n-1$.

Large n. Suppose $n \geq 7$ and $P_n = m!$. By Theorem 1, $m! = P_n > (4n)!$, so $m > 4n$ (factorials increase). By Theorem 2 there is a prime q with $2n < q < 4n$. Then $q < 4n < m$, so $q \mid m! = P_n$. But $q > 2n > 2n - 1$, so this contradicts $(*)$. So there are no solutions with $n \geq 7$.

? **Ask yourself** Why can't we just quote the theorems for every n?

Theorem 1 is only claimed for $n \geq 7$, and in fact it is false for $n = 6$. So $n = 1, \ldots, 6$ need to be checked by hand. We use $(*)$ again to keep the arithmetic short.

Small n.

- $n = 1$: $P_1 = 1 = 1!$, so $m = 1$.
- $n = 2$: $P_2 = 1 \times 6 = 6 = 3!$, so $m = 3$.
- $n = 3$: $P_3 = 6 \times 120 = 720 = 6!$, so $m = 6$.
- $n = 4$: $P_4 = 720 \times 5040$. Since $720 = 6!$ and $5040 = 7 \times 8 \times 9 \times 10$, we get $P_4 = 6! \times 7 \times 8 \times 9 \times 10 = 10!$, so $m = 10$.
- $n = 5$: $P_5 = P_4 \times 9! = 10! \times 9! > 10!$, so $m \geq 11$. Then the prime 11 divides $m!$, but $11 > 9 = 2n - 1$, contradicting $(*)$. No solution.
- $n = 6$: $P_6 = 10! \times 9! \times 11! > 12!$ (since $10! \times 9! > 12$), so $m \geq 13$. Then the prime 13 divides $m!$, but $13 > 11 = 2n - 1$, contradicting $(*)$. No solution.

In each successful case m is unique, because factorials of positive integers are strictly increasing from 2! onwards (and $1! = 1 < 2!$). So the solutions are

$$(n, m) = (1, 1),\ (2, 3),\ (3, 6),\ (4, 10).$$

Checks

The identity $1!\,3!\,5!\,7! = 10!$ is easy to confirm: both sides equal 3 628 800. A computer search over $n \leq 59$ and $m < 400$ found exactly the four pairs above, and confirmed that $n! + 5$ is prime only for $n = 2, 3, 4$ (searching $n \leq 30$). It also confirms that Theorem 1 fails at $n = 6$, which is why the hand check of $n = 5$ and $n = 6$ really is needed.

Common slips

- Forgetting $n = 1$ in part (ii): the product then has the single factor 1!, so $(1, 1)$ is a solution.
- Using Theorem 1 for $n = 5$ or $n = 6$. It is stated only for $n \geq 7$, so those cases need their own argument.
- Saying "$q \mid m!$" without checking $q \leq m$. That is exactly why we need $m > 4n$ from Theorem 1.

Beyond STEP

Theorem 2 is Bertrand's postulate: for every $n \geq 1$ there is a prime p with $n < p \leq 2n$. Chebyshev proved it in 1852, and Erdős later gave a beautiful elementary proof using the central binomial coefficient $\binom{2n}{n}$. Equations such as $n! + 1 = m^2$ (Brocard's problem) are still open. Davenport's *The Higher Arithmetic* is a good next step into the arithmetic of primes and factorials.

WORKED SOLUTION

STEP II 2018, Question 7

Pure • Vectors in Geometry

STEP II 2018, Question 7

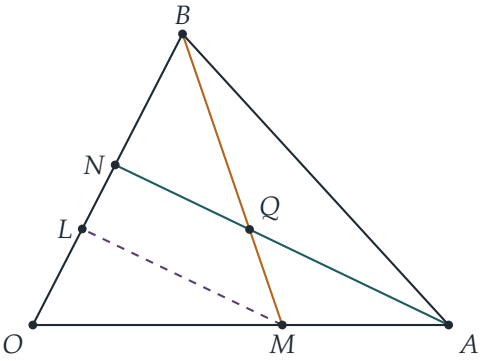
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q7

The key idea

Since M lies on OA and N lies on OB, we can write $\mathbf{m}$ as a multiple of $\mathbf{a}$ and $\mathbf{n}$ as a multiple of $\mathbf{b}$. The point Q is then reached in two ways, once along BM and once along AN, and each route gives $\mathbf{q}$ as a combination of $\mathbf{a}$ and $\mathbf{b}$. Because $\mathbf{a}$ and $\mathbf{b}$ are not parallel, we may compare coefficients, and that gives everything.

Solution



Finding $\mathbf{m}$

Write $\mathbf{m} = m\,\mathbf{a}$ and $\mathbf{n} = n\,\mathbf{b}$, where $0 < m < 1$ and $0 < n < 1$ because M and N lie on the sides.

Along BM. Q lies between M and B with $MQ : QB = \mu : 1$, so Q is the fraction $\frac{\mu}{1+\mu}$ of the way from M to B:

$$\mathbf{q} = \mathbf{m} + \frac{\mu}{1+\mu}(\mathbf{b} - \mathbf{m}) = \frac{1}{1+\mu}\,\mathbf{m} + \frac{\mu}{1+\mu}\,\mathbf{b} = \frac{m}{1+\mu}\,\mathbf{a} + \frac{\mu}{1+\mu}\,\mathbf{b}.$$

Along AN. Similarly $NQ : QA = \nu : 1$, so

$$\mathbf{q} = \mathbf{n} + \frac{\nu}{1+\nu}(\mathbf{a} - \mathbf{n}) = \frac{\nu}{1+\nu}\,\mathbf{a} + \frac{1}{1+\nu}\,\mathbf{n} = \frac{\nu}{1+\nu}\,\mathbf{a} + \frac{n}{1+\nu}\,\mathbf{b}.$$

Ask yourself Why are we allowed to equate the coefficients of **a**, and separately of **b**?

O, A and B are the vertices of a genuine triangle, so **a** and **b** are non-zero and not parallel. If $\alpha\mathbf{a} + \beta\mathbf{b} = \mathbf{0}$ with $\beta \neq 0$, then **b** would be a multiple of **a**; so $\beta = 0$, and then $\alpha = 0$. Hence each vector in the plane has only one expression as a combination of **a** and **b**.

Subtracting the two expressions for **q** and comparing coefficients:

$$\frac{m}{1+\mu} = \frac{\nu}{1+\nu}, \qquad \frac{\mu}{1+\mu} = \frac{n}{1+\nu}.$$

The first gives $m = \dfrac{(1+\mu)\nu}{1+\nu}$, that is,

$$\mathbf{m} = \frac{(1+\mu)\nu}{1+\nu}\,\mathbf{a},$$

as required. The second gives, by the same argument with the roles swapped,

$$\mathbf{n} = \frac{\mu(1+\nu)}{1+\mu}\,\mathbf{b}.$$

Finding λ

Now $\mathbf{l} = \lambda\mathbf{b}$, so

$$\overrightarrow{ML} = \lambda\mathbf{b} - m\,\mathbf{a}, \qquad \overrightarrow{AN} = n\,\mathbf{b} - \mathbf{a}.$$

These are parallel exactly when $\overrightarrow{ML} = k\,\overrightarrow{AN}$ for some scalar k. Comparing coefficients of **a** (allowed, as before) gives $k = m$, and then the **b** coefficients give $\lambda = kn = mn$. (Geometrically: $ML \parallel AN$ makes triangles OML and OAN similar, so $OL/ON = OM/OA$.) Hence

$$\lambda = mn = \frac{(1+\mu)\nu}{1+\nu} \cdot \frac{\mu(1+\nu)}{1+\mu} = \mu\nu.$$

The condition $\mu\nu < 1$

$$m < 1 \iff (1+\mu)\nu < 1+\nu \iff \mu\nu < 1,$$
$$n < 1 \iff \mu(1+\nu) < 1+\mu \iff \mu\nu < 1,$$

using $1+\nu > 0$ and $1+\mu > 0$ to clear denominators. So, given positive μ and ν, the condition $\mu\nu < 1$ is exactly what is needed for M to lie strictly between O and A and N strictly between O and B. Equivalently, the lines AN and BM (with Q dividing them in

these ratios) cross inside the triangle. It also says $\lambda < 1$: the line through M parallel to AN meets OB between O and B, which it must do, since M is on the side OA. If $\mu\nu = 1$, then $M = A$ and $N = B$, and the configuration collapses; if $\mu\nu > 1$, then M and N lie beyond A and B on the extended sides.

Checks

- *Medians.* If M and N are midpoints, then Q is the centroid, which divides each median in the ratio $2 : 1$ from the vertex. So $\mu = \nu = \frac{1}{2}$, and the formula gives $m = \frac{(3/2)(1/2)}{3/2} = \frac{1}{2}$, as it should. Then $\lambda = \frac{1}{4}$: by the midpoint theorem in triangle OAN, L is the midpoint of ON, a quarter of the way along OB.
- *Symmetry.* Swapping $A \leftrightarrow B$ swaps $M \leftrightarrow N$ and $\mu \leftrightarrow \nu$, and the formulas for m and n swap correctly. The answer $\lambda = \mu\nu$ is symmetric too.
- A computer check on about 10^5 random triangles, with random M and N on the sides, measured μ and ν and confirmed m, n and $\lambda = \mu\nu$ to within 10^{-12}. For example, $m = 0.6$ and $n = 0.55$ (as in the diagram) give $\lambda = 0.33$.

Common slips

- Getting the section ratio the wrong way round: Q is $\frac{\mu}{1+\mu}$ of the way from M to B, not from B to M.
- Equating coefficients without saying why. It relies on **a** and **b** not being parallel.
- Expecting the triangle being acute-angled to play a role. It doesn't: the argument works for any triangle.

Beyond STEP

Comparing coefficients works because **a** and **b** are *linearly independent*, so they form a basis of the plane. That is the central idea of a first linear algebra course. The neat answer $\lambda = \mu\nu$ is a ratio result in the same family as Ceva's and Menelaus' theorems, which are cleanest in barycentric coordinates. Strang's *Introduction to Linear Algebra* develops bases and coordinates carefully.

WORKED SOLUTION

STEP II 2018, Question 8

Pure • Bernoulli Differential Equations

STEP II 2018, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q8

The key idea

Both equations are Bernoulli equations: a power of y on the right spoils linearity, but the right power of y as a new variable makes the equation linear. With $y = v^2$ (or $y = v^3$) a common factor of v (or v^2) cancels and what remains is a first-order linear equation with constant coefficients, solved by an integrating factor or by separating variables. For the sketch, the neatest way to compare the two curves is to write both in terms of a single exponential, $u = e^{-t/6}$.

Solution

The question writes $y_1(x)$ and $y_2(x)$, but the independent variable is t throughout, so we write $y_1(t)$ and $y_2(t)$.

Part (i)

Put $y = v^2$ with $v = \sqrt{y} \geq 0$. Then $\dfrac{dy}{dt} = 2v\dfrac{dv}{dt}$ and $y^{1/2} = v$, so the equation becomes

$$2v\frac{dv}{dt} = \alpha v - \beta v^2.$$

One solution is $v \equiv 0$, i.e. the constant solution $y \equiv 0$. Where $v \neq 0$ we can divide by v:

$$\frac{dv}{dt} + \frac{\beta}{2}v = \frac{\alpha}{2}.$$

This is linear with integrating factor $e^{\beta t/2}$:

$$\frac{d}{dt}\left(ve^{\beta t/2}\right) = \frac{\alpha}{2}e^{\beta t/2} \quad \Longrightarrow \quad ve^{\beta t/2} = \frac{\alpha}{\beta}e^{\beta t/2} + C.$$

So the general solution is

$$\sqrt{y} = \frac{\alpha}{\beta} + Ce^{-\beta t/2}, \qquad \text{i.e.} \qquad y = \left(\frac{\alpha}{\beta} + Ce^{-\beta t/2}\right)^2,$$

for a constant C (taken where the bracket is non-negative, since $v = \sqrt{y} \geq 0$), together with the constant solution $y \equiv 0$. Taking $C = 0$ gives the other constant solution $y \equiv \alpha^2/\beta^2$.

For $y_1(0) = 0$ we need $v(0) = 0$, so $C = -\alpha/\beta$, and

$$\boxed{y_1(t) = \frac{\alpha^2}{\beta^2}\left(1 - e^{-\beta t/2}\right)^2}$$

This is not constant, and the bracket $1 - e^{-\beta t/2}$ is non-negative for $t \geq 0$, so it is a genuine solution. (Dividing by v was valid for $t > 0$, where $v > 0$.)

Part (ii)

Now the power is $\frac{2}{3}$, so we want $y^{2/3}$ and y to share a factor. Put $y = v^3$, i.e. $v = y^{1/3} \geq 0$. Then $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 3v^2\dfrac{\mathrm{d}v}{\mathrm{d}t}$ and $y^{2/3} = v^2$, giving

$$3v^2\frac{\mathrm{d}v}{\mathrm{d}t} = \alpha v^2 - \beta v^3.$$

Ask yourself How would you guess the substitution $v = y^{1/3}$?

In part (i), $v = y^{1-\frac{1}{2}}$ made both sides share the factor v. For a Bernoulli equation $\dot{y} = \alpha y^n - \beta y$, the substitution $v = y^{1-n}$ always works, and here $1 - n = \frac{1}{3}$.

Apart from $v \equiv 0$ (the constant solution $y \equiv 0$), divide by v^2:

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\beta}{3}v = \frac{\alpha}{3}.$$

With integrating factor $e^{\beta t/3}$, exactly as before,

$$y^{1/3} = \frac{\alpha}{\beta} + De^{-\beta t/3}, \qquad \text{i.e.} \qquad y = \left(\frac{\alpha}{\beta} + De^{-\beta t/3}\right)^3,$$

for a constant D (with the bracket non-negative). The condition $y_2(0) = 0$ forces $D = -\alpha/\beta$, so

$$\boxed{y_2(t) = \frac{\alpha^3}{\beta^3}\left(1 - e^{-\beta t/3}\right)^3}$$

Part (iii)

With $\alpha = \beta$,

$$y_1(t) = \left(1 - e^{-t/2}\right)^2, \qquad y_2(t) = \left(1 - e^{-t/3}\right)^3.$$

Shape of each curve. Both start at 0 and are increasing, since $1 - e^{-t/2}$ and $1 - e^{-t/3}$ increase from 0 towards 1. Both tend to 1 as $t \to \infty$, the constant solution $y = \alpha^2/\beta^2 = 1$

for (i) and $y = \alpha^3/\beta^3 = 1$ for (ii). Differentiating,

$$y_1' = e^{-t/2}\left(1 - e^{-t/2}\right), \qquad y_2' = e^{-t/3}\left(1 - e^{-t/3}\right)^2,$$

so both have zero gradient at the origin. The gradient $y_1' = w(1-w)$ with $w = e^{-t/2}$ is greatest at $w = \frac{1}{2}$, so y_1 has a point of inflection at $t = 2\ln 2$, $y_1 = \frac{1}{4}$. Similarly $y_2' = w(1-w)^2$ with $w = e^{-t/3}$ is greatest at $w = \frac{1}{3}$, giving an inflection at $t = 3\ln 3$, $y_2 = \frac{8}{27}$. Each curve is an S-shape rising from the origin to the asymptote $y = 1$.

Relative position. Let $u = e^{-t/6}$, so $0 < u < 1$ for $t > 0$, and $e^{-t/2} = u^3$, $e^{-t/3} = u^2$. Then

$$y_1 = (1-u^3)^2 = (1-u)^2(1+u+u^2)^2, \qquad y_2 = (1-u^2)^3 = (1-u)^3(1+u)^3,$$

and so

$$\frac{y_1}{y_2} = \frac{(1+u+u^2)^2}{(1-u)(1+u)^3}.$$

Expanding, $(1+u+u^2)^2 = 1+2u+3u^2+2u^3+u^4$ and $(1-u)(1+u)^3 = (1-u^2)(1+u)^2 = 1+2u-2u^3-u^4$. The numerator minus the denominator is

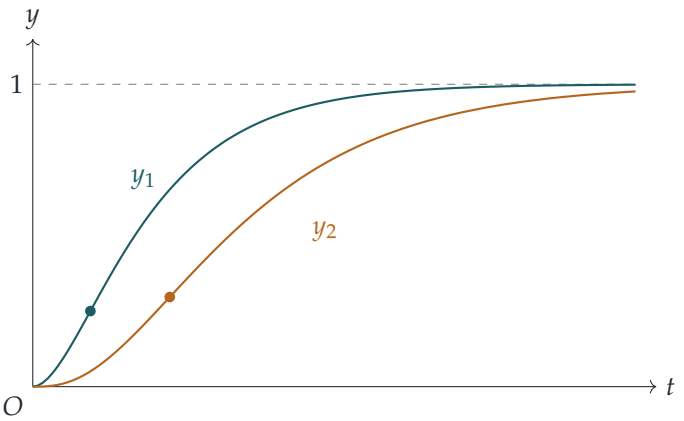
$$3u^2 + 4u^3 + 2u^4 > 0 \qquad (0 < u < 1),$$

and the denominator is positive, so $y_1/y_2 > 1$. Hence

$$y_1(t) > y_2(t) \quad \text{for all } t > 0:$$

the curves meet only at the origin, and y_1 lies above y_2 everywhere else.

This agrees with the behaviour at both ends. Near $t = 0$, $y_1 \approx \frac{t^2}{4}$ while $y_2 \approx \frac{t^3}{27}$, which is much smaller. For large t, $1 - y_1 \approx 2e^{-t/2}$ and $1 - y_2 \approx 3e^{-t/3}$, and $e^{-t/3}$ decays more slowly, so y_2 is further below the asymptote.



The dots mark the points of inflection, $(2\ln 2, \frac{1}{4})$ on y_1 and $(3\ln 3, \frac{8}{27})$ on y_2.

Checks

- Substitute back: for y_1 with $\alpha = \beta = 1$, $y_1^{1/2} - y_1 = (1-w) - (1-w)^2 = w(1-w)$ with $w = e^{-t/2}$, which is exactly y_1'.
- Large t: $y_1 \to \alpha^2/\beta^2$, the non-zero point where $\alpha y^{1/2} = \beta y$; likewise $y_2 \to \alpha^3/\beta^3$, where $\alpha y^{2/3} = \beta y$.
- Numbers with $\alpha = \beta = 1$: at $t = 2$, $y_1 \approx 0.400$ and $y_2 \approx 0.115$; at $t = 10$, $y_1 \approx 0.987$ and $y_2 \approx 0.897$. Both formulas were also confirmed by a computer step-by-step integration of the differential equations.

Common slips

- Dividing by v (or v^2) without noting that $y \equiv 0$ is also a solution. That is why the question asks for the *non-constant* solution through the origin.
- Getting the exponent wrong: the substitution $y = v^3$ gives $e^{-\beta t/3}$, not $e^{-\beta t/2}$.
- Arguing the order of the curves from the gradients alone. Where the curves have the same height $y \in (0,1)$, $y^{2/3} > y^{1/2}$, so y_2 would be rising faster there; this does not say which curve is higher, so a direct comparison is needed.

Beyond STEP

These are Bernoulli equations, a standard first-year family solved by $v = y^{1-n}$. They also illustrate non-uniqueness: since $y^{1/2}$ and $y^{2/3}$ have infinite gradient at $y = 0$, the right-hand side is not Lipschitz there, and $y \equiv 0$ and y_1 both pass through the origin. In fact, staying at 0 until any time t_0 and then following $y_1(t - t_0)$ gives a whole family of solutions. The Picard–Lindelöf theorem explains exactly when this cannot happen. Boyce and DiPrima, *Elementary Differential Equations*, covers both Bernoulli equations and existence and uniqueness.

WORKED SOLUTION

STEP II 2018, Question 9

Mechanics • Collisions Under Gravity

STEP II 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q9

The key idea

Both beads always have the same acceleration g downwards, so *relative* to each other they move at constant velocity. Once A has bounced, the gap between the beads closes at a steady rate, which gives the time to the collision at once. The last part is then pure energy and momentum: knowing how fast A hits the ground tells us how fast it left the collision, and Newton's experimental law does the rest.

Solution

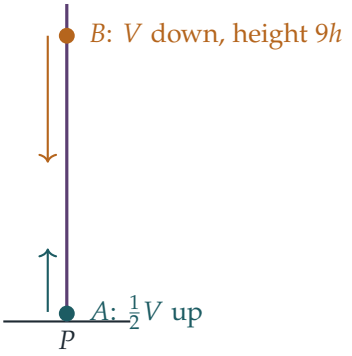
We model the beads as particles of equal mass m on a smooth vertical wire, with no air resistance, so between impacts each bead moves freely under gravity. Take upwards as positive and measure heights from P.

Up to the first bounce

A falls $8h$ from rest, so by conservation of energy

$$V^2 = 2g(8h) = 16gh, \qquad V = 4\sqrt{gh}.$$

In the same time B has also fallen $8h$ from rest (identical motion), so at this instant B is at height $17h - 8h = 9h$, moving downwards with speed V. A rebounds from the ground with speed $\frac{1}{2}V$ upwards.



The collision

From the moment A leaves the ground, both beads accelerate at $-g$, so the velocity of B relative to A stays at its initial value

$$(-V) - \tfrac{1}{2}V = -\tfrac{3}{2}V.$$

Ask yourself Why can we ignore gravity when working out when they meet?

Gravity changes both velocities by the same amount $-gt$, so it cancels in the difference. The gap of $9h$ therefore closes at the constant speed $\frac{3}{2}V$, and the collision happens a time

$$\tau = \frac{9h}{\frac{3}{2}V} = \frac{6h}{V} = \frac{6h}{4\sqrt{gh}} = \frac{3}{2}\sqrt{\frac{h}{g}}$$

after the bounce. (We also need A to be still in the air at that time; we'll see that it is.) A's height then is

$$H = \tfrac{1}{2}V\tau - \tfrac{1}{2}g\tau^2 = 2\sqrt{gh}\cdot\frac{3}{2}\sqrt{\frac{h}{g}} - \frac{1}{2}g\cdot\frac{9h}{4g} = 3h - \frac{9}{8}h = \frac{15}{8}h,$$

as required. Just before the collision,

$$\text{velocity of } A = \tfrac{1}{2}V - g\tau = 2\sqrt{gh} - \tfrac{3}{2}\sqrt{gh} = \tfrac{1}{2}\sqrt{gh},$$
$$\text{velocity of } B = -V - g\tau = -4\sqrt{gh} - \tfrac{3}{2}\sqrt{gh} = -\tfrac{11}{2}\sqrt{gh}.$$

So A is still rising, and

$$u_A = \tfrac{1}{2}\sqrt{gh} \text{ (upwards)}, \qquad u_B = \tfrac{11}{2}\sqrt{gh} \text{ (downwards)}.$$

As a check, B has height $9h - V\tau - \frac{1}{2}g\tau^2 = 9h - 6h - \frac{9}{8}h = \frac{15}{8}h$ too.

The coefficient of restitution

Let e be the coefficient of restitution between the beads, and let v_A, v_B be their velocities (upwards positive) just after the collision. Work in units of $\sqrt{gh}$ for a moment, so $u_A = \frac{1}{2}$ upwards and $u_B = \frac{11}{2}$ downwards.

Momentum (the masses are equal, and the impulse from gravity during the instantaneous collision is negligible):

$$v_A + v_B = \tfrac{1}{2} - \tfrac{11}{2} = -5.$$

Newton's experimental law: the beads approach at speed $\frac{1}{2} + \frac{11}{2} = 6$, and afterwards B, which is above, separates from A at e times this:

$$v_B - v_A = 6e.$$

Subtracting, $v_A = -\frac{1}{2}(5 + 6e)$, so A moves *downwards* with speed $w = \frac{1}{2}(5 + 6e)\sqrt{gh}$.

Now A falls from height $\frac{15}{8}h$ with initial speed w and reaches the ground with speed V (it is below B and moving down faster, since $v_B = -5 - v_A = \frac{1}{2}(6e - 5) > v_A$, so the beads don't meet again first). By energy,

$$V^2 = w^2 + 2g \cdot \tfrac{15}{8}h \quad \Longrightarrow \quad w^2 = 16gh - \tfrac{15}{4}gh = \tfrac{49}{4}gh,$$

so $w = \frac{7}{2}\sqrt{gh}$. Then $\frac{1}{2}(5 + 6e) = \frac{7}{2}$, giving

$$e = \frac{1}{3}.$$

After the collision, A moves down at $\frac{7}{2}\sqrt{gh}$ and B moves down at $\frac{3}{2}\sqrt{gh}$.

Checks

- Kinetic energy: before the collision the total is $\frac{1}{2}m\,gh\,(\frac{1}{4} + \frac{121}{4}) = \frac{61}{4}mgh$; afterwards it is $\frac{1}{2}m\,gh\,(\frac{49}{4} + \frac{9}{4}) = \frac{29}{4}mgh$. Energy is lost, as it must be for $e < 1$.
- With $e = 1$ (and equal masses) the beads would simply swap velocities, so A would leave at $\frac{11}{2}\sqrt{gh}$ and land faster than V; with $e = 0$ they would move together at $\frac{5}{2}\sqrt{gh}$ and A would land slower. The value $\frac{1}{3}$ sits sensibly in between.
- A step-by-step simulation of the whole motion by computer (taking $g = h = 1$) gives a collision at height 1.875 with speeds 0.5 and 5.5, and, with $e = \frac{1}{3}$, a second landing speed of $4 = V$.

Common slips

- Forgetting that B has also been falling while A drops: at the bounce B is at $9h$, not $17h$, and is already moving at speed V.
- Sign errors in Newton's law: write the velocities with one positive direction and check that the separation speed comes out positive.

Beyond STEP

The trick of working in the frame of one falling bead is a first taste of Galilean relativity: in a uniform gravitational field, a freely falling frame sees no gravity at all, which

is the starting point of Einstein's equivalence principle. Collision problems in first-year mechanics are often solved in the centre-of-mass frame, where an impact with coefficient e simply reverses and scales both velocities by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both ideas carefully.

WORKED SOLUTION

STEP II 2018, Question 10

Mechanics • Kinematics on Stretching String

STEP II 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

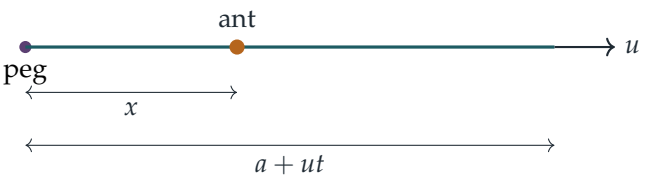
step.maths.org/questions/18-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.12, *The Ant on the Rubber Rope*: https://beyondstep.keithmcnulty.org/ch11/rubber-rope.html

The key idea

Because the string is uniform, it stretches by the same factor everywhere, so each point of the string keeps its *fraction* of the way along. That makes the speed of a point proportional to its distance from the peg. The ant's equation is then linear, and the natural quantity to follow is the ant's fraction $x/(a+ut)$ of the way along the string, rather than x itself.

Solution



Speed of a point of the string

At time t the free end is a distance $a+ut$ from the peg, so that is the length of the string. The string is uniform, so it is stretched by the same factor $(a+ut)/a$ along its whole length. A point of the string that was a distance x_0 from the peg at $t=0$ is therefore always the same fraction x_0/a of the way along:

$$x = \frac{x_0}{a}(a+ut).$$

Differentiating with x_0 fixed, its speed is $\dfrac{x_0 u}{a}$, and writing this in terms of its current position x,

$$\text{speed of the point at distance } x = \frac{ux}{a+ut}.$$

As a check, this is 0 at the peg and u at the free end.

The ant's equation

The ant's speed relative to the peg is v plus the speed of the string beneath it, so

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v + \frac{ux}{a+ut}.$$

By the quotient rule, and then substituting this equation,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) &= \frac{1}{a+ut}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{ux}{(a+ut)^2} \\ &= \frac{1}{a+ut}\left(v + \frac{ux}{a+ut}\right) - \frac{ux}{(a+ut)^2} = \frac{v}{a+ut},\end{aligned}$$

as required.

The time to reach the end

Integrate from 0 to t, using $x = 0$ at $t = 0$:

$$\frac{x}{a+ut} = \int_0^t \frac{v}{a+us}\,\mathrm{d}s = \frac{v}{u}\ln\left(\frac{a+ut}{a}\right) = \frac{1}{k}\ln\left(1+\frac{ut}{a}\right).$$

The left-hand side is the fraction of the string the ant has covered, and the ant reaches the end when it equals 1. The right-hand side increases steadily from 0 without bound, so this happens exactly once, at the time T with

$$\ln\left(1+\frac{uT}{a}\right) = k \iff 1 + \frac{uT}{a} = \mathrm{e}^k \iff uT = a(\mathrm{e}^k - 1),$$

which is the given result.

The journey back

Now the ant walks towards the peg, so its own speed along the string is $-v$, and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -v + \frac{ux}{a+ut}.$$

Ask yourself Do we need to redo the quotient-rule calculation?

No: the working above is unchanged apart from the sign of v, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = -\frac{v}{a+ut}.$$

The journey back starts at $t = T$ with the ant at the end of the string, where $x/(a+ut) = 1$. Integrating from T to t,

$$\frac{x}{a+ut} = 1 - \frac{1}{k}\ln\left(\frac{a+ut}{a+uT}\right).$$

The ant reaches the peg when this fraction is 0, which first happens at the time t_1 with

$$\ln\left(\frac{a+ut_1}{a+uT}\right) = k \iff a+ut_1 = (a+uT)\,\mathrm{e}^k.$$

So the journey back takes

$$t_1 - T = \frac{(a+uT)(\mathrm{e}^k-1)}{u}.$$

From the outward journey, $\mathrm{e}^k - 1 = uT/a$ and $a + uT = a\mathrm{e}^k$, so

$$t_1 - T = \frac{a\mathrm{e}^k}{u}\cdot\frac{uT}{a} = T\mathrm{e}^k.$$

The journey back takes $T\mathrm{e}^k$, longer than the outward trip by the factor e^k.

Checks

- If $u \to 0$ the string does not stretch: $k \to 0$ and $T = a(\mathrm{e}^k-1)/u \to a/v$, while the return time $T\mathrm{e}^k \to a/v$ too, as it should.
- The return time is longer, which makes sense: on the way back the string beneath the ant carries it away from the peg.
- Stepping both differential equations forward numerically for random values of a, u and v (for example $a = 1$, $u = v$, so $k = 1$ and $T = \mathrm{e} - 1 \approx 1.718$) puts the ant at the end at time T and back at the peg after a further $T\mathrm{e} \approx 4.67$, matching the formulas.

Common slips

- Taking every point of the string to move at speed u; only the free end does.
- Forgetting that the return journey starts at $t = T$, not $t = 0$, so the constant of integration changes.
- Leaving the answer as $(a+uT)(\mathrm{e}^k-1)/u$ when it is asked for in terms of T and k alone.

Beyond STEP

The ant always reaches the end, however fast the string stretches, because $\int \mathrm{d}t/(a+ut)$ diverges. Its distance covered as a fraction of the string grows like the logarithm of the

time, and if the stretching were faster (say exponential) the integral would converge and the ant might never arrive. The same idea, of a fixed speed through space that is itself expanding, appears in cosmology, where it decides whether light from a distant galaxy can ever reach us. The linear equation solved here by spotting the right quantity is a case of the integrating-factor method; see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 2018, Question 11

Mechanics • Friction and Moments

STEP II 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q11

The key idea

Three equations pin down the three unknown contact forces: Newton's second law horizontally, no vertical acceleration, and the given condition that the moments about the centre of mass balance. Once both normal reactions are written in terms of the acceleration a, each of the two failures becomes a threshold on a. The front wheel lifts when its reaction falls to zero, and the rear wheel slips when the friction it needs exceeds μ times its reaction. Whichever threshold is lower happens first.

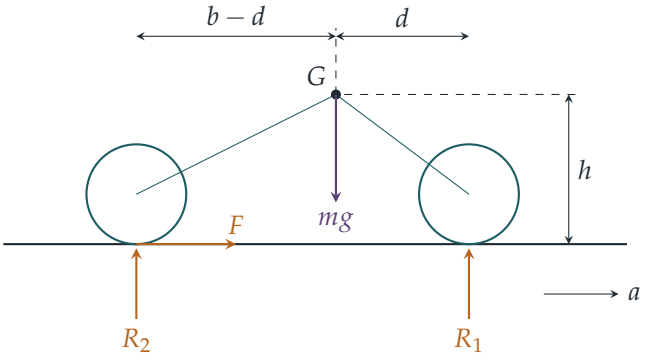
Solution

The model and the forces

Treat the motorbike as a rigid body moving in a straight line with acceleration $a \geq 0$ to the right. The road is horizontal, so the centre of mass G has no vertical acceleration and the bike does not rotate. The forces are:

- the weight mg acting down at G;
- the normal reaction R_1 at the front wheel (which is smooth, so there is no friction there);
- the normal reaction R_2 and the friction F at the rear wheel. The engine drives the rear wheel, so F acts forwards and is the only horizontal force.

The front contact point is a horizontal distance d ahead of G and the rear one is $b - d$ behind it. Both are a distance h below G.



The reactions in terms of a

Horizontally and vertically,

$$F = ma, \qquad R_1 + R_2 = mg.$$

Now take moments about G, with anticlockwise positive (the sense that would lift the front wheel). The weight has no moment about G. R_1, acting up a distance d ahead of G, has anticlockwise moment $R_1 d$. R_2, acting up a distance $b - d$ behind G, has clockwise moment $R_2(b - d)$. And F acts forwards along a line a distance h below G, so it has anticlockwise moment Fh. Setting the total to zero:

$$R_1 d + Fh = R_2(b - d).$$

Substitute $F = ma$ and $R_1 = mg - R_2$:

$$mgd - R_2 d + mah = R_2(b - d) \quad \Longrightarrow \quad R_2 = \frac{m(gd + ah)}{b},$$

and then

$$R_1 = mg - R_2 = \frac{m\big((b - d)g - ah\big)}{b}.$$

As a increases, weight transfers from the front wheel to the rear: R_1 decreases and R_2 increases.

When does each failure happen?

Front wheel. The front wheel stays on the road while $R_1 \geq 0$, that is while

$$a \leq \frac{(b - d)g}{h} =: a_{\text{lift}}.$$

Rear wheel. The rear wheel does not slip while $F \leq \mu R_2$, that is while

$$ma \leq \frac{\mu m(gd + ah)}{b} \iff a(b - \mu h) \leq \mu dg.$$

Ask yourself Why can we divide by $b - \mu h$ without reversing the inequality?

This is exactly what the given condition $\mu < b/h$ is for: it says $b - \mu h > 0$. So the rear wheel grips while

$$a \leq \frac{\mu d g}{b - \mu h} =: a_{\text{slip}}.$$

(Without that condition the friction available would grow at least as fast as the friction needed, and the rear wheel would never slip.)

Both thresholds are positive, and as a increases from zero the rear wheel slips first exactly when $a_{\text{slip}} < a_{\text{lift}}$. Since $b - \mu h > 0$ and $h > 0$ we can multiply out:

$$\begin{aligned} \frac{\mu d g}{b - \mu h} < \frac{(b-d)g}{h} &\iff \mu d h < (b-d)(b - \mu h) \\ &\iff \mu d h + \mu h (b-d) < b(b-d) \\ &\iff \mu h b < b(b-d) \\ &\iff \mu < \frac{b-d}{h}, \end{aligned}$$

dividing by $bh > 0$ at the last step. So the rear wheel slips before the front wheel loses contact if $(*)$ holds, as required. (In fact the chain shows the two statements are equivalent.)

Maximum acceleration when $(*)$ holds

If $(*)$ holds then $a_{\text{slip}} < a_{\text{lift}}$. Any acceleration up to a_{slip} satisfies both conditions, so the front wheel stays down and the rear wheel grips. Any larger acceleration would need $F > \mu R_2$. So the maximum acceleration without slipping is

$$a_{\max} = \frac{\mu d g}{b - \mu h},$$

as required.

When $(*)$ does not hold

Now $\mu \geq \dfrac{b-d}{h}$, so the same chain of equivalences (with $\geq$ in place of $<$) gives $a_{\text{lift}} \leq a_{\text{slip}}$. Every acceleration up to a_{lift} therefore keeps $R_1 \geq 0$ and also satisfies $a \leq a_{\text{slip}}$, so the rear wheel does not slip. Beyond a_{lift} we would need $R_1 < 0$, which is impossible while the front wheel touches the road. So the maximum acceleration with the front wheel in contact is

$$a_{\max} = \frac{(b-d)g}{h}.$$

At this acceleration $R_1 = 0$: the bike is just on the point of a wheelie, with all its weight on the rear wheel.

Checks

- At $a = 0$: $R_2 = mgd/b$ and $R_1 = mg(b-d)/b$, the usual static split by the lever rule. The wheel nearer G carries more weight.
- In the boundary case $\mu = (b-d)/h$ we have $b - \mu h = d$, so $a_{\text{slip}} = \mu g = (b-d)g/h = a_{\text{lift}}$. Both answers agree, as they must.
- A numerical example: $b = 1.5\,\text{m}$, $d = 0.75\,\text{m}$, $h = 0.6\,\text{m}$, $\mu = 0.8$, $g = 9.8\,\text{m}\,\text{s}^{-2}$. Here $(b-d)/h = 1.25 > \mu$, and $a_{\text{slip}} = 5.88/1.02 \approx 5.76\,\text{m}\,\text{s}^{-2}$, well below $a_{\text{lift}} = 12.25\,\text{m}\,\text{s}^{-2}$. A computer check over many random bikes, stepping a upwards and recording which condition fails first, agreed with both formulae.

Common slips

- Getting the sign of the friction moment wrong. Forward friction below G tends to lift the front, which is why acceleration unloads the front wheel.
- Dividing by $b - \mu h$ without saying why it is positive. That is the only job of the condition $\mu < b/h$.
- Taking moments about a contact point and setting the total to zero. The bike is accelerating, so the moments balance about G but not about other points.

Beyond STEP

The given condition about moments is a special case of the rigid-body law that the rate of change of angular momentum about the centre of mass equals the total torque about it. Here the angular momentum is zero, since the bike does not rotate. About any other point an accelerating body acquires an extra "fictitious" torque from ma, which explains the last common slip. The same weight-transfer analysis underlies braking and traction limits in vehicle dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body dynamics carefully.

WORKED SOLUTION

STEP II 2018, Question 12

Probability and Statistics • Optimal Stopping Expectation

STEP II 2018, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q12

The key idea

With a fixed target of h Heads, I win £h exactly when the target is reached before the game ends, and nothing otherwise. So the expected winnings are h times the probability of reaching h Heads, and the whole question is about finding that probability. To maximise over the whole number h, don't differentiate: compare consecutive terms by looking at the ratio E_{h+1}/E_h.

Solution

Throughout, let $q = 1 - p = \frac{1}{N+1}$ be the probability of Tails, and let E_h be my expected winnings with target h (where $h \geq 1$). Since I win £h if I reach h Heads and £0 otherwise,

$$E_h = h \times \mathrm{P}(\text{I reach } h \text{ Heads before the game ends}).$$

Part (i)

In version 1 a single Tail ends the game, so I reach h Heads exactly when the first h tosses are all Heads. The tosses are independent, so this has probability p^h, and

$$E_h = hp^h.$$

To find the best h, compare consecutive values. Since $E_h > 0$,

$$\frac{E_{h+1}}{E_h} = \frac{(h+1)p^{h+1}}{hp^h} = \frac{(h+1)N}{h(N+1)},$$

and

$$\frac{E_{h+1}}{E_h} \geq 1 \iff (h+1)N \geq h(N+1) \iff N \geq h,$$

with equality only when $h = N$. So

$$E_1 < E_2 < \cdots < E_N = E_{N+1} > E_{N+2} > \cdots.$$

The sequence rises strictly up to $h = N$, the next value ties with it, and after that it falls strictly. So the largest value of E_h is E_N, and choosing $h = N$ maximises my expected winnings, as required. (Choosing $h = N+1$ does equally well.) The maximum is $N\left(\frac{N}{N+1}\right)^N$.

Part (ii)

Now the game ends only on two consecutive Tails. Think of the tosses up to and including the hth Head as h blocks, each ending in a Head. If I reach h Heads safely, then each block is either

$$\text{H} \quad \text{or} \quad \text{TH},$$

since two Tails in a row would have ended the game. (A Tail at the very start is fine: there's nothing before it.) Conversely, any sequence of h such blocks never contains two consecutive Tails. Each Tail in it comes straight after a Head or at the start, and straight before a Head. So I reach h Heads exactly when the tosses split into h blocks of these two kinds.

? **Ask yourself** Why can I just multiply the block probabilities together?

After each Head the game is in exactly the same position as at the start: no Tail is pending, and the future tosses are independent of the past. So each block, independently of the others, is safe with probability

$$\text{P(H)} + \text{P(TH)} = p + qp = p(1+q) = \frac{N}{N+1} \cdot \frac{N+2}{N+1} = \frac{N(N+2)}{(N+1)^2}.$$

(These two events are disjoint, because they differ at the first toss.) Hence the probability of reaching h Heads is $\left(\frac{N(N+2)}{(N+1)^2}\right)^h$, and

$$E_h = \frac{hN^h(N+2)^h}{(N+1)^{2h}},$$

as required.

The case $N = 2$. Here

$$E_h = h\left(\frac{2 \cdot 4}{9}\right)^h = h\left(\frac{8}{9}\right)^h.$$

As in part (i),

$$\frac{E_{h+1}}{E_h} = \frac{8(h+1)}{9h} \geq 1 \iff 8h + 8 \geq 9h \iff h \leq 8,$$

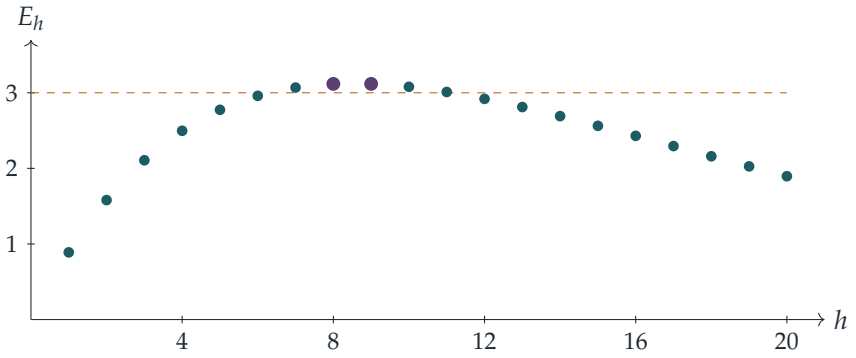
with equality only at $h = 8$. So E_h increases strictly up to $h = 8$, then $E_9 = E_8$, and then it decreases strictly. The maximum is

$$E_8 = 8\left(\frac{8}{9}\right)^8 = \frac{2^3 \cdot 2^{24}}{3^{16}} = \frac{2^{27}}{3^{16}}.$$

To estimate this, take logarithms to base 3 and use $\log_3 2 \approx 0.63$:

$$\log_3 E_8 = 27\log_3 2 - 16 \approx 27(0.63) - 16 = 17.01 - 16 = 1.01.$$

So $E_8 \approx 3^{1.01}$, which is very close to $3^1 = 3$. The maximum expected winnings are approximately £3.



The values $E_h = h(8/9)^h$ in version 2 with $N = 2$; the joint maximum at $h = 8, 9$ is just above the dashed line at 3.

Checks

- Version 2 can only be kinder than version 1, since every all-Heads start is safe in both. Indeed $\frac{N(N+2)}{(N+1)^2} = 1 - \frac{1}{(N+1)^2} > \frac{N}{N+1}$.
- With $N = 1$, version 1 gives $E_1 = E_2 = \frac{1}{2}$: one Head with probability $\frac{1}{2}$, or two with probability $\frac{1}{4}$.
- Exactly, $E_8 = 2^{27}/3^{16} = 134\,217\,728/43\,046\,721 \approx 3.118$, so the estimate £3 is good. A computer simulation of 2×10^5 games with $N = 2$ and $h = 8$ gave average winnings of about £3.11.

Common slips

- In version 2, forgetting that a single Tail at the start is allowed, or wrongly thinking a Tail may follow the hth Head. I stop as soon as the hth Head appears.
- Differentiating hp^h and setting the result to zero. This gives $h = -1/\ln p$, which usually isn't an integer, and turning it into the best integer needs care. The ratio test is cleaner and exact.
- Missing the tie: $h = N$ and $h = N + 1$ give the same expected winnings, so the "show that" asks only that $h = N$ achieves the maximum.

Beyond STEP

This is a simple *optimal stopping* problem. Here I fix the target in advance, but you can check that nothing is lost by doing so: whenever I hold H Heads with no Tail pending, the gain from one more toss is positive exactly when H is small enough. Problems like this, where one decides when to stop by comparing "stop now" with "one more step", lead to martingales and dynamic programming. The block argument in part (ii) is a first taste of renewal ideas for Markov chains. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

WORKED SOLUTION

STEP II 2018, Question 13

Probability and Statistics • Recurrence Relations

STEP II 2018, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

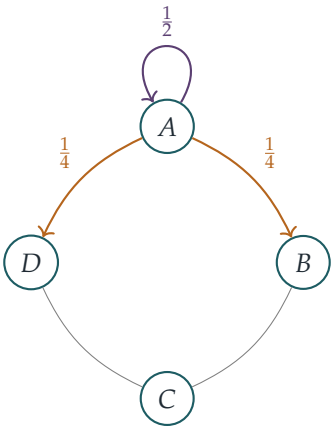
step.maths.org/questions/18-s2-q13

The key idea

Wherever the parcel is, the next whistle moves it by a fixed rule, so each of A_{n+1}, B_{n+1}, C_{n+1}, D_{n+1} is a fixed combination of A_n, B_n, C_n, D_n. Four coupled recurrences look daunting, but the right *combinations* (sums and differences of opposite children) each satisfy a one-line recurrence of their own, and those we can solve at once.

Solution

Label the children around the circle. On each whistle the holder keeps the parcel with probability $\frac{1}{2}$ and passes it to each neighbour with probability $\frac{1}{4}$; note that, by the question's set-up, clockwise and anticlockwise passes are equally likely.



The figure shows the moves from A; the same pattern applies at every child.

Part (i)

Starting from A, after one whistle

$$A_1 = \tfrac{1}{2}, \qquad B_1 = \tfrac{1}{4}, \qquad C_1 = 0, \qquad D_1 = \tfrac{1}{4},$$

since C is two places from A and cannot be reached in a single move.

To be at a given child after the second whistle, the parcel was either already there and stayed (probability $\frac{1}{2}$), or was at a neighbour and was passed the right way (probability $\frac{1}{4}$ from each neighbour). So

$$\begin{aligned} A_2 &= \tfrac{1}{2}A_1 + \tfrac{1}{4}B_1 + \tfrac{1}{4}D_1 = \tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{16} = \tfrac{3}{8}, \\ B_2 &= \tfrac{1}{2}B_1 + \tfrac{1}{4}A_1 + \tfrac{1}{4}C_1 = \tfrac{1}{8} + \tfrac{1}{8} + 0 = \tfrac{1}{4}, \\ C_2 &= \tfrac{1}{2}C_1 + \tfrac{1}{4}B_1 + \tfrac{1}{4}D_1 = 0 + \tfrac{1}{16} + \tfrac{1}{16} = \tfrac{1}{8}, \\ D_2 &= \tfrac{1}{2}D_1 + \tfrac{1}{4}C_1 + \tfrac{1}{4}A_1 = \tfrac{1}{8} + 0 + \tfrac{1}{8} = \tfrac{1}{4}. \end{aligned}$$

As a check, $\frac{3}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{4} = 1$.

Part (ii)

The argument used for $n = 2$ works for every $n \geq 0$, by the law of total probability (conditioning on who holds the parcel after the nth whistle):

$$\begin{aligned} A_{n+1} &= \tfrac{1}{2}A_n + \tfrac{1}{4}(B_n + D_n), & B_{n+1} &= \tfrac{1}{2}B_n + \tfrac{1}{4}(A_n + C_n), \\ C_{n+1} &= \tfrac{1}{2}C_n + \tfrac{1}{4}(B_n + D_n), & D_{n+1} &= \tfrac{1}{2}D_n + \tfrac{1}{4}(A_n + C_n), \end{aligned}$$

with $A_0 = 1$ and $B_0 = C_0 = D_0 = 0$.

Finding B_n and D_n. Adding the equations for B_{n+1} and D_{n+1},

$$B_{n+1} + D_{n+1} = \tfrac{1}{2}(B_n + D_n) + \tfrac{1}{2}(A_n + C_n) = \tfrac{1}{2}(A_n + B_n + C_n + D_n) = \tfrac{1}{2},$$

because someone always holds the parcel, so the four probabilities sum to 1.

? **Ask yourself** The sum $B + D$ is now known, but how do we split it between B and D?

Subtract instead of adding: the $A_n + C_n$ terms cancel, leaving

$$B_{n+1} - D_{n+1} = \tfrac{1}{2}(B_n - D_n).$$

Since $B_0 - D_0 = 0$, this gives $B_n - D_n = 0$ for every n. (This is the mirror symmetry of the game: passing clockwise and anticlockwise are equally likely, and B and D sit symmetrically either side of A.) Combining with $B_n + D_n = \frac{1}{2}$,

$$B_n = D_n = \tfrac{1}{4} \qquad \text{for all } n \geq 1,$$

while of course $B_0 = D_0 = 0$. This agrees with part (i).

Finding A_n and C_n. For $n \geq 1$ we now know $A_n + C_n = 1 - (B_n + D_n) = \frac{1}{2}$. Subtracting the equations for A_{n+1} and C_{n+1}, the $B_n + D_n$ terms cancel:

$$A_{n+1} - C_{n+1} = \tfrac{1}{2}(A_n - C_n),$$

and this holds for all $n \geq 0$. As $A_0 - C_0 = 1$, we get $A_n - C_n = \left(\frac{1}{2}\right)^n$ for all $n \geq 0$. Solving with $A_n + C_n = \frac{1}{2}$:

$$\boxed{A_n = \frac{1}{4} + \frac{1}{2^{n+1}}, \qquad C_n = \frac{1}{4} - \frac{1}{2^{n+1}} \qquad (n \geq 1).}$$

These formulas fail at $n = 0$ (they would give $A_0 = \frac{3}{4}$), because the relation $A_n + C_n = \frac{1}{2}$ only starts after the first whistle.

Checks

- $n = 1$: $A_1 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and $C_1 = 0$; $n = 2$: $A_2 = \frac{3}{8}$ and $C_2 = \frac{1}{8}$. Both match part (i).
- $n = 3$: the formulas give $(A_3, B_3, C_3, D_3) = (\frac{5}{16}, \frac{1}{4}, \frac{3}{16}, \frac{1}{4})$, which sums to 1 and agrees with stepping the recurrences directly; a computer simulation of 200 000 games gave $A_3 \approx 0.313$ and $C_3 \approx 0.188$.
- As $n \to \infty$ every probability tends to $\frac{1}{4}$: after many whistles the parcel is equally likely to be anywhere, as symmetry suggests.

Common slips

- Writing B_{n+1} in terms of the probabilities of passing *from* B, rather than *into* B. Ask "where could the parcel have been just before?"
- Stating $A_n = \frac{1}{4} + 2^{-(n+1)}$ for all $n \geq 0$; it is wrong at $n = 0$.
- Assuming $B_n = D_n$ "by symmetry" without a word of reason; the difference recurrence makes it precise.

Beyond STEP

This is a *Markov chain*: a random walk on a cycle of four states with transition matrix P, and $(A_n, B_n, C_n, D_n) = (1, 0, 0, 0)P^n$. The combinations $B_n - D_n$, $A_n - C_n$ and $A_n + C_n - B_n - D_n$ come from the eigenvectors of P (with eigenvalues $\frac{1}{2}$, $\frac{1}{2}$ and 0), and the convergence to the uniform distribution at rate 2^{-n} is governed by the second-largest eigenvalue. Grimmett and Welsh, *Probability: An Introduction*, has a friendly treatment of Markov chains and their long-run behaviour.

WORKED SOLUTION

STEP III 2018, Question 1

Pure • Quadratic Roots and Inequalities

STEP III 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q1

The key idea

Vieta's formulas turn both symmetric expressions in u and v into simple functions of the coefficients α and β. Once one expression is fixed, α can be eliminated, and the other expression becomes a function of β alone: exactly one of the two functions sketched in part (i). The condition that u and v are real (discriminant ≥ 0) then tells us which stretch of the curve we are allowed to use, and the sketch does the rest.

Solution

Part (i)

The curve $y = \mathrm{f}(\beta)$. Differentiating,

$$\mathrm{f}'(\beta) = 1 + \frac{1}{\beta^2} + \frac{2}{\beta^3} = \frac{\beta^3 + \beta + 2}{\beta^3} = \frac{(\beta+1)(\beta^2 - \beta + 2)}{\beta^3},$$

where we spotted the root $\beta = -1$ of the cubic. The quadratic factor has discriminant $1 - 8 < 0$, so it is always positive. Hence the only stationary point is at $\beta = -1$, where

$$\mathrm{f}(-1) = -1 + 1 - 1 = -1.$$

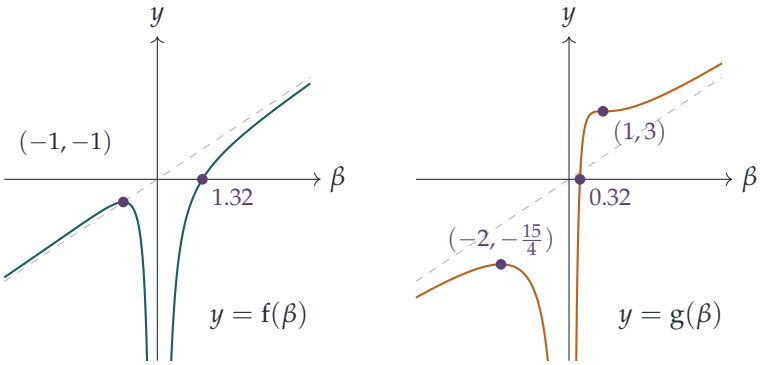
Signs: for $-1 < \beta < 0$ the numerator is positive and $\beta^3 < 0$, so $\mathrm{f}' < 0$; for $\beta < -1$ both are negative, so $\mathrm{f}' > 0$. So $(-1, -1)$ is a *maximum*, and $\mathrm{f}(\beta) \leq -1$ for every $\beta < 0$. For $\beta > 0$ everything is positive, so f is increasing there.

Behaviour at the edges: as $\beta \to 0$ from either side the term $-1/\beta^2$ dominates, so $\mathrm{f} \to -\infty$. As $\beta \to \pm\infty$, $\mathrm{f}(\beta) - \beta \to 0$, so $y = \beta$ is an oblique asymptote. The curve crosses the axis once, where $\beta^3 - \beta - 1 = 0$, at $\beta \approx 1.32$.

The curve $y = \mathrm{g}(\beta)$. Similarly,

$$\mathrm{g}'(\beta) = 1 - \frac{3}{\beta^2} + \frac{2}{\beta^3} = \frac{\beta^3 - 3\beta + 2}{\beta^3} = \frac{(\beta-1)^2(\beta+2)}{\beta^3}.$$

The stationary points are at $\beta = -2$, with $g(-2) = -2 - \frac{3}{2} - \frac{1}{4} = -\frac{15}{4}$, and at $\beta = 1$, with $g(1) = 3$. For $-2 < \beta < 0$ we have $g' < 0$, and for $\beta < -2$ we have $g' > 0$, so $(-2, -\frac{15}{4})$ is a maximum and $g(\beta) \leq -\frac{15}{4}$ for all $\beta < 0$. For $\beta > 0$, $g' \geq 0$ with equality only at $\beta = 1$: the derivative does not change sign there, so $(1, 3)$ is a stationary point of inflection and g is increasing on $\beta > 0$. Again $g \to -\infty$ as $\beta \to 0$, $y = \beta$ is an asymptote, and there is one zero, at $\beta \approx 0.32$ (the root of $\beta^3 + 3\beta - 1 = 0$).



The dashed line is the asymptote $y = \beta$.

Part (ii)

By Vieta's formulas, $u + v = -\alpha$ and $uv = \beta$ (and $uv \neq 0$ since $\beta \neq 0$). So

$$u + v + \frac{1}{uv} = -\alpha + \frac{1}{\beta}, \qquad \frac{1}{u} + \frac{1}{v} + uv = \frac{u+v}{uv} + uv = \beta - \frac{\alpha}{\beta}.$$

Part (iii)

The condition $-\alpha + \frac{1}{\beta} = -1$ gives $\alpha = 1 + \frac{1}{\beta}$. Substituting into the second expression,

$$\frac{1}{u} + \frac{1}{v} + uv = \beta - \frac{1}{\beta}\left(1 + \frac{1}{\beta}\right) = \beta - \frac{1}{\beta} - \frac{1}{\beta^2} = f(\beta).$$

So we must show $f(\beta) \leq -1$ for every *allowed* β.

Ask yourself The curve $y = f(\beta)$ goes off to $+\infty$, so the inequality can't hold for all β. What stops β from being large?

The reality of u and v. The roots are real exactly when $\alpha^2 \geq 4\beta$, that is,

$$\left(1 + \frac{1}{\beta}\right)^2 \geq 4\beta \iff (\beta+1)^2 \geq 4\beta^3 \iff 4\beta^3 - \beta^2 - 2\beta - 1 \leq 0,$$

where we multiplied by $\beta^2 > 0$. Now $\beta = 1$ is a root of the cubic, and

$$4\beta^3 - \beta^2 - 2\beta - 1 = (\beta - 1)(4\beta^2 + 3\beta + 1).$$

The quadratic factor has discriminant $9 - 16 < 0$, so it is always positive. Hence the roots are real if and only if $\beta \leq 1$ (with $\beta \neq 0$).

- If $\beta < 0$: from part (i), $\mathrm{f}(\beta) \leq \mathrm{f}(-1) = -1$.
- If $0 < \beta \leq 1$: f is increasing on $\beta > 0$, so $\mathrm{f}(\beta) \leq \mathrm{f}(1) = 1 - 1 - 1 = -1$.

In every case $\frac{1}{u} + \frac{1}{v} + uv \leq -1$, as required. Equality occurs at $\beta = -1$ ($\alpha = 0$, roots ± 1) and at $\beta = 1$ ($\alpha = 2$, double root -1).

Part (iv)

Now $-\alpha + \frac{1}{\beta} = 3$, so $\alpha = \frac{1}{\beta} - 3$ and

$$\frac{1}{u} + \frac{1}{v} + uv = \beta - \frac{1}{\beta}\left(\frac{1}{\beta} - 3\right) = \beta + \frac{3}{\beta} - \frac{1}{\beta^2} = \mathrm{g}(\beta).$$

The roots are real when $\alpha^2 \geq 4\beta$:

$$\left(\frac{1}{\beta} - 3\right)^2 \geq 4\beta \iff (1 - 3\beta)^2 \geq 4\beta^3 \iff 4\beta^3 - 9\beta^2 + 6\beta - 1 \leq 0.$$

Again $\beta = 1$ is a root, and this time it is a double root:

$$4\beta^3 - 9\beta^2 + 6\beta - 1 = (\beta - 1)(4\beta^2 - 5\beta + 1) = (\beta - 1)^2(4\beta - 1).$$

Since $(\beta - 1)^2 \geq 0$, the cubic is ≤ 0 exactly when $\beta \leq \frac{1}{4}$ *or* $\beta = 1$. So the allowed values are $\beta < 0$, $0 < \beta \leq \frac{1}{4}$, and the isolated value $\beta = 1$.

> **Ask yourself** Is the greatest value simply $\mathrm{g}(\frac{1}{4})$, the right-hand end of the interval?

Not quite: don't lose the isolated point. On the three pieces:

- $\beta < 0$: $\mathrm{g}(\beta) \leq -\frac{15}{4}$ by part (i).
- $0 < \beta \leq \frac{1}{4}$: g is increasing, so $\mathrm{g}(\beta) \leq \mathrm{g}(\frac{1}{4}) = \frac{1}{4} + 12 - 16 = -\frac{15}{4}$.
- $\beta = 1$: $\mathrm{g}(1) = 3$.

So the greatest value of $\frac{1}{u} + \frac{1}{v} + uv$ is $\boxed{3}$, attained when $\beta = 1$, $\alpha = -2$, that is $u = v = 1$. (Indeed $1 + 1 + 1 = 3$ for both expressions.) Without that double root, nothing larger than $-\frac{15}{4}$ would be possible.

Checks

- Put $u = v = 1$ into both expressions: each is 3, matching part (iv). Put $u = v = -1$: each is -1, the equality case in part (iii).
- Try $\beta = \frac{1}{4}$ in part (iv): $\alpha = 1$, and $x^2 + x + \frac{1}{4} = (x + \frac{1}{2})^2$, so $u = v = -\frac{1}{2}$. Then $u + v + \frac{1}{uv} = -1 + 4 = 3$ and $\frac{1}{u} + \frac{1}{v} + uv = -4 + \frac{1}{4} = -\frac{15}{4}$, as claimed.

- A computer search over thousands of random β giving real roots found nothing above -1 in part (iii), and nothing above -3.75 in part (iv) apart from $\beta = 1$.

Common slips

- Forgetting the discriminant condition, and concluding in part (iii) that there is no upper bound.
- Solving $(\beta - 1)^2(4\beta - 1) \leq 0$ as just $\beta \leq \frac{1}{4}$ and missing $\beta = 1$, which gives the answer $-\frac{15}{4}$ instead of 3.
- Multiplying the inequality $\alpha^2 \geq 4\beta$ by β (which may be negative) rather than by β^2.

Beyond STEP

This is a small constrained optimisation problem: we maximise a function over the set of coefficients that give real roots, and that set can include isolated points as well as intervals. The maximum of a function on a set need not sit at a stationary point or at the end of an interval, which is why careful descriptions of the feasible set matter so much in analysis and optimisation. Abbott's *Understanding Analysis* gives a careful treatment of extreme values of continuous functions on general sets.

WORKED SOLUTION

STEP III 2018, Question 2

Pure • Hermite Polynomials

STEP III 2018, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q2

The key idea

Turn the definition round: it says that the nth derivative of z is $(-1)^n y_n z$. Differentiating this once more and comparing with the same statement for $n+1$ gives part (i) in a line. After that, part (i) is the tool that lets us differentiate a recurrence, which drives the induction in (ii). The final inequality then follows from the identity in (ii) by a second, very short induction.

Solution

Throughout, $z = \mathrm{e}^{-x^2}$, so $z' = -2xz$, and $z > 0$ for every x. Rearranging the definition,

$$\frac{\mathrm{d}^n z}{\mathrm{d}x^n} = (-1)^n y_n z \qquad (n \geq 0),$$

which also holds for $n = 0$ because $y_0 = 1$.

Part (i)

Differentiate the displayed equation, using the product rule and $z' = -2xz$:

$$\frac{\mathrm{d}^{n+1} z}{\mathrm{d}x^{n+1}} = (-1)^n \left(y_n' z + y_n z'\right) = (-1)^n \left(y_n' - 2x y_n\right) z.$$

But the same equation with $n+1$ in place of n says the left-hand side is $(-1)^{n+1} y_{n+1} z$. Equating the two and dividing by $(-1)^n z$, which is never zero,

$$y_n' - 2x y_n = -y_{n+1}, \qquad \text{that is} \qquad \frac{\mathrm{d}y_n}{\mathrm{d}x} = 2x y_n - y_{n+1},$$

as required. Nothing here needed $n \geq 1$, so the result also holds for $n = 0$; we shall use that below.

For later use: $z' = -2xz$ and $z'' = (4x^2 - 2)z$, so $y_1 = 2x$ and $y_2 = 4x^2 - 2$.

Part (ii)

Base case. For $n = 1$ we need $y_2 = 2xy_1 - 2y_0$, and indeed $2x \cdot 2x - 2 = 4x^2 - 2 = y_2$.

Inductive step. Suppose that, for some $k \geq 1$,

$$y_{k+1} = 2xy_k - 2ky_{k-1}. \qquad (*)$$

We want the same relation one step further on, so we differentiate $(*)$:

$$y'_{k+1} = 2y_k + 2xy'_k - 2ky'_{k-1}.$$

? **Ask yourself** How do we get rid of the derivatives, and is part (i) allowed for y_{k-1} when $k = 1$?

Replace every derivative using part (i). For y'_{k-1} with $k = 1$ this is the $n = 0$ case, which we noted holds as well. This gives

$$2xy_{k+1} - y_{k+2} = 2y_k + 2x\big(2xy_k - y_{k+1}\big) - 2k\big(2xy_{k-1} - y_k\big),$$

so

$$y_{k+2} = 4xy_{k+1} - 4x^2y_k - 2(k+1)y_k + 4kx\,y_{k-1}.$$

The term in y_{k-1} is removed with the hypothesis $(*)$ once more: $2ky_{k-1} = 2xy_k - y_{k+1}$, so $4kx\,y_{k-1} = 4x^2y_k - 2xy_{k+1}$. Substituting,

$$\begin{aligned} y_{k+2} &= 4xy_{k+1} - 4x^2y_k - 2(k+1)y_k + 4x^2y_k - 2xy_{k+1} \\ &= 2xy_{k+1} - 2(k+1)y_k, \end{aligned}$$

which is the relation with $k+1$ in place of k. By induction, $y_{n+1} = 2xy_n - 2ny_{n-1}$ for all $n \geq 1$.

The deduction. Use the recurrence twice, once at n and once at $n+1$, to make the term $2xy_ny_{n+1}$ appear in both products:

$$\begin{aligned} y_{n+1}^2 &= y_{n+1}\big(2xy_n - 2ny_{n-1}\big) = 2xy_ny_{n+1} - 2ny_{n-1}y_{n+1}, \\ y_ny_{n+2} &= y_n\big(2xy_{n+1} - 2(n+1)y_n\big) = 2xy_ny_{n+1} - 2(n+1)y_n^2. \end{aligned}$$

Subtracting, the x terms cancel:

$$y_{n+1}^2 - y_ny_{n+2} = 2(n+1)y_n^2 - 2ny_{n-1}y_{n+1} = 2n\big(y_n^2 - y_{n-1}y_{n+1}\big) + 2y_n^2,$$

which is the required identity.

Part (iii)

Write $D_n = y_n^2 - y_{n-1}y_{n+1}$. The identity from (ii) reads

$$D_{n+1} = 2nD_n + 2y_n^2 \qquad (n \geq 1).$$

We prove $D_n > 0$ for all $n \geq 1$ by induction. For $n = 1$,

$$D_1 = y_1^2 - y_0 y_2 = 4x^2 - (4x^2 - 2) = 2 > 0$$

for every x. If $D_n > 0$ for some $n \geq 1$, then $2nD_n > 0$ because $n \geq 1$, and $2y_n^2 \geq 0$, so $D_{n+1} > 0$. Hence $y_n^2 - y_{n-1}y_{n+1} > 0$ for all $n \geq 1$ and all real x, as required.

Note that we needed D_n strictly positive to get a strict inequality: y_n^2 alone can be zero, at the zeros of y_n.

Checks

- From the recurrence, $y_3 = 2x(4x^2 - 2) - 4 \cdot 2x = 8x^3 - 12x$. Differentiating $z'' = (4x^2 - 2)z$ directly gives $z''' = (-8x^3 + 12x)z$, matching $-y_3 z$.
- Part (i) at $n = 1$: $y_1' = 2$, and $2x \cdot 2x - (4x^2 - 2) = 2$. ✓
- Part (iii) at $n = 2$: $D_2 = (4x^2 - 2)^2 - 2x(8x^3 - 12x) = 8x^2 + 4$, which is positive, and equals $2D_1 + 2y_1^2 = 4 + 8x^2$, as the identity predicts.
- A computer check of the recurrences and the inequality up to $n = 14$, at many rational values of x (including near the zeros of each y_n), agreed exactly.

Common slips

- Forgetting that the inductive step uses part (i) for y_{k-1}, which at $k = 1$ means y_0: part (i) was only asked for $n \geq 1$, so check the case $n = 0$ separately.
- Concluding only $D_{n+1} \geq 0$ in part (iii): the strict inequality comes from $D_n > 0$, not from y_n^2.

Beyond STEP

The functions y_n are the Hermite polynomials $H_n(x)$, and the relation $\mathrm{d}^n z/\mathrm{d}x^n = (-1)^n H_n z$ is their Rodrigues formula. They are orthogonal with weight e^{-x^2} on the whole real line, and they give the energy states of the quantum harmonic oscillator. The inequality in (iii) is a Turán-type inequality, a family of results that hold for many classical orthogonal polynomials. Boyce and DiPrima, *Elementary Differential Equations*, meets Hermite's equation through series solutions.

WORKED SOLUTION

STEP III 2018, Question 3

Pure • Euler Differential Equations

STEP III 2018, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q3

The key idea

The left-hand side is an Euler (equidimensional) equation: each derivative comes with a matching power of x. Expanding the nested form $(*)$ and comparing coefficients pins down a, b and c. Once the equation is in that form it can be solved by two straightforward integrations, peeling off one derivative at a time; the cases arise exactly when one of those integrals is of x^{-1} and produces a logarithm.

Solution

Throughout we work on $x > 0$, so that all the powers of x are defined.

Finding a, b and c

By the product rule,

$$(x^c y)' = x^c y' + c x^{c-1} y, \qquad \text{so} \qquad x^b (x^c y)' = x^{b+c} y' + c x^{b+c-1} y.$$

Differentiating again,

$$\left(x^b (x^c y)'\right)' = x^{b+c} y'' + (b+c) x^{b+c-1} y' + c x^{b+c-1} y' + c(b+c-1) x^{b+c-2} y.$$

Multiplying by x^a and choosing $a + b + c = 2$ gives

$$x^a \left(x^b (x^c y)'\right)' = x^2 y'' + (b + 2c)\, x y' + c(b + c - 1)\, y.$$

This matches the given left-hand side provided

$$b + 2c = 1 - 2p, \qquad c(b + c - 1) = p^2 - q^2.$$

The first gives $b + c - 1 = -2p - c$, so the second becomes $-c^2 - 2pc = p^2 - q^2$, that is $(c + p)^2 = q^2$. Taking $c + p = q$,

$$c = q - p, \qquad b = 1 - 2p - 2c = 1 - 2q, \qquad a = 2 - b - c = 1 + p + q.$$

So the equation can be written as

$$x^{1+p+q}\left(x^{1-2q}\left(x^{q-p}y\right)'\right)' = \mathrm{f}(x),$$

as required. (The other root $c + p = -q$ gives the same thing with q replaced by $-q$, which is no different, since only q^2 appears in the equation.)

Part (i)

With $\mathrm{f} = 0$ we can divide by $x^{1+p+q} \neq 0$:

$$\left(x^{1-2q}\left(x^{q-p}y\right)'\right)' = 0 \quad\Longrightarrow\quad x^{1-2q}\left(x^{q-p}y\right)' = A \quad\Longrightarrow\quad \left(x^{q-p}y\right)' = Ax^{2q-1},$$

for a constant A. The next integration depends on whether the power $2q - 1$ is -1.

Ask yourself Why does the value of p not create separate cases here?

The exponent we integrate is $2q - 1$, which does not involve p; p only enters through the factor x^{q-p}, which we divide by at the end. So the cases depend on q alone.

Case $q \neq 0$. Integrating, $x^{q-p}y = \dfrac{A}{2q}x^{2q} + B$, and multiplying by x^{p-q} (and renaming the constant $\frac{A}{2q}$ as A),

$$y = Ax^{p+q} + Bx^{p-q}.$$

Case $q = 0$. Now $\left(x^{-p}y\right)' = Ax^{-1}$, so $x^{-p}y = A\ln x + B$ and

$$y = x^p\left(A\ln x + B\right).$$

In each case there are two arbitrary constants, as expected for a second-order equation.

Part (ii)

This is the case $q = 0$ with $\mathrm{f}(x) = x^n$, so $a = 1 + p$, $b = 1$, $c = -p$, and the equation reads

$$x^{1+p}\left(x\left(x^{-p}y\right)'\right)' = x^n \quad\Longleftrightarrow\quad \left(x\left(x^{-p}y\right)'\right)' = x^{n-p-1}.$$

The first integration involves x^{n-p-1}, which is x^{-1} exactly when $n = p$.

Case $n \neq p$. Integrating once,

$$x\left(x^{-p}y\right)' = \frac{x^{n-p}}{n-p} + A \quad\Longrightarrow\quad \left(x^{-p}y\right)' = \frac{x^{n-p-1}}{n-p} + \frac{A}{x}.$$

Since $n - p \neq 0$ again, integrating gives

$$x^{-p}y = \frac{x^{n-p}}{(n-p)^2} + A\ln x + B, \qquad \text{so} \qquad y = \frac{x^n}{(n-p)^2} + x^p\left(A\ln x + B\right).$$

Case $n = p$. Now the first integration gives a logarithm:

$$x\left(x^{-p}y\right)' = \ln x + A \quad \Longrightarrow \quad \left(x^{-p}y\right)' = \frac{\ln x}{x} + \frac{A}{x}.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{2}(\ln x)^2 = \frac{\ln x}{x}$,

$$x^{-p}y = \tfrac{1}{2}(\ln x)^2 + A\ln x + B, \qquad \text{so} \qquad y = x^p\left(\tfrac{1}{2}(\ln x)^2 + A\ln x + B\right).$$

In both cases the part with A and B is the complementary function from part (i) with $q = 0$, as it should be.

Checks

- Take $p = q = 0$: the equation is $x^2y'' + xy' = 0$, i.e. $(xy')' = 0$, so $y = A\ln x + B$, matching part (i).
- Try $y = x^m$ in the homogeneous equation: $m(m-1) + (1-2p)m + p^2 - q^2 = (m-p)^2 - q^2$, so $m = p \pm q$, a repeated root when $q = 0$, just as found.
- In part (ii) with $n = p = 1$, $y = \frac{1}{2}x(\ln x)^2$ has $y' = \frac{1}{2}(\ln x)^2 + \ln x$ and $y'' = \frac{\ln x}{x} + \frac{1}{x}$, so $x^2y'' - xy' + y = x\ln x + x - \frac{1}{2}x(\ln x)^2 - x\ln x + \frac{1}{2}x(\ln x)^2 = x$, as required.
- All the solutions were also substituted back into the equation by computer at many random values of p, q, n and x; the residuals were zero to rounding error (around 10^{-6}).

Common slips

- Forgetting the case $q = 0$ in (i), or the case $n = p$ in (ii), where $\int x^{-1}\,\mathrm{d}x = \ln x$ replaces the power rule.
- Losing the constant of integration at the first step: it is what produces the second independent solution.
- Assuming $a + b + c = 2$ without saying why: it is needed so that y'' is multiplied by x^2.

Beyond STEP

Writing a second-order operator as a composition of first-order ones is called *factorising* the operator, and it is the idea behind reduction of order and, in quantum mechanics, ladder operators. The substitution $x = \mathrm{e}^t$ turns any Euler equation into one with constant coefficients, which explains the repeated root $m = p$ and the $\ln x$ (that is, t) factor. Boyce and DiPrima, *Elementary Differential Equations*, treats Euler equations and the method of reduction of order.

WORKED SOLUTION

STEP III 2018, Question 4

Pure • Tangents to a Hyperbola

STEP III 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q4

The key idea

The tangent is quickest from the standard result that the tangent to $x^2/a^2 - y^2/b^2 = 1$ at (x_1, y_1) is $xx_1/a^2 - yy_1/b^2 = 1$. After that, everything is two straight lines meeting, so it is a matter of solving pairs of linear equations and tidying the trigonometry. In part (ii) the perpendicularity condition is a single relation between $\sin\theta$ and $\sin\phi$ involving b, and it is exactly what lets us remove b from the expressions and then bring it back as $a^2 - b^2$. Sum-to-product formulae make the final algebra short.

Solution

The tangent at P

Differentiating $x^2/a^2 - y^2/b^2 = 1$ implicitly gives $\dfrac{2x}{a^2} - \dfrac{2y}{b^2}\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at (x_1, y_1) is $\dfrac{b^2x_1}{a^2y_1}$ and the tangent is

$$y - y_1 = \frac{b^2x_1}{a^2y_1}(x - x_1) \iff \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1,$$

using the fact that (x_1, y_1) is on the curve. (This form also covers the vertices, where $y_1 = 0$ and the tangent is vertical.) With $x_1 = a\sec\theta$, $y_1 = b\tan\theta$:

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1.$$

Multiplying through by $ab\cos\theta$ (which is non-zero, since $\sec\theta$ exists) gives

$$bx - ay\sin\theta = ab\cos\theta,$$

as required.

Part (i)

The two given lines are the asymptotes $y = \pm\frac{b}{a}x$.

At S, $ay = bx$. Substituting $ay = bx$ into the tangent gives $bx(1-\sin\theta) = ab\cos\theta$, so

$$S = \left(\frac{a\cos\theta}{1-\sin\theta},\ \frac{b\cos\theta}{1-\sin\theta}\right).$$

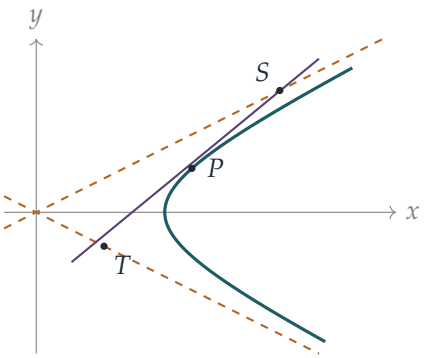
At T, $ay = -bx$, which gives $bx(1+\sin\theta) = ab\cos\theta$, so

$$T = \left(\frac{a\cos\theta}{1+\sin\theta},\ -\frac{b\cos\theta}{1+\sin\theta}\right).$$

Both denominators are non-zero because $\cos\theta \neq 0$ means $\sin\theta \neq \pm 1$. Adding, and using $(1-\sin\theta)(1+\sin\theta) = \cos^2\theta$:

$$x_S + x_T = a\cos\theta \cdot \frac{(1+\sin\theta)+(1-\sin\theta)}{\cos^2\theta} = \frac{2a}{\cos\theta} = 2a\sec\theta,$$
$$y_S + y_T = b\cos\theta \cdot \frac{(1+\sin\theta)-(1-\sin\theta)}{\cos^2\theta} = \frac{2b\sin\theta}{\cos\theta} = 2b\tan\theta.$$

So the mid-point of ST is $(a\sec\theta,\ b\tan\theta)$: **the mid-point of ST is P itself**. The tangent at any point is bisected at the point of contact by the two asymptotes.



Part (ii)

The perpendicularity condition. Written as $ay\sin\theta = bx - ab\cos\theta$, the tangent at P has gradient $\dfrac{b}{a\sin\theta}$ when $\sin\theta \neq 0$. (If $\sin\theta = 0$ the tangent is vertical, so the tangent at Q would have to be horizontal; but no tangent is horizontal, since $b/(a\sin\phi) \neq 0$. So $\sin\theta \neq 0$, and likewise $\sin\phi \neq 0$.) Perpendicular tangents need the product of the gradients to be -1:

$$\frac{b^2}{a^2\sin\theta\sin\phi} = -1 \qquad \Longleftrightarrow \qquad \sin\theta\sin\phi = -\frac{b^2}{a^2}. \tag{$*$}$$

In particular $\sin\theta$ and $\sin\phi$ have opposite signs, so $\sin\phi - \sin\theta \neq 0$. Note that $(*)$ needs $b^2/a^2 < 1$, which is where $a > b$ is used.

The intersection. The point (x, y) satisfies

$$bx - ay\sin\theta = ab\cos\theta, \qquad bx - ay\sin\phi = ab\cos\phi.$$

Subtracting the second from the first gives $ay(\sin\phi - \sin\theta) = ab(\cos\theta - \cos\phi)$. Multiplying the first by $\sin\phi$, the second by $\sin\theta$, and subtracting, gives $bx(\sin\phi - \sin\theta) = ab(\sin\phi\cos\theta - \cos\phi\sin\theta) = ab\sin(\phi - \theta)$. Hence

$$x = \frac{a\sin(\phi-\theta)}{\sin\phi - \sin\theta}, \qquad y = \frac{b(\cos\theta - \cos\phi)}{\sin\phi - \sin\theta}.$$

Ask yourself These still involve b. How can we write x^2 and y^2 using only a, θ and ϕ, and can the fractions be simplified?

The condition $(*)$ says $b^2 = -a^2\sin\theta\sin\phi$, so squaring removes b. To simplify, write $\alpha = \frac{1}{2}(\phi + \theta)$ and $\beta = \frac{1}{2}(\phi - \theta)$. The sum-to-product formulae give

$$\sin\phi - \sin\theta = 2\cos\alpha\sin\beta, \qquad \cos\theta - \cos\phi = 2\sin\alpha\sin\beta, \qquad \sin(\phi-\theta) = 2\sin\beta\cos\beta,$$

and $\cos\alpha\sin\beta \neq 0$ because $\sin\phi - \sin\theta \neq 0$. Cancelling $2\sin\beta$,

$$x = \frac{a\cos\beta}{\cos\alpha}, \qquad y = b\tan\alpha,$$

and therefore

$$x^2 = \frac{a^2\cos^2\left(\frac{1}{2}(\phi-\theta)\right)}{\cos^2\left(\frac{1}{2}(\phi+\theta)\right)},$$
$$y^2 = -a^2\sin\theta\sin\phi\,\tan^2\left(\tfrac{1}{2}(\phi+\theta)\right).$$

Equivalently, without half-angles,

$$x^2 = \frac{a^2\sin^2(\phi-\theta)}{(\sin\phi-\sin\theta)^2}, \qquad y^2 = -\frac{a^2\sin\theta\sin\phi\,(\cos\theta-\cos\phi)^2}{(\sin\phi-\sin\theta)^2}.$$

The circle. We also need $\sin\theta\sin\phi$ in terms of α and β. Since $\theta = \alpha - \beta$ and $\phi = \alpha + \beta$,

$$\begin{aligned}\sin\theta\sin\phi &= \sin^2\alpha\cos^2\beta - \cos^2\alpha\sin^2\beta \\ &= (1-\cos^2\alpha)\cos^2\beta - \cos^2\alpha(1-\cos^2\beta) = \cos^2\beta - \cos^2\alpha.\end{aligned}$$

Write $k = \sin\theta\sin\phi = \cos^2\beta - \cos^2\alpha$. Then, with $\tan^2\alpha = (1-\cos^2\alpha)/\cos^2\alpha$,

$$\begin{aligned}x^2 + y^2 &= \frac{a^2}{\cos^2\alpha}\Big[\cos^2\beta - k\,(1-\cos^2\alpha)\Big] \\ &= \frac{a^2}{\cos^2\alpha}\Big[(\cos^2\beta - k) + k\cos^2\alpha\Big] \\ &= \frac{a^2}{\cos^2\alpha}\Big[\cos^2\alpha + k\cos^2\alpha\Big] \qquad \text{since } \cos^2\beta - k = \cos^2\alpha \\ &= a^2(1+k) = a^2 + a^2\sin\theta\sin\phi = a^2 - b^2,\end{aligned}$$

using $(*)$ in the last step. So the point of intersection of any two perpendicular tangents lies on the circle $x^2 + y^2 = a^2 - b^2$, as required.

Checks

- Take $a = 2$, $b = 1$, $\theta = \frac{\pi}{6}$, $\phi = -\frac{\pi}{6}$. Then $\sin\theta\sin\phi = -\frac{1}{4} = -b^2/a^2$, so the tangents are perpendicular (gradients 1 and -1). Here $\alpha = 0$ and $\beta = -\frac{\pi}{6}$, so $x = 2\cos\frac{\pi}{6} = \sqrt{3}$ and $y = 0$, and $x^2 + y^2 = 3 = a^2 - b^2$. Directly: the tangents $x - y = \sqrt{3}$ and $x + y = \sqrt{3}$ do meet at $(\sqrt{3}, 0)$.
- In part (i), as $\theta \to 0$ the tangent becomes the vertical line $x = a$, which meets the asymptotes at $(a, \pm b)$, with mid-point $(a, 0) = P$.
- All the formulae above, including the mid-point in (i) and $x^2 + y^2 = a^2 - b^2$, were confirmed by computer for thousands of random choices of $a > b > 0$ and perpendicular pairs θ, ϕ, solving the two tangent equations directly.

Common slips

- Leaving b in the expressions for x^2 and y^2: the question asks for a, θ and ϕ only, so $(*)$ must be used to eliminate b^2.
- Taking the gradient of the tangent as $-b/(a\sin\theta)$ by mis-rearranging; check that it is positive at a point in the first quadrant.
- Dividing by $\sin\phi - \sin\theta$ without saying why it is non-zero.

Beyond STEP

The circle in part (ii) is the *director circle* of the hyperbola: the locus of points from which the two tangents are perpendicular. For an ellipse the director circle is $x^2 + y^2 = a^2 + b^2$, and for a hyperbola with $a < b$ there is no real director circle at all, which is why the question insists on $a > b$. The bisection property in part (i) is a first glimpse of projective and affine geometry of conics, where tangents, asymptotes and poles and polars are treated uniformly. For problems in exactly this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2018, Question 5

Pure • AM-GM Inequality

STEP III 2018, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q5

The key idea

This is a proof of the AM–GM inequality by induction, but with a twist: instead of showing $A_n \geq G_n$ directly, we show that the weighted gap $n(A_n - G_n)$ can never decrease when a new number is added. The gap starts at zero when $n = 1$, so it stays non-negative. The quantity λ_k is chosen so that G_{k+1} becomes a simple multiple of G_k, which turns the whole comparison into a one-variable polynomial inequality.

Solution

Part (i)

Two facts link step k to step $k+1$. For the sums,

$$(k+1)A_{k+1} = a_1 + \cdots + a_k + a_{k+1} = kA_k + a_{k+1}.$$

For the products, $G_k^k = a_1 a_2 \cdots a_k$, and the definition of λ_k gives $a_{k+1} = G_k \lambda_k^{k+1}$. So

$$G_{k+1}^{k+1} = G_k^k \, a_{k+1} = G_k^k \cdot G_k \lambda_k^{k+1} = (G_k \lambda_k)^{k+1}, \qquad \text{so} \qquad G_{k+1} = G_k \lambda_k,$$

taking positive $(k+1)$th roots, since $G_k > 0$ and $\lambda_k > 0$. This is exactly why λ_k was defined as it was.

Now we compare the two sides:

$$\begin{aligned}(k+1)(A_{k+1} - G_{k+1}) - k(A_k - G_k) &= kA_k + a_{k+1} - (k+1)G_{k+1} - kA_k + kG_k \\ &= G_k \lambda_k^{k+1} - (k+1)G_k \lambda_k + kG_k \\ &= G_k\left(\lambda_k^{k+1} - (k+1)\lambda_k + k\right).\end{aligned}$$

Since $G_k > 0$, the left-hand side is ≥ 0 if and only if the bracket is ≥ 0. This is the required equivalence.

Part (ii)

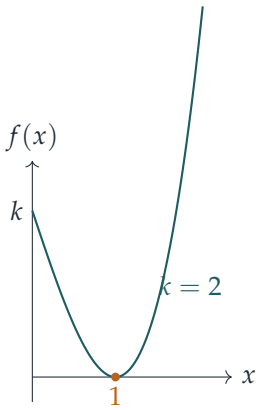
Differentiating,

$$f'(x) = (k+1)x^k - (k+1) = (k+1)(x^k - 1).$$

For $0 < x < 1$ we have $x^k < 1$, so $f'(x) < 0$, and for $x > 1$ we have $x^k > 1$, so $f'(x) > 0$. Hence f is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$, and its least value on $x > 0$ is

$$f(1) = 1 - (k+1) + k = 0.$$

So $f(x) \geq 0$ for all $x > 0$. Moreover, if $x \neq 1$ then $f(x) > f(1) = 0$ by the strict monotonicity on either side of 1. So $f(x) = 0$ only when $x = 1$, and $f(1) = 0$ gives the converse.



(For $k = 2$, $f(x) = x^3 - 3x + 2 = (x-1)^2(x+2)$; the double root at $x = 1$ is where the graph just touches the axis.)

Part (iii)

Write $d_n = n(A_n - G_n)$ for each positive integer n.

(a) By part (ii) with $x = \lambda_k > 0$, we have $\lambda_k^{k+1} - (k+1)\lambda_k + k \geq 0$, so by part (i)

$$d_{k+1} \geq d_k \qquad \text{for every } k \geq 1.$$

Also $A_1 = a_1 = G_1$, so $d_1 = 0$. Hence

$$0 = d_1 \leq d_2 \leq d_3 \leq \cdots \leq d_n,$$

so $d_n \geq 0$, and dividing by $n > 0$ gives $A_n \geq G_n$ for all n.

(b)

Ask yourself If $A_n = G_n$, what does that tell us about the earlier steps of the chain, not just the last one?

If $A_n = G_n$ then $d_n = 0$, and the chain becomes $0 = d_1 \le d_2 \le \cdots \le d_n = 0$. A non-decreasing chain that starts and ends at 0 must be constant, so $d_{k+1} = d_k$ for each $k = 1, \ldots, n-1$. By the identity in part (i) this means $G_k f(\lambda_k) = 0$, and as $G_k > 0$, $f(\lambda_k) = 0$. By part (ii), $\lambda_k = 1$, which says $a_{k+1} = G_k$, for each $k = 1, \ldots, n-1$.

Now we argue by induction on k. For $k = 1$: $a_2 = G_1 = a_1$. If $a_1 = \cdots = a_k = a$, then $G_k = (a^k)^{1/k} = a$, so $a_{k+1} = G_k = a$. Hence $a_1 = a_2 = \cdots = a_n$, as required.

Checks

Take $a_1, a_2, a_3 = 1, 4, 16$. Then $A_2 = 2.5$, $G_2 = 2$, $A_3 = 7$, $G_3 = 4$, so $d_1, d_2, d_3 = 0, 1, 9$: increasing, as promised. Here $\lambda_2 = (16/2)^{1/3} = 2$, and $G_2 f(\lambda_2) = 2(8 - 6 + 2) = 8 = d_3 - d_2$, which confirms the identity from part (i). A computer check of the identity and of $d_{k+1} \ge d_k$ on many thousands of random positive sequences found no exceptions.

Common slips

- Proving only one direction in part (i). Writing the difference as $G_k f(\lambda_k)$ gives both directions at once, because $G_k > 0$.
- In (iii)(b), concluding only that $\lambda_{n-1} = 1$. You need every step of the chain to be an equality, which comes from the chain starting at $d_1 = 0$.
- Forgetting to say why $\lambda_k > 0$ before applying part (ii), which only covers $x > 0$.

Beyond STEP

The inequality $x^{k+1} - (k+1)x + k \ge 0$ is a special case of Bernoulli's inequality, and AM–GM itself is a special case of Jensen's inequality for the concave function $\ln x$. These convexity inequalities are basic tools in first-year analysis. Several other proofs of AM–GM, including Cauchy's "forward–backward" induction, are a good exercise; Bartle and Sherbert, *Introduction to Real Analysis*, covers the analysis background.

WORKED SOLUTION

STEP III 2018, Question 6

Pure • Complex Numbers and Geometry

STEP III 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q6

The key idea

A complex number is real exactly when it equals its own conjugate, so collinearity of three points turns into an equation between z's and z^*'s. On the unit circle, $a^* = 1/a$, which lets us clear the conjugates of the circle points and leaves a *linear* equation for any point q on a chord. Each later result comes from writing down two such chord equations and adding or subtracting them.

Solution

Part (i)

Since A, Q, C are collinear and $A \neq C$, the point Q is A plus a real multiple of the vector from A to C: $q - a = t(c - a)$ for some real t. As $c \neq a$ we may divide, so

$$\frac{q-a}{c-a} = t \quad \text{is real.}$$

A real number equals its conjugate, so

$$\frac{q-a}{c-a} = \frac{q^*-a^*}{c^*-a^*},$$

where $c^* - a^* \neq 0$ because $c \neq a$. Cross-multiplying gives $(c-a)(q^*-a^*) = (c^*-a^*)(q-a)$, as required.

Now suppose $aa^* = cc^* = 1$, so $a^* = 1/a$ and $c^* = 1/c$. Then

$$c^* - a^* = \frac{1}{c} - \frac{1}{a} = \frac{a-c}{ac} = -\frac{c-a}{ac},$$

and the previous equation becomes

$$(c-a)\left(q^* - \frac{1}{a}\right) = -\frac{c-a}{ac}(q-a).$$

Dividing by $c - a \neq 0$ and multiplying by $-ac$,

$$-ac\,q^* + c = q - a, \qquad \text{that is,} \qquad q + ac\,q^* = a + c.$$

This is the equation of the chord AC: every point q on the line through A and C satisfies it.

Part (ii)

Because A, B, C, D are in anticlockwise order, AC and BD are the diagonals of a convex quadrilateral inscribed in the circle, so they cross at a point Q strictly inside the circle. In particular Q is different from all four points, so part (i) applies to the lines AC and BD:

$$q + ac\,q^* = a + c,$$
$$q + bd\,q^* = b + d.$$

Subtracting the second from the first removes q:

$$(ac - bd)\,q^* = (a + c) - (b + d). \tag{1}$$

Ask yourself We have q^*; how do we get an equation for q to add to it?

Take the conjugate of (1) and again use $a^* = 1/a$ and so on. The left side becomes

$$\Big(\frac{1}{ac} - \frac{1}{bd}\Big)q = \frac{bd - ac}{abcd}\,q,$$

and the right side becomes

$$\frac{1}{a} + \frac{1}{c} - \frac{1}{b} - \frac{1}{d} = \frac{a + c}{ac} - \frac{b + d}{bd} = \frac{bd(a + c) - ac(b + d)}{abcd}.$$

Multiplying by $-abcd$,

$$(ac - bd)\,q = ac(b + d) - bd(a + c). \tag{2}$$

Adding (1) and (2),

$$(ac - bd)(q + q^*) = a + c - b - d + abc + acd - abd - bcd.$$

Grouping the terms on the right,

$$(a - b) + (acd - bcd) + (c - d) + (abc - abd) = (a - b)(1 + cd) + (c - d)(1 + ab),$$

which gives the required result.

Part (iii)

The lines AB and CD are opposite sides of the inscribed quadrilateral, so where they meet, at P, is outside the circle. So P is distinct from A and B, and from C and D. Applying part (i) to the collinear points A, P, B gives $p + ab\,p^* = a + b$. Since p is real, $p^* = p$, and so

$$p(1 + ab) = a + b.$$

In the same way, from C, P, D,

$$p(1 + cd) = c + d.$$

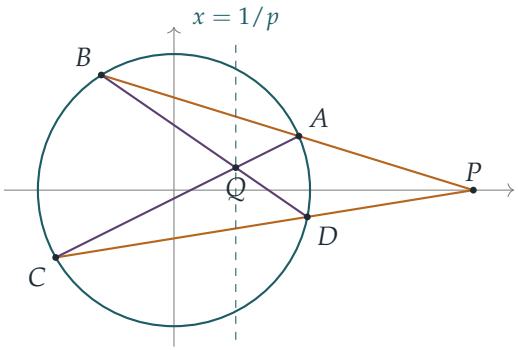
Now multiply the final result of part (ii) by p and substitute these two equations:

$$\begin{aligned} p\,(ac - bd)(q + q^*) &= (a - b)\,p(1 + cd) + (c - d)\,p(1 + ab) \\ &= (a - b)(c + d) + (c - d)(a + b) \\ &= (ac + ad - bc - bd) + (ac + bc - ad - bd) \\ &= 2(ac - bd). \end{aligned}$$

Since $ac - bd \neq 0$, we may divide by it, giving $p(q + q^*) = 2$, as required.

In fact $ac - bd$ can never be zero here. If $ac = bd$, then (1) forces $a + c = b + d$ as well, so $z^2 - (a + c)z + ac$ and $z^2 - (b + d)z + bd$ are the same quadratic. Then $\{a, c\} = \{b, d\}$, which contradicts the points being distinct.

Since $q + q^* = 2\,\mathrm{Re}\,q$, the result says $\mathrm{Re}\,q = 1/p$. However the two lines through P are chosen, the crossing point Q of the diagonals lies on the fixed vertical line $x = 1/p$.



Checks

- Symmetric case: take A, B at angles $20°$, $70°$ and C, D at $-70°$, $-20°$. By the reflection in the real axis, P and Q are both real, and the result says $pq = 1$. With $a = e^{i\alpha}$, $b = e^{i\beta}$, the formula $p = (a + b)/(1 + ab)$ gives $p = \cos\frac{\alpha-\beta}{2} / \cos\frac{\alpha+\beta}{2} = \cos 25° / \cos 45° \approx 1.2817$. Find q directly as the point where AC crosses the real axis: $q \approx 0.7802$, and $pq \approx 1.0000$.

- The diagram uses $p = 2.2$. The diagonals cross at $x \approx 0.4545 = 1/2.2$, as predicted. All three identities were also confirmed by computer on thousands of random configurations.

Common slips

- Forgetting that conjugating $ac - bd$ gives $1/(ac) - 1/(bd)$, not $ac - bd$ again; the factor $abcd$ must be cleared carefully.
- Dividing by $c - a$ or $ac - bd$ without saying why it is non-zero.

Beyond STEP

Here Q lies on the *polar* of P with respect to the circle: for a point P outside the unit circle at distance p from the centre, its polar is the line perpendicular to OP at distance $1/p$ from O. This is part of Brocard's theorem in projective geometry, and the map $P \mapsto 1/p^*$ behind it is *inversion* in the circle, a key tool in complex analysis and geometry. Needham's *Visual Complex Analysis* develops inversion and the geometry of the Argand diagram beautifully.

WORKED SOLUTION

STEP III 2018, Question 7

Pure • De Moivre and Series

STEP III 2018, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 2.4, $\pi^2/6$: https://beyondstep.keithmcnulty.org/ch02/basel.html

The key idea

The expression $\cot\theta + \mathrm{i}$ is $(\cos\theta + \mathrm{i}\sin\theta)/\sin\theta$ in disguise, so De Moivre's theorem turns its powers into multiple angles. Expanding the same powers with the binomial theorem gives a polynomial in $\cot^2\theta$, whose roots we can read off from where $\sin(2n+1)\theta$ vanishes. Vieta's formula then sums the roots for free, and squeezing $1/\theta^2$ between $\cot^2\theta$ and $1+\cot^2\theta$ traps the Basel sum between two bounds that tend to the same limit.

Solution

Part (i)

Since $\sin\theta \neq 0$,

$$\cot\theta \pm \mathrm{i} = \frac{\cos\theta \pm \mathrm{i}\sin\theta}{\sin\theta}.$$

By De Moivre's theorem, $(\cos\theta \pm \mathrm{i}\sin\theta)^{2n+1} = \cos(2n+1)\theta \pm \mathrm{i}\sin(2n+1)\theta$, so

$$\frac{(\cot\theta+\mathrm{i})^{2n+1} - (\cot\theta-\mathrm{i})^{2n+1}}{2\mathrm{i}} = \frac{2\mathrm{i}\sin(2n+1)\theta}{2\mathrm{i}\sin^{2n+1}\theta} = \frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta},$$

which is the required identity.

Now expand the left-hand side with the binomial theorem instead. Writing $c = \cot\theta$ and $N = 2n+1$,

$$(c+\mathrm{i})^N - (c-\mathrm{i})^N = \sum_{k=0}^{N} \binom{N}{k} c^{N-k}\,\mathrm{i}^k\left(1-(-1)^k\right).$$

The bracket is 0 for even k and 2 for odd k. Put $k = 2j+1$ with $j = 0, 1, \ldots, n$; then $\mathrm{i}^k = \mathrm{i}(-1)^j$ and $c^{N-k} = c^{2(n-j)}$. Dividing by 2i, and writing $x = \cot^2\theta$,

$$\frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta} = \sum_{j=0}^{n}(-1)^j\binom{2n+1}{2j+1}x^{n-j} = \binom{2n+1}{1}x^n - \binom{2n+1}{3}x^{n-1} + \cdots + (-1)^n.$$

Call this polynomial $P(x)$. It has degree n, with leading coefficient $2n+1 \neq 0$.

Take $\theta_m = \dfrac{m\pi}{2n+1}$ for $m = 1, \ldots, n$. Then $0 < \theta_m < \frac{1}{2}\pi$, so $\sin\theta_m \neq 0$, and $\sin(2n+1)\theta_m = \sin m\pi = 0$. Hence

$$P(\cot^2\theta_m) = 0, \qquad m = 1, 2, \ldots, n.$$

Ask yourself How do we know these are *all* the solutions, and not just some of them?

On $(0, \frac{1}{2}\pi)$, $\cot\theta$ is positive and strictly decreasing, so $\cot^2\theta$ is strictly decreasing too. The angles $\theta_1 < \theta_2 < \cdots < \theta_n$ all lie in this interval, so the n numbers $\cot^2\theta_m$ are distinct. A polynomial of degree n has at most n roots, so these are exactly the solutions of $P(x) = 0$.

Part (ii)

The sum of the roots of $a_n x^n + a_{n-1}x^{n-1} + \cdots$ is $-a_{n-1}/a_n$. Here $a_n = \binom{2n+1}{1}$ and $a_{n-1} = -\binom{2n+1}{3}$, so

$$\sum_{m=1}^{n}\cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{\binom{2n+1}{3}}{\binom{2n+1}{1}} = \frac{(2n+1)(2n)(2n-1)}{6\,(2n+1)} = \frac{n(2n-1)}{3},$$

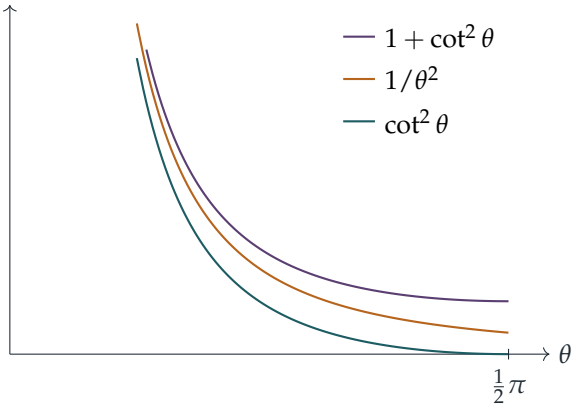
as required.

Part (iii)

For $0 < \theta < \frac{1}{2}\pi$ we are given $0 < \sin\theta < \theta < \tan\theta$. All three are positive, so squaring keeps the order, and taking reciprocals reverses it:

$$\frac{1}{\tan^2\theta} < \frac{1}{\theta^2} < \frac{1}{\sin^2\theta}, \qquad \text{that is} \qquad \cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta,$$

using $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$.



Put $\theta = \theta_m = \dfrac{m\pi}{2n+1}$, so that $\dfrac{1}{\theta_m^2} = \dfrac{(2n+1)^2}{\pi^2}\cdot\dfrac{1}{m^2}$, and add the inequalities for $m = 1, \ldots, n$. Writing $S_n = \sum_{m=1}^{n} \frac{1}{m^2}$ and using part (ii),

$$\frac{n(2n-1)}{3} < \frac{(2n+1)^2}{\pi^2} S_n < n + \frac{n(2n-1)}{3} = \frac{2n(n+1)}{3}.$$

Multiplying through by $\pi^2/(2n+1)^2 > 0$,

$$\frac{\pi^2}{3}\cdot\frac{n(2n-1)}{(2n+1)^2} < S_n < \frac{\pi^2}{3}\cdot\frac{2n(n+1)}{(2n+1)^2}.$$

Dividing top and bottom by n^2, both fractions tend to $\frac{2}{4} = \frac{1}{2}$ as $n \to \infty$, so both bounds tend to $\frac{\pi^2}{6}$. By the squeeze (sandwich) theorem, $S_n \to \frac{\pi^2}{6}$, which means

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6}.$$

Checks

- For $n = 1$ the polynomial is $3x - 1$, with root $\frac{1}{3} = \cot^2\frac{\pi}{3}$, and the formula in (ii) gives $1 \times 1/3 = \frac{1}{3}$ too. For $n = 2$, $\cot^2\frac{\pi}{5} + \cot^2\frac{2\pi}{5}$ should be 2; numerically $1.894 + 0.106 = 2.000$.
- With $n = 50$ the bounds in (iii) are $1.5964 < S_{50} = 1.6251 < 1.6448$, and $\pi^2/6 \approx 1.6449$. The bounds close in on each other, as they must, since their difference is $\pi^2 n/(2n+1)^2 \approx \pi^2/(4n)$.

Common slips

- Showing that each $\cot^2\theta_m$ is a root but forgetting to argue that they are distinct, which is what makes them *all* the solutions.

- Getting the sign in Vieta's formula wrong: the coefficient of x^{n-1} is $-\binom{2n+1}{3}$, so the sum of roots is $+\binom{2n+1}{3}/\binom{2n+1}{1}$.
- Reversing the inequalities without noting that all quantities are positive, which is what allows squaring and taking reciprocals.

Beyond STEP

Euler first found $\sum 1/m^2 = \pi^2/6$ in 1734 by treating $\sin x/x$ as an "infinite polynomial" and comparing coefficients, the same Vieta idea used here, but without the rigour that the squeeze supplies. At university you will meet this sum again through Fourier series (Parseval's identity) and as $\zeta(2)$, the first value of the Riemann zeta function; the values $\zeta(2k)$ are all rational multiples of π^{2k}, while very little is known about $\zeta(3)$. For the analysis behind limits and the squeeze theorem, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2018, Question 8

Pure • Integrals of Fractional Parts

STEP III 2018, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q8

The key idea

The awkward thing in the integrand is x^{-1}, so substitute $y = 1/x$: this sends $[0,1]$ to $[1,\infty)$, which splits naturally into unit intervals. If f has period 1, each unit interval can be shifted back to $[0,1]$, and the leftover rational factors form a telescoping sum. Part (ii) is then a matter of spotting that $\{x\}$ and $\{2x\}$ both have period 1.

Solution

Part (i)

Put $x = 1/y$, so $\mathrm{d}x = -\frac{1}{y^2}\,\mathrm{d}y$; as x runs from 0 to 1, y runs from ∞ down to 1. Also $1 + x = \frac{y+1}{y}$, so

$$I = \int_{\infty}^{1} \frac{\mathrm{f}(y)}{(y+1)/y}\left(-\frac{1}{y^2}\right)\mathrm{d}y = \int_{1}^{\infty} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y.$$

Splitting $[1,\infty)$ into the intervals $[n, n+1]$ for $n = 1, 2, 3, \ldots$ gives

$$I = \sum_{n=1}^{\infty} \int_{n}^{n+1} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y,$$

as required.

Now suppose $\mathrm{f}(x) = \mathrm{f}(x+1)$ for all x, so that $\mathrm{f}(x+n) = \mathrm{f}(x)$ for every integer n. In the nth integral put $y = x + n$, so that x runs over $[0,1]$:

$$\int_{n}^{n+1} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y = \int_{0}^{1} \frac{\mathrm{f}(x)}{(x+n)(x+n+1)}\,\mathrm{d}x.$$

? **Ask yourself** How do we add up infinitely many integrals like this?

Every integral has the same $f(x)$ and range, so (ignoring convergence, as we are told to) we can swap the sum and the integral and add up the rational factors first. By partial fractions,

$$\frac{1}{(x+n)(x+n+1)} = \frac{1}{x+n} - \frac{1}{x+n+1},$$

so the sum telescopes: for fixed $x \in [0,1]$,

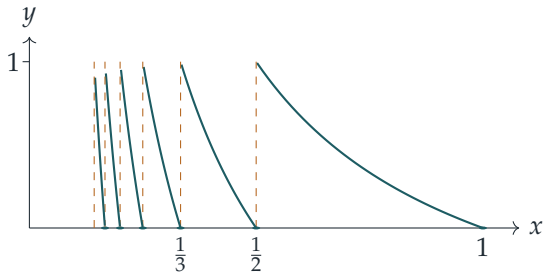
$$\sum_{n=1}^{N} \left(\frac{1}{x+n} - \frac{1}{x+n+1} \right) = \frac{1}{x+1} - \frac{1}{x+N+1} \longrightarrow \frac{1}{1+x} \quad \text{as } N \to \infty.$$

Hence

$$I = \int_0^1 f(x) \sum_{n=1}^{\infty} \frac{1}{(x+n)(x+n+1)} \, dx = \int_0^1 \frac{f(x)}{1+x} \, dx,$$

which is the result we wanted.

Part (ii)



The graph of $\{x^{-1}\}$ above shows why a direct attack is hopeless: on each interval $\left(\frac{1}{n+1}, \frac{1}{n}\right]$ it equals $\frac{1}{x} - n$, and there are infinitely many such pieces crowding towards 0. Part (i) avoids all of this.

The first integral. Take $f(x) = \{x\}$. Adding 1 to x adds 1 to $\lfloor x \rfloor$, so $\{x+1\} = \{x\}$ and f has period 1. On $[0,1)$ we have $\{x\} = x$ (the value at the single point $x = 1$ doesn't affect the integral). So by part (i),

$$\int_0^1 \frac{\{x^{-1}\}}{1+x} \, dx = \int_0^1 \frac{x}{1+x} \, dx = \int_0^1 \left(1 - \frac{1}{1+x}\right) dx = 1 - \ln 2.$$

The second integral. Take $f(x) = \{2x\}$, so that $f(x^{-1}) = \{2x^{-1}\}$ as needed.

Ask yourself The function $\{2x\}$ has period $\frac{1}{2}$. Can we still use part (i)?

Yes: anything with period $\frac{1}{2}$ also has period 1, since $\{2(x+1)\} = \{2x+2\} = \{2x\}$. So part (i) applies. On $[0, \frac{1}{2})$, $2x \in [0,1)$ and $\{2x\} = 2x$; on $[\frac{1}{2}, 1)$, $2x \in [1,2)$ and $\{2x\} = 2x - 1$. Therefore

$$\int_0^1 \frac{\{2x^{-1}\}}{1+x} \, dx = \int_0^{1/2} \frac{2x}{1+x} \, dx + \int_{1/2}^1 \frac{2x-1}{1+x} \, dx.$$

Writing $\frac{2x}{1+x} = 2 - \frac{2}{1+x}$ and $\frac{2x-1}{1+x} = 2 - \frac{3}{1+x}$,

$$\int_0^{1/2} \frac{2x}{1+x}\,\mathrm{d}x = 1 - 2\ln\tfrac{3}{2},$$
$$\int_{1/2}^{1} \frac{2x-1}{1+x}\,\mathrm{d}x = 1 - 3\ln\frac{2}{3/2} = 1 - 3\ln\tfrac{4}{3}.$$

Adding, and collecting logarithms,

$$\begin{aligned}\int_0^1 \frac{\{2x^{-1}\}}{1+x}\,\mathrm{d}x &= 2 - 2\ln 3 + 2\ln 2 - 6\ln 2 + 3\ln 3 \\ &= 2 + \ln 3 - 4\ln 2 = 2 + \ln\tfrac{3}{16}.\end{aligned}$$

Checks

- Both answers should lie between 0 and $\int_0^1 \frac{\mathrm{d}x}{1+x} = \ln 2 \approx 0.693$, since $0 \le \{\,\cdot\,\} < 1$. We get $1 - \ln 2 \approx 0.3069$ and $2 + \ln\frac{3}{16} \approx 0.3260$, which are sensible: the fractional part averages about $\frac{1}{2}$, so the answers should be roughly half of 0.693.
- Summing the integrals over the pieces $\left(\frac{1}{n+1}, \frac{1}{n}\right]$ (and halves of them for the second integral) for many thousands of pieces, by computer, gives 0.306853 and 0.326024, matching both answers.

Common slips

- Getting the limits the wrong way round after $y = 1/x$ and losing the minus sign; the minus from $\mathrm{d}x$ and the reversed limits cancel.
- Assuming $\{2x\} = 2x$ on all of $[0,1]$: it drops by 1 at $x = \frac{1}{2}$.
- Thinking part (i) needs period exactly 1; any function with period $\frac{1}{k}$ (for a positive integer k) also has period 1.

Beyond STEP

Swapping an infinite sum with an integral, which we did here without comment, is exactly the kind of step that needs justification at university; theorems such as monotone and dominated convergence say when it is safe (here every term is non-negative when $f \ge 0$, which is the easy case). The same substitution idea gives famous integrals of the fractional part, such as $\int_0^1 \{1/x\}\,\mathrm{d}x = 1 - \gamma$, where γ is Euler's constant. For the careful treatment of limits, series and integrals, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2018, Question 9

Mechanics • Collisions and Recurrence Relations

STEP III 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q9

The key idea

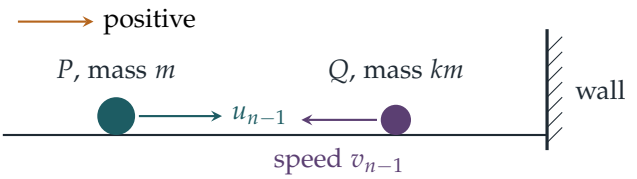
Each collision between P and Q gives two linear equations, one from conservation of momentum and one from Newton's law of restitution, linking the velocities before and after. The wall simply reverses Q's velocity, so the state before one collision is fixed by the state after the previous one. That gives a pair of first-order recurrences for u_n and v_n, and eliminating v leaves a single second-order recurrence for u_n. Part (ii) is then the usual business of fitting a solution of a linear recurrence to its first two terms.

Solution

We model P and Q as particles moving along one line on a smooth floor, so no horizontal forces act between collisions and each particle keeps a constant velocity. Take the direction towards the wall as positive throughout.

What happens between collisions. After the nth collision Q has velocity v_n (towards the wall, since it collides with the wall next). The wall collision is perfectly elastic, so Q comes back with velocity $-v_n$, while P still has velocity u_n. So just before collision $n+1$ the velocities are u_n (for P) and $-v_n$ (for Q).

This also holds for the first collision if we read v_0 the right way: before it, Q moves away from the wall with speed v_0, so its velocity is $-v_0$, exactly the pattern above with $n = 0$.



Part (i)

For the nth collision ($n \geq 1$), conservation of momentum (dividing through by m) gives

$$u_{n-1} - kv_{n-1} = u_n + kv_n. \tag{1}$$

The speed of approach is $u_{n-1} - (-v_{n-1}) = u_{n-1} + v_{n-1}$. Afterwards Q, which is nearer the wall, moves towards the wall faster than P, so the speed of separation is $v_n - u_n$. Newton's law of restitution gives

$$v_n - u_n = e\,(u_{n-1} + v_{n-1}). \tag{2}$$

Substituting v_n from (2) into (1):

$$u_{n-1} - kv_{n-1} = (1+k)u_n + ke\,(u_{n-1} + v_{n-1}),$$

so

$$k(1+e)\,v_{n-1} = (1-ke)\,u_{n-1} - (1+k)\,u_n \qquad (n \geq 1). \tag{3}$$

This expresses each v in terms of two consecutive u's, which is exactly what we need to eliminate v.

Ask yourself Equation (3) gets rid of v_{n-1}, but (2) also contains v_n. How do we deal with that?

Equation (3) holds for every $n \geq 1$, so we may also use it with $n+1$ in place of n:

$$k(1+e)\,v_n = (1-ke)\,u_n - (1+k)\,u_{n+1}.$$

Now multiply (2) by $k(1+e)$ and replace both $k(1+e)v_n$ and $k(1+e)v_{n-1}$ using these:

$$(1-ke)u_n - (1+k)u_{n+1} - k(1+e)u_n = ke(1+e)\,u_{n-1} + e\big[(1-ke)u_{n-1} - (1+k)u_n\big].$$

Collect terms. The coefficient of u_{n-1} on the right is $ke + ke^2 + e - ke^2 = e(1+k)$. Moving the u_n terms to the left, their coefficient is

$$1 - ke - k - ke + e + ke = 1 - k + e - ke = (1+e)(1-k).$$

Hence

$$(1+k)u_{n+1} - (1-k)(1+e)u_n + e(1+k)u_{n-1} = 0 \qquad (n \geq 1),$$

and replacing n by $n-1$ gives $(*)$ for $n \geq 2$, as required.

Part (ii)

With $e = \frac{1}{2}$ and $k = \frac{1}{34}$, multiplying $(*)$ by 68 gives

$$70u_n - 99u_{n-1} + 35u_{n-2} = 0,$$

whose auxiliary equation $70t^2 - 99t + 35 = (10t-7)(7t-5) = 0$ has roots $\frac{7}{10}$ and $\frac{5}{7}$, matching the given form.

Two conditions fix A and B, and we need the first two terms. We know u_0. For u_1, use (3) with $n = 1$:

$$(1+k)u_1 = (1-ke)u_0 - k(1+e)v_0,$$
$$\text{so} \quad \tfrac{35}{34}u_1 = \tfrac{67}{68}u_0 - \tfrac{3}{68}v_0, \quad \text{giving} \quad u_1 = \frac{67u_0 - 3v_0}{70}.$$

Putting $n = 0$ and $n = 1$ into the given form,

$$A + B = u_0, \qquad \tfrac{7}{10}A + \tfrac{5}{7}B = \frac{67u_0 - 3v_0}{70}.$$

Multiplying the second equation by 70 gives $49A + 50B = 67u_0 - 3v_0$. Subtracting 49 times the first,

$$B = 18u_0 - 3v_0, \qquad A = u_0 - B = 3v_0 - 17u_0.$$

Sign of u_n for large n. Since $\frac{5}{7} > \frac{7}{10}$ (as $50 > 49$), the B term dominates. Precisely,

$$u_n = \left(\tfrac{5}{7}\right)^n \left[\, B + A\left(\tfrac{49}{50}\right)^n \right],$$

and $\left(\frac{49}{50}\right)^n \to 0$, so the bracket tends to B. If $0 < 6u_0 < v_0$ then $B = 3(6u_0 - v_0) < 0$, so the bracket is negative for all sufficiently large n. Since $\left(\frac{5}{7}\right)^n > 0$, u_n is negative for large n, as required. (Concretely, the bracket is negative once $|A|(\frac{49}{50})^n < |B|$.)

Physically, P is eventually knocked back away from the wall: Q has enough initial speed relative to P to keep driving it back, even though Q is much lighter.

Checks

- Take $e = 1$, $k = 1$ (equal masses, perfectly elastic): $(*)$ becomes $2u_n + 2u_{n-2} = 0$, so $u_n = -u_{n-2}$. Indeed equal masses colliding elastically swap velocities, so $u_1 = -v_0$ and $u_2 = -v_1$; then Q carries P's old velocity u_0 to the wall and back, giving $u_2 = -u_0$.
- At $n = 0$ the formula gives $A + B = u_0$, and at $n = 1$ it gives

  $$\tfrac{7}{10}(3v_0 - 17u_0) + \tfrac{5}{7}(18u_0 - 3v_0) = \tfrac{1}{70}(67u_0 - 3v_0),$$

  as it should.
- With $u_0 = 1$, $v_0 = 7$ we get $A = 4$, $B = -3$. Stepping through the collisions one at a time confirms the formula exactly, and u_n first becomes negative at $n = 15$, where $u_{15} \approx -0.00029$.

Common slips

- Forgetting that Q's velocity just before a collision is $-v_{n-1}$, not v_{n-1}: the wall reverses it.

- Getting the sign of the separation speed wrong in the restitution equation; after the collision Q is ahead of P in the positive direction, so it is $v_n - u_n$.
- Using $u_1 = \frac{7}{10}A + \frac{5}{7}B$ without having found u_1 from the collision equations; $(*)$ alone cannot give it.

Beyond STEP

The pair of equations (1) and (2) is a linear map taking the state (u_{n-1}, v_{n-1}) to (u_n, v_n), so the whole motion is repeated multiplication by a 2×2 matrix. The roots $\frac{7}{10}$ and $\frac{5}{7}$ are its eigenvalues, and long-term behaviour is governed by the largest one: this is the start of the theory of linear dynamical systems and difference equations. For more on the mechanics of collisions, see Kleppner and Kolenkow, *An Introduction to Mechanics*; for the matrix viewpoint, Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 2018, Question 10

Mechanics • Compound Pendulum

STEP III 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q10

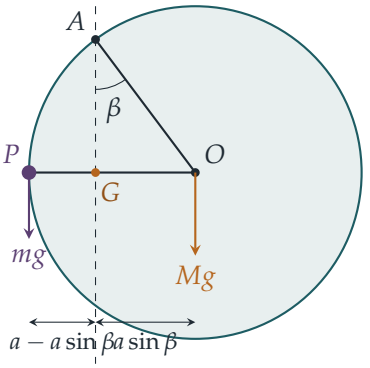
The key idea

Disc and particle together form a single rigid body turning about the fixed axis L, so everything is controlled by one angle. In equilibrium the combined centre of mass must lie directly below A, which is a moment balance about A. For the motion, energy is conserved because the axis is smooth and the reaction there does no work. Once the energy is written as a function of θ and $\dot{\theta}$, differentiating it gives the equation of motion, and small oscillations follow by replacing $\sin\theta$ with θ.

Solution

We model the disc as a uniform rigid lamina, the particle as a point mass fixed to its rim, and the axis L as smooth.

Equilibrium. The only forces with a moment about A are the two weights: Mg acting at O and mg acting at P. The reaction at A passes through A. Since AO makes angle β with the downward vertical, O is a horizontal distance $a\sin\beta$ to one side of A, say to the right. The weight Mg then turns the disc clockwise about A, and only mg can balance it, so P must be on the *left* of the vertical through A. Because OP is horizontal and has length a, the horizontal distance of P to the left of A is $a - a\sin\beta$. (Here $a\sin\beta < a$, so P really is on the left of the vertical.)



Taking moments about A,

$$Mg\,a\sin\beta = mg\,(a - a\sin\beta) \quad\Longrightarrow\quad \sin\beta = \frac{m}{M+m}.$$

Both O and P are a vertical distance $a\cos\beta$ below A. So, in the triangle with AP as hypotenuse, the horizontal side is $a(1-\sin\beta)$ and the vertical side is $a\cos\beta$:

$$AP^2 = a^2(1-\sin\beta)^2 + a^2\cos^2\beta = a^2(2-2\sin\beta)$$
$$= 2a^2\Big(1 - \frac{m}{M+m}\Big) = \frac{2Ma^2}{M+m},$$

so $AP/a = \sqrt{2M/(M+m)}$, as required.

It's worth noting where the combined centre of mass G is. It lies on the segment OP, because it is a weighted average of O and P, so it is at depth $a\cos\beta$ below A. In equilibrium it is directly below A, so

$$AG = a\cos\beta.$$

Conservation of energy. The disc and particle move as one rigid body. In equilibrium OP is horizontal, and OP turns with the body, so when OP makes angle θ with the horizontal, the whole body has turned through θ from its equilibrium position. Its angular speed is therefore $|\dot\theta|$.

Ask yourself How fast is the particle moving?

Every point of the body moves on a circle centred on the axis. The particle stays a fixed distance AP from A, so its speed is $AP\,|\dot\theta|$ and its kinetic energy is

$$\tfrac12 m\,AP^2\,\dot\theta^2 = \tfrac12 m\cdot 2a^2(1-\sin\beta)\,\dot\theta^2 = (1-\sin\beta)\,ma^2\dot\theta^2,$$

using $AP^2 = 2a^2(1-\sin\beta)$ from above. The disc's kinetic energy is $\tfrac12 I\dot\theta^2$.

For the potential energy, treat the total mass $M+m$ as concentrated at G. The line AG is vertical in equilibrium, and it has also turned through θ, so G is now a depth $a\cos\beta\cos\theta$ below A. Measuring potential energy from the equilibrium position,

$$\text{PE} = (M+m)g\big(a\cos\beta - a\cos\beta\cos\theta\big) = (M+m)ga\cos\beta\,(1-\cos\theta).$$

The axis is smooth and the reaction at A acts at a point that doesn't move, so it does no work. Gravity is the only other force, and it is conservative. So kinetic plus potential energy is constant:

$$\tfrac{1}{2}I\dot\theta^2 + (1-\sin\beta)\,ma^2\dot\theta^2 + (m+M)ga\cos\beta\,(1-\cos\theta) = \text{constant},$$

which is the required result.

Small oscillations. Differentiating with respect to t:

$$\dot\theta\Big[\big(I + 2(1-\sin\beta)ma^2\big)\ddot\theta + (M+m)ga\cos\beta\,\sin\theta\Big] = 0.$$

The bracket must vanish whenever $\dot\theta \neq 0$, and by continuity it vanishes at every instant. For small θ, $\sin\theta \approx \theta$, so

$$\ddot\theta = -\omega^2\theta, \qquad \omega^2 = \frac{(M+m)ga\cos\beta}{I + 2(1-\sin\beta)ma^2},$$

which is simple harmonic motion with period $2\pi/\omega$.

Now put $m = \tfrac{3}{2}M$. Then $\sin\beta = \dfrac{3/2}{5/2} = \tfrac{3}{5}$, and since β is acute, $\cos\beta = \tfrac{4}{5}$. The numerator is

$$\tfrac{5}{2}M \cdot ga \cdot \tfrac{4}{5} = 2Mga,$$

and with $I = \tfrac{3}{2}Ma^2$ the denominator is

$$\tfrac{3}{2}Ma^2 + 2\cdot\tfrac{2}{5}\cdot\tfrac{3}{2}Ma^2 = \tfrac{3}{2}Ma^2 + \tfrac{6}{5}Ma^2 = \tfrac{27}{10}Ma^2.$$

So $\omega^2 = \dfrac{20g}{27a}$, and the period is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{27a}{20g}} = 2\pi\cdot\frac{3}{2}\sqrt{\frac{3a}{5g}} = 3\pi\sqrt{\frac{3a}{5g}},$$

using $\sqrt{27/20} = \sqrt{9\cdot 3/(4\cdot 5)} = \tfrac{3}{2}\sqrt{3/5}$. This is the required period.

Checks

- With $m = 0$ we get $\sin\beta = 0$ and $AP/a = \sqrt{2}$. The disc hangs with O below A, and a rim point level with O is indeed $\sqrt{2}\,a$ from A. The period becomes $2\pi\sqrt{(3a/2)/g}$, the familiar result for a disc pivoted at its rim.
- The kinetic-energy terms add to $\tfrac{1}{2}(I + m\,AP^2)\dot\theta^2$, the moment of inertia of the whole body about L, as they should.
- A step-by-step computer simulation of the rigid body, with $m = \tfrac{3}{2}M$ and $a = 0.5\,\text{m}$, gave a small-oscillation period of $1.64815\,\text{s}$. The formula gives $3\pi\sqrt{1.5/49.05} \approx 1.64815\,\text{s}$. The simulation also showed that the energy expression stays constant for large swings.

Common slips

- Putting P on the same side of the vertical as O. Then the two weights turn the body the same way and there can be no equilibrium.
- Using a instead of AP for the particle's distance from the axis. The particle moves on a circle about A, not about O.
- Finding the potential energy of O and P separately and getting lost in the trigonometry. Using the combined centre of mass G, at distance $a\cos\beta$ from A, is much cleaner.

Beyond STEP

This is a *compound pendulum*: any rigid body swinging about a fixed axis oscillates like a simple pendulum of length $I_{\text{total}}/\big((M+m)h\big)$, where h is the distance from the axis to the centre of mass. Our energy method is the one-variable case of Lagrangian mechanics, in which the equations of motion come from the kinetic and potential energies written in a single generalised coordinate. Kleppner and Kolenkow's *An Introduction to Mechanics* treats rigid-body rotation and the physical pendulum carefully.

WORKED SOLUTION

STEP III 2018, Question 11

Mechanics • Slack String Projectile

STEP III 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

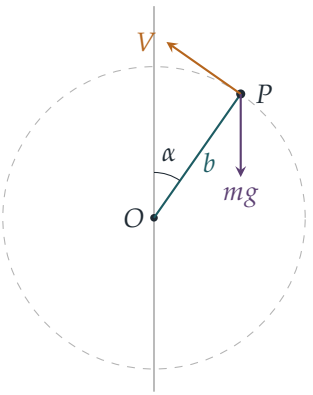
step.maths.org/questions/18-s3-q11

The key idea

The string goes slack at the moment its tension falls to zero, so the radial equation of motion at that instant gives V. After that the particle is an ordinary projectile, and the string becomes taut again when the particle's distance from O returns to b. Working in coordinates centred on O, that condition collapses beautifully, because the projectile starts on the circle moving along the tangent. The final part is then a question of whether the velocity at the jerk points straight along the string.

Solution

We model the particle as a point mass m on a light inextensible string, with no air resistance. Take O as origin, with x horizontal and y vertically upwards, and suppose the particle swings anticlockwise so that it goes slack at the point P on the right of the upward vertical.



At P the string makes angle α with the upward vertical, so $P = (b\sin\alpha,\ b\cos\alpha)$. The

velocity there is perpendicular to the string and directed up and over the top:

$$\mathbf{v}_P = V(-\cos\alpha,\ \sin\alpha).$$

The speed when the string goes slack

Resolve towards O along the string. The tension R acts towards O, and since P is above O the weight has component $mg\cos\alpha$ towards O as well. The particle is moving in a circle of radius b, so

$$R + mg\cos\alpha = \frac{mV^2}{b}.$$

The string becomes slack when $R = 0$, so

$$V^2 = gb\cos\alpha, \qquad V = \sqrt{gb\cos\alpha}.$$

When the string is taut again

Once slack, the particle is a projectile. After time t its position is

$$x = b\sin\alpha - Vt\cos\alpha, \qquad y = b\cos\alpha + Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

The string is taut again when $x^2 + y^2 = b^2$. Expanding,

$$\begin{aligned} x^2 + y^2 &= b^2\sin^2\alpha - 2bVt\sin\alpha\cos\alpha + V^2t^2\cos^2\alpha \\ &\quad + b^2\cos^2\alpha + V^2t^2\sin^2\alpha + \tfrac{1}{4}g^2t^4 + 2bVt\sin\alpha\cos\alpha - bgt^2\cos\alpha - gVt^3\sin\alpha \\ &= b^2 + \left(V^2 - gb\cos\alpha\right)t^2 - gVt^3\sin\alpha + \tfrac{1}{4}g^2t^4. \end{aligned}$$

The terms in t cancel because the particle starts out moving along the tangent, and the terms in t^2 cancel because $V^2 = gb\cos\alpha$. Hence

$$x^2 + y^2 - b^2 = gt^3\left(\tfrac{1}{4}gt - V\sin\alpha\right).$$

For $0 < t < 4V\sin\alpha/g$ this is negative, so the particle is strictly inside the circle and the string stays slack. It is zero again (and becomes positive afterwards) at the time T with

$$gT = 4V\sin\alpha,$$

as required.

The direction of motion just before

At time T the velocity components are

$$\dot{x} = -V\cos\alpha, \qquad \dot{y} = V\sin\alpha - gT = V\sin\alpha - 4V\sin\alpha = -3V\sin\alpha.$$

So the particle is moving down and to the left, and the angle β its path makes with the horizontal satisfies

$$\tan\beta = \frac{|\dot{y}|}{|\dot{x}|} = \frac{3V\sin\alpha}{V\cos\alpha} = 3\tan\alpha,$$

as required.

Coming to rest

Ask yourself What has to be true of the velocity for the particle to stop dead?

The jerk destroys only the component of momentum along the string. The component perpendicular to the string is untouched, so the particle comes to rest if and only if that perpendicular component is already zero, that is, if and only if the velocity just before is parallel to the string $\overrightarrow{OQ}$, where Q is the particle's position at time T. (The velocity is not zero, since $\dot{x} = -V\cos\alpha \neq 0$.)

Using $V^2 = gb\cos\alpha$ and $T = 4V\sin\alpha/g$,

$$x(T) = b\sin\alpha - \frac{4V^2\sin\alpha\cos\alpha}{g} = b\sin\alpha\,(1 - 4\cos^2\alpha),$$
$$y(T) = b\cos\alpha + \frac{4V^2\sin^2\alpha}{g} - \frac{8V^2\sin^2\alpha}{g} = b\cos\alpha\,(1 - 4\sin^2\alpha).$$

The vectors (x, y) and $(\dot{x}, \dot{y})$ are parallel if and only if $x\dot{y} - y\dot{x} = 0$. Dividing by $-bV$,

$$3\sin^2\alpha\,(1 - 4\cos^2\alpha) - \cos^2\alpha\,(1 - 4\sin^2\alpha) = 0.$$

Write $S = \sin^2\alpha$, so $\cos^2\alpha = 1 - S$ and $1 - 4\cos^2\alpha = 4S - 3$:

$$3S(4S - 3) - (1 - S)(1 - 4S) = 0$$
$$12S^2 - 9S - (1 - 5S + 4S^2) = 0$$
$$8S^2 - 4S - 1 = 0,$$

so $S = \frac{1}{16}\left(4 \pm \sqrt{16 + 32}\right) = \frac{1}{4}\left(1 \pm \sqrt{3}\right)$. Since $S = \sin^2\alpha > 0$, the negative root is impossible, and the remaining root lies in $(0, 1)$, so it corresponds to a genuine acute angle. Every step above is reversible, so the particle comes instantaneously to rest when the string becomes taut if and only if

$$\sin^2\alpha = \frac{1 + \sqrt{3}}{4}.$$

Checks

- The formula $V^2 = gb\cos\alpha$ gives $V \to 0$ as $\alpha \to \frac{1}{2}\pi$: the string goes slack at the level of O only if the particle has stopped there, which is right.
- With $\sin^2\alpha = \frac{1}{4}(1 + \sqrt{3})$ we get $\alpha \approx 55.7°$ and $\cos^2\alpha = \frac{1}{4}(3 - \sqrt{3})$, so $Q = \left(b\sin\alpha\,(\sqrt{3} - 2),\ -\sqrt{3}\,b\cos\alpha\right)$, below and to the left of O. The slope of OQ is $\sqrt{3}\cot\alpha/(2 - \sqrt{3})$ and the slope of the path is $3\tan\alpha$. These agree, because

$$3(2 - \sqrt{3})\tan^2\alpha = \frac{3(2 - \sqrt{3})(1 + \sqrt{3})}{3 - \sqrt{3}} = \frac{3(\sqrt{3} - 1)}{\sqrt{3}(\sqrt{3} - 1)} = \sqrt{3}.$$

  A computer simulation of the motion confirmed all three results.

Common slips

- Using energy alone to find V: energy links V to the initial speed, but it is the zero-tension condition that fixes V in terms of b, g and α.
- Getting the velocity direction at P wrong. It is perpendicular to OP, so it makes angle α with the horizontal, not with the vertical.
- Forgetting to justify that T is the *first* time the string is taut; the factorisation $gt^3(\frac{1}{4}gt - V\sin\alpha)$ shows it is.

Beyond STEP

A string can pull but not push, so a slack-string problem is really a constrained system whose constraint switches on and off; the same idea of one-sided constraints appears in contact mechanics and in Lagrangian mechanics with inequality constraints. The sudden jerk is an impulsive constraint force, which is treated in first-year courses alongside collisions and the coefficient of restitution. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both circular motion and impulses carefully.

WORKED SOLUTION

STEP III 2018, Question 12

Probability and Statistics • Order Statistics

STEP III 2018, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q12

The key idea

The event "the kth smallest number is at most y" is the same as "at least k of the n numbers are at most y", and the number of values at most y is binomial. That gives the distribution function straight away. Differentiating it produces two sums, and the two binomial identities in (ii) are exactly what makes them telescope. Once we know the density, the fact that it integrates to 1 evaluates an integral for free, and the same integral (with n and k shifted) gives the mean.

Solution

Call the n numbers $X_1, \ldots, X_n$. They are independent, each uniform on $[0,1]$.

Part (i)

Fix y with $0 \le y \le 1$ and let N be the number of the X_i with $X_i \le y$. Each X_i satisfies $X_i \le y$ with probability y, independently of the others, so $N \sim \mathrm{B}(n, y)$.

Now $Y_k \le y$ exactly when at least k of the numbers are at most y. If $Y_k \le y$, then Y_k and the $k-1$ numbers below it are all at most y, so $N \ge k$. Conversely, if $N \ge k$, then the k smallest numbers are all at most y, and in particular $Y_k \le y$. Hence

$$\mathrm{P}(Y_k \le y) = \mathrm{P}(N \ge k) = \sum_{m=k}^{n} \binom{n}{m} y^m (1-y)^{n-m},$$

which is $(*)$.

Part (ii)

Writing out the factorials, for $1 \le m \le n$,

$$m\binom{n}{m} = \frac{n!}{(m-1)!\,(n-m)!} = n \cdot \frac{(n-1)!}{(m-1)!\,\big((n-1)-(m-1)\big)!} = n\binom{n-1}{m-1},$$

and similarly, for $0 \le m \le n-1$,

$$(n-m)\binom{n}{m} = \frac{n!}{m!\,(n-m-1)!} = n \cdot \frac{(n-1)!}{m!\,\big((n-1)-m\big)!} = n\binom{n-1}{m}.$$

(When $m = n$ the left-hand side is 0, and so is the right if we read $\binom{n-1}{n}$ as 0.)

The density. The density $f(y)$ of Y_k is the derivative of $(*)$. Differentiating each term by the product rule,

$$f(y) = \sum_{m=k}^{n} \binom{n}{m} \Big[m\, y^{m-1}(1-y)^{n-m} - (n-m)\, y^{m}(1-y)^{n-m-1} \Big].$$

In the second piece the $m = n$ term vanishes because of the factor $n - m$, so that sum really runs to $m = n-1$. Using the two identities,

$$f(y) = n \sum_{m=k}^{n} \binom{n-1}{m-1} y^{m-1}(1-y)^{n-m} \;-\; n \sum_{m=k}^{n-1} \binom{n-1}{m} y^{m}(1-y)^{n-1-m}.$$

> **Ask yourself** These two sums look almost the same. How can we make them line up?

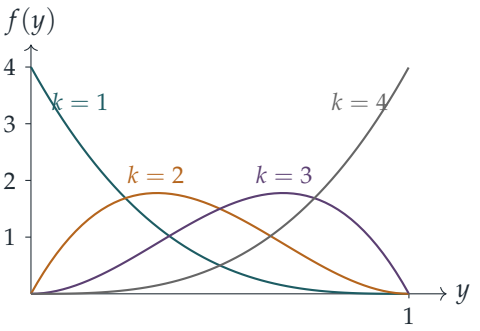
Put $j = m+1$ in the second sum. It becomes

$$n \sum_{j=k+1}^{n} \binom{n-1}{j-1} y^{j-1}(1-y)^{n-j},$$

which is the first sum term for term, except that it starts at $k+1$ rather than k. Everything cancels except the $m = k$ term of the first sum:

$$f(y) = n\binom{n-1}{k-1} y^{k-1}(1-y)^{n-k},$$

as required. (If $k = n$ the second sum is empty and the first has only the term $m = n$, so the result still holds.)



The densities for $n = 4$: the kth smallest of four numbers piles up near $k/5$.

The integral. A density integrates to 1 over its range, which here is $[0,1]$. Therefore

$$n\binom{n-1}{k-1}\int_0^1 y^{k-1}(1-y)^{n-k}\,\mathrm{d}y = 1,$$

so

$$\int_0^1 y^{k-1}(1-y)^{n-k}\,\mathrm{d}y = \frac{1}{n\binom{n-1}{k-1}} = \frac{(k-1)!\,(n-k)!}{n!}.$$

This holds for all integers $1 \le k \le n$.

Part (iii)

By definition,

$$\mathrm{E}(Y_k) = \int_0^1 y\,f(y)\,\mathrm{d}y = n\binom{n-1}{k-1}\int_0^1 y^k(1-y)^{n-k}\,\mathrm{d}y.$$

> **Ask yourself** This integral isn't quite the one we just found. Which values of n and k turn it into that form?

The integrand is $y^{K-1}(1-y)^{N-K}$ with $K = k+1$ and $N = n+1$, and $1 \le K \le N$, so the formula from (ii) applies:

$$\int_0^1 y^k(1-y)^{n-k}\,\mathrm{d}y = \frac{k!\,(n-k)!}{(n+1)!}.$$

Since $n\binom{n-1}{k-1} = \dfrac{n!}{(k-1)!\,(n-k)!}$,

$$\mathrm{E}(Y_k) = \frac{n!}{(k-1)!\,(n-k)!}\cdot\frac{k!\,(n-k)!}{(n+1)!} = \frac{k}{n+1}.$$

Checks

- With $n = 1$, $k = 1$ we have a single uniform number, and $\mathrm{E}(Y_1) = \frac{1}{2}$, as it should be.
- Symmetry: replacing every X_i by $1 - X_i$ turns the kth smallest into the kth largest, so $\mathrm{E}(Y_k) + \mathrm{E}(Y_{n+1-k}) = 1$. Indeed $\frac{k}{n+1} + \frac{n+1-k}{n+1} = 1$.
- The n numbers cut $[0,1]$ into $n+1$ gaps, and the answer says each gap has mean length $\frac{1}{n+1}$, which is very plausible by symmetry.
- With $n = 5$, $k = 2$: $(*)$ at $y = 0.3$ gives 0.47178, and a computer simulation of 200 000 samples gave 0.4726 and a sample mean of Y_2 of 0.33333, against $\frac{1}{3}$.

Common slips

- Writing $\mathrm{P}(Y_k \le y) = \binom{n}{k}y^k(1-y)^{n-k}$, the probability of *exactly* k values below y, instead of *at least* k.

- Forgetting that the $m = n$ term of the second sum is zero, which leaves the telescoping looking as if it fails by one term.
- Shifting only k (and not n) when reusing the integral in (iii).

Beyond STEP

The Y_k are the *order statistics* of the sample, and Y_k has a Beta distribution with parameters k and $n-k+1$. The integral in (ii) is the Beta function $B(a,b) = \int_0^1 y^{a-1}(1-y)^{b-1}\,\mathrm{d}y = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$, which you will meet again alongside the Gamma function. Order statistics underlie the sample median and non-parametric statistics. Blitzstein and Hwang, *Introduction to Probability*, treats order statistics and the Beta distribution beautifully, including a "story" proof of the Beta integral much like the argument here.

WORKED SOLUTION

STEP III 2018, Question 13

Probability and Statistics • Poisson Distribution

STEP III 2018, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q13

The key idea

Substituting $t = -1$ into a probability generating function attaches a sign $(-1)^r$ to each probability, so averaging $G(1)$ and $G(-1)$ cancels the odd terms and keeps the even ones. The same filtering with the fourth roots of unity $1, -1, \mathrm{i}, -\mathrm{i}$ picks out the multiples of 4. Once each new generating function is in closed form, the mean is just its derivative at $t = 1$, and the question becomes a comparison of elementary functions.

Solution

Write $p_r = \mathrm{P}(X = r)$, so that $G(t) = \sum_{r \geq 0} p_r t^r$.

Even values. Adding the series at $t = 1$ and $t = -1$,

$$G(1) + G(-1) = \sum_{r \geq 0} p_r \left(1 + (-1)^r\right) = 2 \sum_{r \text{ even}} p_r,$$

because $1 + (-1)^r$ is 2 for even r and 0 for odd r. Dividing by 2 gives the required result. (Both series converge absolutely, since $\sum p_r = 1$.)

The Poisson generating function. With $p_r = \mathrm{e}^{-\lambda} \lambda^r / r!$,

$$G(t) = \mathrm{e}^{-\lambda} \sum_{r \geq 0} \frac{(\lambda t)^r}{r!} = \mathrm{e}^{-\lambda} \mathrm{e}^{\lambda t} = \mathrm{e}^{-\lambda(1-t)},$$

using the exponential series, which converges for every t.

Part (i)

The probabilities of Y must sum to 1, so k is the reciprocal of $\mathrm{P}(X \text{ even})$. By the first result,

$$\mathrm{P}(X \text{ even}) = \tfrac{1}{2}\left(1 + \mathrm{e}^{-2\lambda}\right) = \mathrm{e}^{-\lambda} \cdot \tfrac{1}{2}\left(\mathrm{e}^{\lambda} + \mathrm{e}^{-\lambda}\right) = \mathrm{e}^{-\lambda} \cosh \lambda, \qquad \text{so} \qquad k = \frac{\mathrm{e}^{\lambda}}{\cosh \lambda}.$$

The generating function of Y keeps only the even terms of G:

$$G_Y(t) = k \sum_{r \text{ even}} p_r t^r = k \cdot \tfrac{1}{2}\big(G(t) + G(-t)\big)$$
$$= \frac{\mathrm{e}^{\lambda}}{\cosh\lambda} \cdot \mathrm{e}^{-\lambda} \cdot \tfrac{1}{2}\big(\mathrm{e}^{\lambda t} + \mathrm{e}^{-\lambda t}\big) = \frac{\cosh \lambda t}{\cosh \lambda},$$

where $\frac{1}{2}\big(G(t) + G(-t)\big)$ is the even part of G, exactly as in the first result with t in place of 1. This is the required form.

The mean is $G_Y'(1)$:

$$\mathrm{E}(Y) = \frac{\lambda \sinh \lambda}{\cosh \lambda} = \lambda \tanh \lambda.$$

For $\lambda > 0$ we have $\mathrm{e}^{-\lambda} > 0$, so $\sinh\lambda = \frac{1}{2}(\mathrm{e}^{\lambda} - \mathrm{e}^{-\lambda}) < \frac{1}{2}(\mathrm{e}^{\lambda} + \mathrm{e}^{-\lambda}) = \cosh\lambda$, which gives $0 < \tanh\lambda < 1$. Hence $\mathrm{E}(Y) < \lambda$, as required.

Part (ii)

Now we need to keep only the terms with r a multiple of 4.

> **Ask yourself** What plays the role of $t = -1$ when we want to filter by multiples of 4 rather than 2?

Use all four fourth roots of unity. For a whole number r,

$$1 + (-1)^r + \mathrm{i}^r + (-\mathrm{i})^r = \begin{cases} 4 & \text{if } 4 \mid r, \\ 0 & \text{otherwise.} \end{cases}$$

If $4 \mid r$, every term is 1. If r is odd, the terms cancel in pairs: $(-1)^r = -1$ and $(-\mathrm{i})^r = -\mathrm{i}^r$. If $r \equiv 2 \pmod 4$, the terms are $1 + 1 - 1 - 1 = 0$. So

$$\sum_{4 \mid r} \frac{x^r}{r!} = \tfrac{1}{4}\big(\mathrm{e}^{x} + \mathrm{e}^{-x} + \mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x}\big) = \tfrac{1}{2}\big(\cosh x + \cos x\big).$$

(You can also see this directly: $\cosh x = \sum_{r \text{ even}} x^r/r!$, while $\cos x$ has the same terms with signs $+, -, +, \ldots$, so adding them doubles the terms with $4 \mid r$ and cancels the rest.)

Taking $x = \lambda t$ and multiplying by $\mathrm{e}^{-\lambda}$, and then fixing c by requiring $G_Z(1) = 1$, we get

$$G_Z(t) = \frac{\cosh \lambda t + \cos \lambda t}{\cosh \lambda + \cos \lambda}.$$

The denominator is positive for $\lambda > 0$, because $\cosh\lambda > 1 \geq -\cos\lambda$. Differentiating and setting $t = 1$,

$$\mathrm{E}(Z) = G_Z'(1) = \frac{\lambda(\sinh \lambda - \sin \lambda)}{\cosh \lambda + \cos \lambda}.$$

Since $\lambda > 0$ and the denominator is positive,

$$\begin{aligned} \mathrm{E}(Z) < \lambda &\iff \sinh\lambda - \sin\lambda < \cosh\lambda + \cos\lambda \\ &\iff \sin\lambda + \cos\lambda > \sinh\lambda - \cosh\lambda = -\mathrm{e}^{-\lambda} \\ &\iff \sin\lambda + \cos\lambda + \mathrm{e}^{-\lambda} > 0. \end{aligned}$$

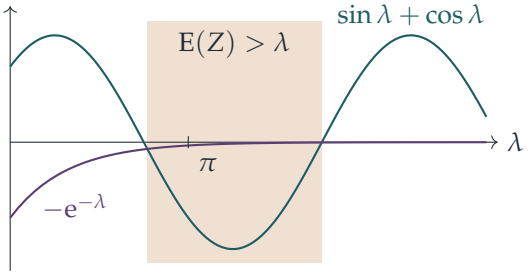
This fails at $\lambda = \pi$: there $\sin\pi + \cos\pi + \mathrm{e}^{-\pi} = -1 + \mathrm{e}^{-\pi} < 0$ because $\mathrm{e}^{-\pi} < 1$. In fact

$$\mathrm{E}(Z)\big|_{\lambda=\pi} = \frac{\pi\sinh\pi}{\cosh\pi - 1} > \pi,$$

since $\sinh\pi > \cosh\pi - 1$ (equivalently $1 > \mathrm{e}^{-\pi}$).

So the answer is **no**: $\mathrm{E}(Z) < \lambda$ does not hold for all positive λ.

For a fuller picture, write $\sin\lambda + \cos\lambda = \sqrt{2}\sin(\lambda + \frac{\pi}{4})$. The condition fails whenever this curve dips below $-\mathrm{e}^{-\lambda}$. The first such interval runs from about $\lambda \approx 2.42$ to $\lambda \approx 5.49$, and later intervals get closer and closer to $\left(\frac{3\pi}{4} + 2n\pi,\ \frac{7\pi}{4} + 2n\pi\right)$ as $\mathrm{e}^{-\lambda} \to 0$.



Checks

- As $\lambda \to 0$, $\mathrm{E}(Y) = \lambda\tanh\lambda \approx \lambda^2$. This makes sense: Y is almost always 0, and $\mathrm{P}(Y = 2) \approx \lambda^2/2$.
- As $\lambda \to \infty$, $\tanh\lambda \to 1$, so $\mathrm{E}(Y)$ is just below λ. For large λ the Poisson distribution is spread out, so throwing away the odd values hardly moves the mean.
- The computer summed the Poisson series directly and confirmed every generating function and both means. For example, at $\lambda = \pi$ it gives $\mathrm{E}(Y) \approx 3.1299 < \pi$ and $\mathrm{E}(Z) \approx 3.4254 > \pi$. It also located the sign changes of $\sin\lambda + \cos\lambda + \mathrm{e}^{-\lambda}$ at $\lambda \approx 2.419$ and 5.495.

Common slips

- Forgetting the normalising constant: $\cosh\lambda t$ alone is not a generating function, because its value at $t = 1$ is not 1.
- Assuming that the answer to (ii) must be "yes" by analogy with (i). Keeping only multiples of 4 can remove more mass below λ than above it, for instance when $\lambda = \pi$: the nearest allowed value 4 is above π and is much heavier than 0.

- Dividing an inequality by $\cosh\lambda + \cos\lambda$ without checking that it is positive.

Beyond STEP

Averaging a generating function over roots of unity to pick out the terms in one residue class is called the *roots-of-unity filter* (or *multisection* of a series). It is a basic tool in combinatorics, and it is the simplest case of the discrete Fourier transform. Conditioning a distribution on an event and then recomputing its mean, as here, comes up throughout probability. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions in depth.

Beyond STEP

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